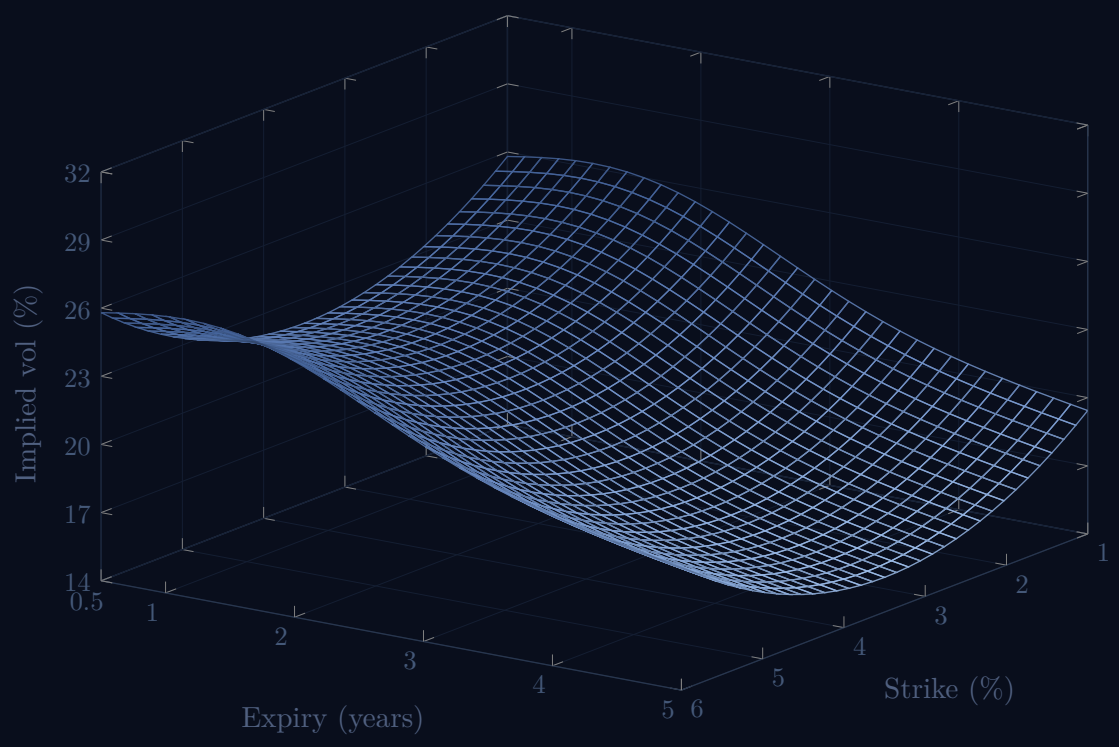
Foundations of Fixed Income Valuation

A Mathematical Introduction



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Preface

This booklet provides a rigorous yet accessible introduction to fixed income mathematics, covering everything from basic interest rate mechanics to advanced derivative pricing. It is designed for students and practitioners entering quantitative finance who seek a comprehensive foundation built on clear definitions, worked examples, and mathematical precision.

What this booklet covers. The fifteen chapters and four appendices are organised in a deliberate sequence. Chapters 1 and 2 lay the foundations: interest rates, compounding conventions, the time value of money, and the discount factor. Chapters 3 and 4 introduce the core cash instruments: fixed-rate bonds and zero-coupon bonds. Chapter 5 introduces spot rates and forward rates. Chapter 6 presents floating-rate bonds. Chapter 7 develops the family of interest rate curves (yield, spot, forward, OIS, Treasury, swap, inflation, par). Chapter 8 covers annuities, deposits, and amortised loans. Chapter 9 introduces duration and convexity as measures of interest rate risk. Chapter 10 presents the general framework of financial derivatives — forwards, futures, swaps, and options — along with the Greeks. Chapters 11, 12, and 13 cover specific derivative instruments: forward rate agreements, interest rate swaps, FX forwards, and cross-currency swaps. Chapter 14 introduces interest rate options (caps, floors, swaptions) and the Black model. Chapter 15 presents portfolio risk measurement through Value at Risk and Expected Shortfall.
The appendices provide essential reference material: probability and statistics fundamentals (Appendix A), the Nelson–Siegel and Svensson parametric curve models (Appendix B), a comprehensive table of notation and symbols (Appendix C), and an alphabetical glossary of all definitions (Appendix D).

What this booklet assumes. A working knowledge of calculus (derivatives, integrals, Taylor series), linear algebra, and basic probability and statistics. No prior exposure to finance is required — every financial concept is defined from scratch. Stochastic calculus is *not* assumed; the few places where it naturally arises (the Black model) are treated at a descriptive level, with pointers to rigorous references.

How to read this booklet. The chapters are meant to be read in order: later chapters reference definitions and results from earlier ones extensively. However, a reader already comfortable with bond pricing could begin at Chapter 5 (spot and forward rates) without difficulty, referring back as needed. The glossary (Appendix D) provides quick access to any definition by term.

Errors and feedback. Despite careful review, errors inevitably creep into any technical manuscript. If you spot a mistake — mathematical, typographical, or conceptual — or if you have suggestions for improvement, I would be grateful to hear from you. Your feedback helps make this booklet better for future readers.

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CHAPTER 1

Interest Rates

1.1 What is an interest rate?

When you lend money to someone, you forgo the ability to use that money yourself during the loan period. The interest rate is the compensation you receive for this sacrifice. Conversely, the borrower pays for the privilege of using money that is not theirs. Before defining the interest rate precisely, we introduce some basic terminology.

Definition 1: Cash flow

A **cash flow** is any payment of money from one party to another at a specified point in time. A cash flow can be **positive** (money received) or **negative** (money paid). A financial contract is fully characterised by its schedule of cash flows: the amounts, their directions, and their timing.

Definition 2: Loan and investment

A **loan** is a financial arrangement in which one party (the **lender**) provides a sum of money to another party (the **borrower**), who is obligated to repay it — typically with interest — according to agreed terms. From the lender's perspective, providing a loan is an **investment**: the lender commits capital today in exchange for future cash flows (repayment of principal plus interest). The terms "lending" and "investing" describe the same transaction from the lender's point of view.

We now introduce the basic quantity that interest rates act on.

Definition 3: Principal, nominal, and face value

The **principal** (also called **nominal**, **face value**, or **par value**) is the original amount of money lent or borrowed in a financial transaction. It is denoted by N throughout this booklet.

In the context of bonds and other debt securities, these terms are used interchangeably:

- **Principal** emphasises the role as the amount lent or invested.
- **Nominal** or **face value** emphasises the amount printed on the security, which determines coupon payments and the final redemption amount.
- **Par value** refers to the same quantity; a security trading "**at par**" means its market price equals N.

Definition 4: Interest rate and interest payment

An **interest rate** r is the fraction of a principal amount N that a borrower pays to a lender as compensation for the use of money over a specified period of time. It is conventionally expressed as a percentage *per annum* (p.a.).
The **interest payment** is the actual monetary amount paid to the lender, determined by applying the interest rate to the principal over the loan period.

Think of it this way: if you deposit 10,000 € in a bank account, the bank is *borrowing* your money (it uses it to make loans). The interest rate is the "rental fee" the bank pays you for using your money. Every financial contract must specify not only the principal and interest rate, but also *when* the money must be repaid.

Definition 5: Maturity

The **maturity** (also called **term** or **tenor**) is the date at which a financial contract expires and the principal must be repaid. The **time to maturity** T is measured in years from today, where today is $t = 0$.

1.1.1 Time notation

Throughout this booklet, time periods are expressed using a number followed by a unit letter:

Symbol	Meaning	Example
`y`	Years	`3y` (3 years)
`m`	Months	`6m` (6 months)
`w`	Weeks	`2w` (2 weeks)
`d`	Days	`90d` (90 days)

When the unit is clear from context (for instance, inside a formula where t is defined in years), we may write simply $t = 3$ instead of $t = 3\mathtt{y}$. In tables and schedules, we always include the unit for clarity.

Example 1: A simple loan

You lend $N = 10{,}000$ € to a friend for `1y` at an annual interest rate of $r = 3.5\%$. After one year the friend returns:

$$10{,}000 + 10{,}000 \cdot 0.035 = 10{,}350\,€.$$

The 350 € is the interest payment. You are 350 € richer than if you had kept the money under your mattress. From your perspective, this loan is an investment: you committed 10,000 € today (a negative cash flow) and received 10,350 € in `1y` (a positive cash flow).

1.2 Why do interest rates exist?

Interest rates exist for three fundamental reasons:

1. **Time preference.** People generally prefer consuming today rather than tomorrow. To persuade someone to postpone consumption, you must offer a reward.
2. **Inflation.** Prices tend to rise over time. A euro today buys more than a euro next year, so lenders demand compensation for loss of purchasing power.

3. **Risk.** The borrower might *default* (fail to repay part or all of the debt). Lenders demand a premium (surcharge) for bearing this risk.

Definition 6: Default

A **default** (or **credit event**) occurs when a borrower fails to meet the contractual obligations of a debt, such as missing an interest payment, failing to repay the principal at maturity, or violating other terms of the agreement. Default can be **partial** (the borrower pays less than promised) or **total** (the borrower pays nothing).

Remark (The Fisher equation). The relationship between time preference and inflation is formalised by the **Fisher equation**:

$$r_{\text{nom}} \approx r_{\text{real}} + \pi,$$

where r_{nom} is the **nominal** (observed) rate, r_{real} is the **real rate** (compensation for time preference), and π is the **expected inflation**. The exact relationship is $(1 + r_{\text{nom}}) = (1 + r_{\text{real}})(1 + \pi)$, but for small rates the linear approximation above is excellent.

1.3 Can interest rates be negative?

Yes. The European Central Bank (ECB) maintained a negative deposit facility rate from 2014 to 2022. In that regime, banks effectively *paid* the ECB to hold their money. This seems paradoxical — why would a lender accept receiving *less* than they deposited? The answer involves two observations:

1. If inflation is expected to be negative (**deflation**), even a negative nominal rate can yield a positive real return.
2. Since most money is electronic, it is impractical (and costly) to convert it to physical cash. This practical barrier allows central banks to push rates modestly below zero.

Example 2: Negative interest rate in practice

You deposit 10,000 € at a rate of $r = -0.50\%$ for 1y. What amount do you receive at maturity, and what is the real return if consumer prices fell by 1% during the year?

Solution. After one year you receive:

$$10{,}000 - 10{,}000 \cdot 0.005 = 9{,}950\,€.$$

You lost 50 € in nominal terms. However, if consumer prices fell by 1%, the real purchasing power of your 9,950 € is equivalent to $9{,}950/0.99 \approx 10{,}051$ € at today's prices. So in *real* terms, you gained approximately 51 €.

1.4 Compounding conventions

Given a principal N, a rate r, and a time period t (in years), the amount of money at the end depends on *how often* interest is calculated and added to the principal. Finance uses three main conventions.

Definition 7: Simple interest

Under **simple interest**, interest is computed only on the original principal N, never on previously accumulated interest. The **future value** after t years is:

$$\mathrm{FV} = N\,(1 + r\,t).$$

This is the natural convention for short-term instruments (maturities $\leq 1\mathtt{y}$).

Under simple interest, growth is *linear* in t: there is no "interest on interest". The formula follows directly from the fact that the interest each year is $N \cdot r$, so over t years we accumulate $N \cdot r \cdot t$ in interest, giving $\mathrm{FV} = N + N \cdot r \cdot t = N(1 + rt)$.

Definition 8: Compound interest

Under **compound interest** with **compounding frequency** n times per year, interest is added to the principal n times each year. Each sub-period, the interest is computed on the *current balance* (original principal plus any accumulated interest). The future value after t years is:

$$\mathrm{FV} = N\left(1 + \frac{r}{n}\right)^{n\,t}.$$

Common choices are $n = 1$ (**annual**), $n = 2$ (**semi-annual**), $n = 4$ (**quarterly**), and $n = 12$ (**monthly**).

The key difference is that under compounding, you earn "**interest on interest**". Each period's interest is added to the principal, and in the next period you earn interest on this larger amount. Growth is therefore *exponential* rather than linear. To see why: if we compound n times per year, each compounding period is $1/n$ years long, earning interest at rate r/n per period. After one period we have $N(1 + r/n)$; after two periods $N(1 + r/n)^2$; after nt periods (corresponding to t years) we have $N(1 + r/n)^{nt}$.

Definition 9: Continuous compounding

Continuous compounding is the mathematical limit of compound interest as the compounding frequency $n \to \infty$:

$$\mathrm{FV} = \lim_{n\to\infty} N\left(1 + \frac{r}{n}\right)^{n\,t} = N\,e^{r\,t}.$$

This is the convention preferred in theoretical work because of its mathematical simplicity.

In practice, continuous compounding can be closely approximated by compounding at very short discrete intervals. A natural way to think about it is to imagine compounding every second: since a year contains 31,536,000 seconds, setting $n = 31{,}536{,}000$ in the compound interest formula yields a result that is, for all practical purposes, indistinguishable from Ne^{rt}. Continuous compounding is therefore best understood not as a physically realised process, but as the limiting case that arises when the compounding interval becomes vanishingly small — a convenient and elegant idealisation of increasingly frequent discrete compounding.

Example 3: Comparing the three conventions

Lend $N = 1{,}000\,€$ for $t = 1\mathtt{y}$ at a quoted rate of $r = 5\%$. We compute the terminal amount under each compounding convention.

Convention	Formula	Terminal amount
Simple	$1{,}000 \cdot (1 + 0.05 \cdot 1)$	$1{,}050.00\,€$
Annual compound ($n = 1$)	$1{,}000 \cdot (1.05)^1$	$1{,}050.00\,€$
Semi-annual ($n = 2$)	$1{,}000 \cdot (1.025)^2$	$1{,}050.63\,€$
Quarterly ($n = 4$)	$1{,}000 \cdot (1.0125)^4$	$1{,}050.95\,€$
Monthly ($n = 12$)	$1{,}000 \cdot (1 + 0.05/12)^{12}$	$1{,}051.16\,€$
Continuous	$1{,}000 \cdot e^{0.05}$	$1{,}051.27\,€$

Observe that, for the *same quoted rate*, more frequent compounding yields a higher terminal amount. Moreover, simple and annual compound coincide for $t = 1\mathtt{y}$, but they differ for $t \neq 1\mathtt{y}$.

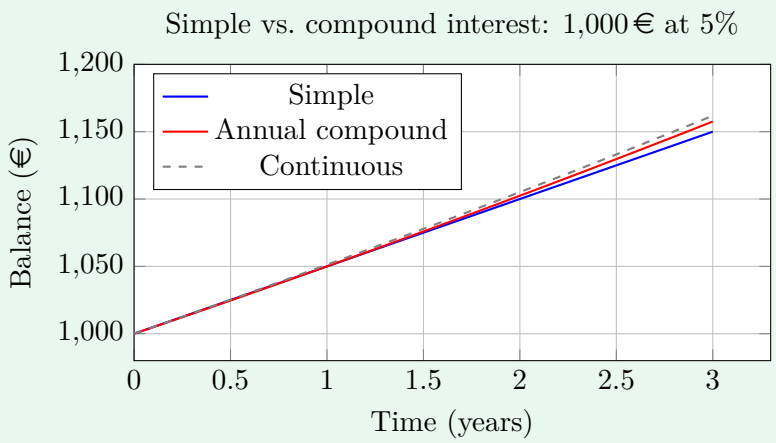
Example 4: Simple vs. compound over multiple years

Lend $1{,}000\,€$ at $r = 5\%$ for $t = 3\mathtt{y}$. Compare the terminal amounts under simple interest, annual compounding, and continuous compounding.

Solution.

- **Simple:** $\mathrm{FV} = 1{,}000\,(1 + 0.05 \cdot 3) = 1{,}150.00\,€$. Growth is linear: exactly $+50\,€$ each year.
- **Annual compound:** $\mathrm{FV} = 1{,}000\,(1.05)^3 = 1{,}157.63\,€$. The extra $7.63\,€$ beyond simple interest is "interest on interest".
- **Continuous:** $\mathrm{FV} = 1{,}000\,e^{0.05 \cdot 3} = 1{,}000\,e^{0.15} = 1{,}161.83\,€$. This is the limiting case as we compound more and more frequently.

Continuous compounding gives the highest return ($1{,}161.83\,€$), followed by annual compounding ($1{,}157.63\,€$), then simple interest ($1{,}150.00\,€$). The more frequently interest is compounded, the more "interest on interest" accumulates.



1.4.1 Equivalent rates

Since different compounding conventions produce different terminal amounts for the same quoted rate, we need a way to compare "apples to apples".

Definition 10: Equivalent interest rates

Two interest rates are **equivalent** if they produce the same future value when applied to the same principal over the same time period.
For compound interest, rates r_1 (compounded n_1 times per year) and r_2 (compounded n_2 times per year) are equivalent over one year if:

$$\left(1+\frac{r_1}{n_1}\right)^{n_1} = \left(1+\frac{r_2}{n_2}\right)^{n_2}.$$

In particular, the continuously compounded rate r_c equivalent to an annually compounded rate r_a satisfies:

$$e^{r_c} = 1 + r_a.$$

Since more frequent compounding generates more "free" interest on interest, to match a given terminal amount the quoted rate must be *lower* when compounding is more frequent.

Example 5: Finding equivalent rates

A bank quotes an annual rate of $r_a = 5\%$ (compounded annually). What semi-annual rate r_2 and what continuous rate r_c produce the same terminal amount on a 1y deposit of 1,000 €?

Solution. The annual compound terminal amount is $\text{FV} = 1{,}000 \cdot 1.05 = 1{,}050$ €.
Semi-annual rate r_2: We need $(1 + r_2/2)^2 = 1.05$:

$$1 + \frac{r_2}{2} = 1.05^{1/2} = 1.02470 \quad \Longrightarrow \quad r_2 = 2 \cdot 0.02470 = 4.939\%.$$

Continuous rate r_c:

$$e^{r_c} = 1.05 \quad \Longrightarrow \quad r_c = \ln(1.05) = 4.879\%.$$

As expected: $r_c < r_2 < r_a$. The more frequent the compounding, the lower the equivalent quoted rate.

Remark (Maturity dependence). When converting between compound and continuous rates, the equivalent rates do *not* depend on the maturity t (because both are "time-additive"). However, when simple interest is involved, the equivalent rate *does* depend on t, because simple interest is *not* time-additive.

1.5 Arbitrage

Before proceeding to value more complex instruments, we introduce one of the central organising principles of financial mathematics: the absence of arbitrage.

Definition 11: Arbitrage

An **arbitrage** (also called an **arbitrage opportunity** or **free lunch**) is a trading strategy that:

1. requires no net investment of capital (zero or negative initial cost), and
2. has no probability of loss, yet
3. has a positive probability of profit.

In short, it is "riskless profit from nothing". A market in which no arbitrage opportunities exist is said to be **arbitrage-free**.

The **no-arbitrage principle** (sometimes called the **law of one price**) is the foundational assumption of modern financial pricing: in a well-functioning market, arbitrage opportunities cannot persist, because as soon as one appears, traders exploit it, and their trading activity drives prices back into line. This principle allows us to derive pricing relationships by demanding that economically equivalent cash flows must have the same price.

No-arbitrage principle

If two portfolios produce *identical cash flows* in all future states, they must have the *same price* today:

$$\text{Cash flows}_1 = \text{Cash flows}_2 \quad \Longrightarrow \quad \text{Price}_1 = \text{Price}_2.$$

Equivalently: if $\text{Price}_1 \neq \text{Price}_2$, buy the cheaper portfolio and sell the more expensive one to lock in riskless profit.

Virtually every pricing formula in this booklet — from discount factors to bond prices to forward rates — is derived by assuming the absence of arbitrage and applying this principle.

Example 6: A simple arbitrage

Suppose two banks offer 1y deposits on 100,000 €:

- Bank A: borrow at $r_A = 3\%$ (you can borrow 100,000 € and repay 103,000 € in 1y).
- Bank B: lend at $r_B = 4\%$ (you can deposit 100,000 € and receive 104,000 € in 1y).

Construct an arbitrage strategy and compute the riskless profit.

Solution. Execute the following strategy at zero initial cost:

1. Borrow 100,000 € from Bank A at 3%.
2. Deposit 100,000 € at Bank B at 4%.
3. In 1y: receive 104,000 € from Bank B, repay 103,000 € to Bank A.

Net profit: $104{,}000 - 103{,}000 = 1{,}000$ € with zero initial investment and zero risk. This is an arbitrage. In practice, such mispricing is quickly eliminated as traders borrow from A and lend to B, pushing A's rate up and B's rate down until the opportunity vanishes.

1.6 Day-count conventions

So far we have treated "one year" as if it were a perfectly defined concept. In practice, what counts as "one year" depends on a legal and market convention called a day-count convention.

Definition 12: Day-count convention

A **day-count convention** is a rule that specifies, for two dates d_1 and d_2, the **year fraction**:

$$t = \frac{\text{number of days between } d_1 \text{ and } d_2 \text{ (by some counting rule)}}{\text{number of days in the “year” (by the same rule)}}.$$

The most common conventions include:

Convention	Description
30/360	Pretend every month has exactly 30 days and every year has exactly 360 days. Simple and predictable; widely used in the bond market.
Act/360	Count the actual calendar days between the two dates; divide by 360. Used for money-market instruments (deposits, FRAs). Note: a full year (365 or 366 days) gives $t > 1$!
Act/365	Count actual calendar days; divide by 365 (ignoring leap years). Common in the UK and some other markets.
Act/Act	Count actual calendar days in the period; divide by the actual number of days in the year (365 or 366 for leap years). The most precise convention; used for government bonds (US Treasuries, German Bunds) and interest rate swaps.

Why does this matter? Because the interest payment is $I = N\,r\,t$, and different rules for computing t give different amounts of interest.

Example 7: Impact of day-count conventions

A deposit of $N = 1{,}000{,}000$ € at $r = 4\%$ p.a. accrues interest from 31 March 2011 to 31 March 2012. The year 2012 is a leap year, so there are 366 actual calendar days between the two dates. Compute the interest $I = N\,r\,t$ under each convention.

Solution.

Convention	Numerator	Year fraction t	Interest I
30/360	360	$360/360 = 1.000$	40,000 €
Act/360	366	$366/360 = 1.017$	40,667 €
Act/365	366	$366/365 = 1.003$	40,110 €
Act/Act	366	$366/366 = 1.000$	40,000 €

The Act/360 vs. 30/360 difference amounts to approximately 667 € per million — on a 1,000,000,000 € (one billion) portfolio this scales to roughly 667,000 €.

Note: Act/Act gives the same result as 30/360 in this case because the period is exactly one calendar year. The differences between conventions become more pronounced for partial-year periods or when the period spans leap-year boundaries.

1.6.1 Date-rolling conventions

Definition 13: Date-rolling convention

A **date-rolling convention** (or **business day convention**) is a rule that specifies how to adjust a payment date when it falls on a weekend or public holiday. Common conventions include:

Convention	Description
Previous	Move to the *preceding* business day.
Following	Move to the *next* business day.
Modified Following	Next business day, *unless* it falls in the next month — then move to the *previous* business day instead.

"Modified Following" is the most common convention in practice because it avoids shifting cash flows across month boundaries (which would change the accrual period significantly).

Example 8: Date rolling in practice

A payment is due on Saturday, 31 March 2012 (a non-business day). The preceding loan started on 31 March 2011 with $N = 1{,}000{,}000$ €, $r = 4\%$, and day-count convention Act/360. Determine the adjusted payment date and the interest under each rolling convention.

Solution.

Convention	Adjusted date	Reasoning
Previous	30 March 2012	Previous business day
Following	2 April 2012	Next business day
Modified Following	30 March 2012	Next business day is in April → use previous instead

Impact on interest calculation (Act/360):

Convention	Payment date	Days	Interest
Previous	30 Mar 2012	365	40,556 €
Following	2 Apr 2012	368	40,889 €
Modified Following	30 Mar 2012	365	40,556 €

The interest is computed as $I = N \cdot r \cdot \frac{\text{days}}{360}$.

CHAPTER 2

Discount Factor and Time Value of Money

2.1 The fundamental principle

The single most important idea in finance is:

The Time Value of Money

A euro today is worth **more** than a euro in the future, because a euro today can be invested to earn interest.

This means that cash flows occurring at different times cannot simply be added together. To compare or aggregate them, we must first convert them to a common point in time — usually "today" ($t = 0$).

2.2 Future value and present value

This principle leads to two complementary concepts for comparing cash flows at different points in time. Future value answers: "If I invest N euros today, how much will I have later?" Present value answers the reverse: "How much is a future payment of C euros worth today?"

Definition 14: Future value

The **future value** FV of an amount N invested today for t years at rate r is the amount you will have at time t:

$$\mathrm{FV} = N\,(1+r)^t \quad \text{(annual)}, \qquad \mathrm{FV} = N\,e^{r\,t} \quad \text{(continuous)}.$$

See Definitions 8 and 9 for the compounding conventions.

Definition 15: Present value

The **present value** PV of a cash flow C to be received at time t is the amount that,

if invested today at rate r, would grow to exactly C:

$$\mathrm{PV} = \frac{C}{(1+r)^t} \quad \text{(annual)}, \qquad \mathrm{PV} = C\,e^{-r\,t} \quad \text{(continuous)}.$$

Example 9: Future and present value

(a) You deposit 10,000 € for 2y at $r = 4\%$ (annual compounding). What is the future value?

$$\mathrm{FV} = 10{,}000 \cdot 1.04^2 = 10{,}816\,€.$$

(b) You are promised 10,816 € in 2y. At a rate of $r = 4\%$ (annual compounding), what is the present value?

$$\mathrm{PV} = \frac{10{,}816}{1.04^2} = 10{,}000\,€.$$

Notice that these two operations are exact inverses: one "undoes" the other.

2.3 The discount factor

We have seen that present value requires dividing a future cash flow by $(1+r)^t$ (or multiplying by $e^{-r\,t}$ under continuous compounding). When valuing bonds, swaps, or any instrument with multiple cash flows, we must repeat this calculation for each payment date — potentially dozens of times. This repetitive discounting motivates introducing a unified concept that captures the "time value of money" for each maturity in a single number.

Definition 16: Discount factor

The **discount factor** $\mathrm{DF}(t)$ for maturity t is the present value of 1 € to be received at time t:

$$\mathrm{DF}(t) = \frac{1}{(1+r)^t} \quad \text{(annual)}, \qquad \mathrm{DF}(t) = e^{-r\,t} \quad \text{(continuous)}.$$

The discount factor lets you price *any* cash flow by simple multiplication:

$$\mathrm{PV}(\text{cash flow } C \text{ at time } t) = C \cdot \mathrm{DF}(t).$$

This is simply the present value formula rewritten: instead of computing $C/(1+r)^t$ each time, we pre-compute $\mathrm{DF}(t) = 1/(1+r)^t$ once and then multiply by C. The advantage becomes clear when pricing instruments with many cash flows — we compute the discount function once and reuse it.

Remark (Why discount factors are useful). The discount factor is *convention-free*: a discount factor of $\mathrm{DF}(2) = 0.920$ means "1 € in 2y is worth 0.92 € today", regardless of whether the underlying rate is quoted with annual, semi-annual, or continuous compounding. This makes discount factors the *lingua franca* of fixed income.

Properties of the discount factor

For positive interest rates $r > 0$ and $t \geq 0$:
1. $0 < \mathrm{DF}(t) \leq 1$ (a future euro is worth less than a euro today).
2. $\mathrm{DF}(0) = 1$ (a euro today is worth one euro today).
3. $\mathrm{DF}(t_1) > \mathrm{DF}(t_2)$ whenever $t_1 < t_2$ (more distant cash flows are worth less).

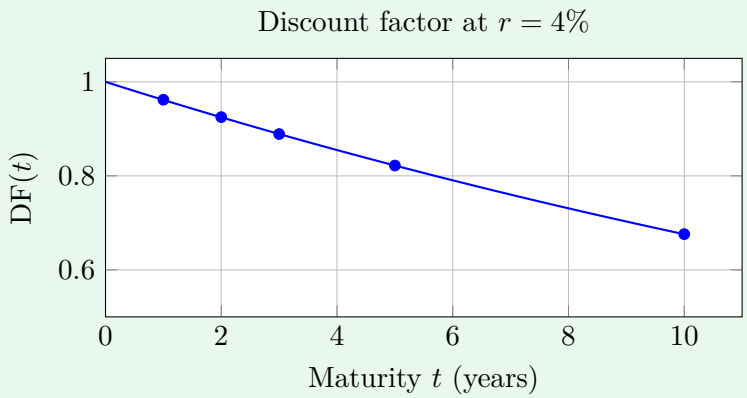
Indeed, the discount factor takes the form of a decreasing hyperbola under annual compounding and a decreasing exponential under continuous compounding, with both satisfying the above properties.

Example 10: Computing discount factors

At an annual rate of $r = 4\%$, compute the discount factors $\mathrm{DF}(t)$ for $t = 1\mathrm{y}, 2\mathrm{y}, 3\mathrm{y}, 5\mathrm{y}, 10\mathrm{y}$ and interpret each result.

Solution.

t	$(1.04)^t$	$\mathrm{DF}(t) = 1/(1.04)^t$	**Interpretation**
1y	1.040	0.962	1 € in 1y = 0.96 € today
2y	1.082	0.925	1 € in 2y = 0.92 € today
3y	1.125	0.889	1 € in 3y = 0.89 € today
5y	1.217	0.822	1 € in 5y = 0.82 € today
10y	1.480	0.676	1 € in 10y = 0.68 € today



As expected from the properties above, $\mathrm{DF}(t)$ is strictly decreasing: the further into the future a cash flow lies, the less it is worth today.

Example 11: Extracting a rate from a discount factor

Suppose a 2y discount factor is $\mathrm{DF}(2) = 0.950$. What are the corresponding annual and continuous rates?

Solution. We use the definition $\mathrm{DF}(t) = 1/(1+r)^t$ for annual compounding and $\mathrm{DF}(t) = e^{-r\,t}$ for continuous compounding.

Annual rate r_a: Starting from $(1+r_a)^2 = 1/\mathrm{DF}(2) = 1/0.950$, we take the square

root:

$$1 + r_a = \left(\frac{1}{0.950}\right)^{1/2} = 1.02598 \quad \Longrightarrow \quad r_a = 2.60\%.$$

Continuous rate r_c: Starting from $e^{2r_c} = 1/\,\mathrm{DF}(2) = 1/0.950$, we take the natural logarithm and divide by 2:

$$2r_c = \ln\left(\frac{1}{0.950}\right) = -\ln(0.950) = 0.05129 \quad \Longrightarrow \quad r_c = 2.56\%.$$

As expected, $r_c < r_a$: the continuously compounded rate is lower because continuous compounding generates more "interest on interest."

2.4 Net present value

Most financial instruments generate multiple cash flows at different times. To value the whole package, we discount each cash flow individually and sum them up.

Definition 17: Net present value

Given an initial investment I_0 at time $t = 0$, and cash flows $c_1, c_2, \ldots, c_n$ at times $t_1, t_2, \ldots, t_n$ with discount factors $\mathrm{DF}(t_i)$, the **net present value** is:

$$\mathrm{NPV} = -I_0 + \sum_{i=1}^{n} c_i \cdot \mathrm{DF}(t_i).$$

The net present value aggregates all cash flows — both inflows and outflows — into a single number expressed in today's euros. This allows us to answer the fundamental question: "Is this investment worth undertaking?" The sign and magnitude of the NPV provide immediate guidance:

- NPV > 0: the investment **creates value**. The present value of what you receive exceeds what you pay today, so you should accept the investment.
- NPV < 0: the investment **destroys value**. You pay more today than the present value of future cash flows, so you should reject it.
- NPV $= 0$: the investment is **fairly priced**. The present value of benefits exactly equals the cost, so you are indifferent — the investment neither creates nor destroys value. This is the equilibrium condition in competitive markets.

Example 12: NPV of a simple project

You invest 100,000 € today and receive three payments: 40,000 € in 1y, 50,000 € in 2y, and 30,000 € in 3y. The annual rate is $r = 5\%$. Is the project worth undertaking?

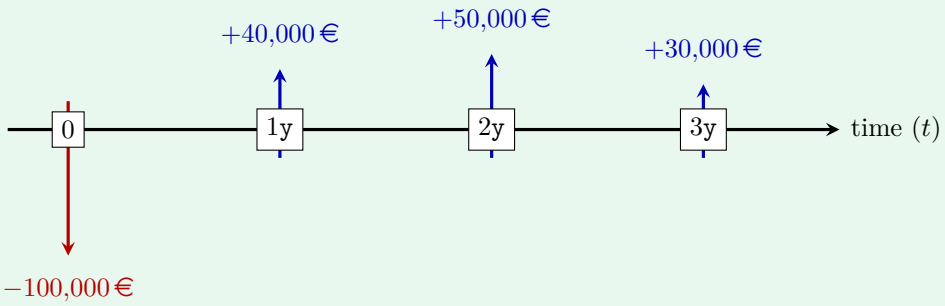
Solution. First, compute the discount factors:

$$\mathrm{DF}(1) = \frac{1}{1.05} = 0.952, \quad \mathrm{DF}(2) = \frac{1}{1.05^2} = 0.907, \quad \mathrm{DF}(3) = \frac{1}{1.05^3} = 0.864.$$

Then:

$$\begin{aligned}\mathrm{NPV} &= -100{,}000 + 40{,}000 \cdot 0.952 + 50{,}000 \cdot 0.907 + 30{,}000 \cdot 0.864 \\ &= -100{,}000 + 38{,}095 + 45{,}351 + 25{,}916 \;=\; +9{,}362\,€.\end{aligned}$$

Since NPV > 0, the project is worth undertaking: it creates 9,362 € of value in today's terms.



2.5 Cash flow projection

The NPV formula takes a set of cash flows as given and discounts them. But in practice, the first — and often the hardest — step in valuing any financial instrument is determining *what those cash flows will be*. This process is called cash flow projection.

Definition 18: Cash flow projection

Cash flow projection is the process of determining the amounts and timing of all future payments generated by a financial instrument. It is the necessary first step before any present value calculation can be performed.

For the instruments considered so far — such as deposits or simple investments with fixed payments — projecting cash flows is straightforward: the amounts, dates, and directions of payments are specified in the contract and known with certainty at inception.
Many financial instruments, however, generate cash flows whose amounts are *unknown today*. For example, loans with interest payments linked to periodically resetting market rates depend on future rate levels that have not yet been observed. To apply the NPV framework, we must therefore *estimate* these future cash flows. Forward rates — implied by today's term structure — provide a market-based estimate of future rates, allowing us to construct a projected cash flow schedule that can then be discounted.
In more complex instruments, cash flows may also depend on contingent events, such as early repayment by the borrower or contractual clauses triggered by specific market conditions. In such cases, projection requires models that account for this optionality.

Remark (Projection and discounting: two separate steps). Regardless of the instrument's complexity, valuation always follows the same two-step logic:

1. **Project:** Determine (or estimate) every future cash flow — its amount, timing, and direction.
2. **Discount:** Convert each projected cash flow to its present value using the appropriate discount factor.

Later we will see that modern practice often uses different rates for projection and discounting. For now, the key insight is that these are logically independent operations: the first asks "how much will be paid?" and the second asks "what is that payment worth today?"

CHAPTER 3

Fixed-Rate Bonds

3.1 What is a bond?

A bond is the most basic fixed income instrument. It is essentially a formalised loan (Definition 2): the issuer (borrower) promises to make specific payments to the holder (lender) according to a predetermined schedule.

Definition 19: Issuer and holder

The **issuer** (also called the **borrower** or **obligor**) is the entity that creates and sells a bond in order to raise capital. The issuer is contractually obligated to make all promised payments.
The **holder** (also called the **bondholder**, **lender**, or **investor**) is the party that purchases the bond and, in return, is entitled to receive the promised cash flows.

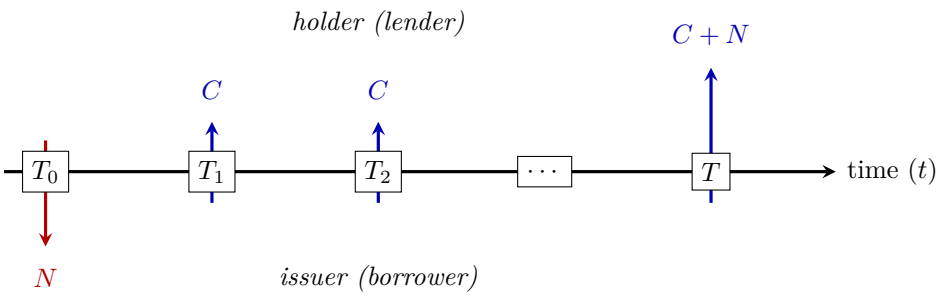
Bonds are transferable: the holder can sell the bond to another investor in the **secondary market**. Common types of issuers include governments (**sovereign bonds**), corporations (**corporate bonds**), municipalities, and supranational organisations.

Definition 20: Bond

A **bond** (or **fixed income security**, or **debt instrument**) is a contract in which the issuer is obligated to pay the holder:

1. **Principal** (or **face value**, **nominal**, **par value**; see Definition 3): the amount N repaid at **maturity**, the final date T.
2. **Coupons:** periodic interest payments made at regular intervals (typically every `6m` or `12m`). For a bond with face value N, **coupon rate** c (expressed per annum), and m payments per year, each **coupon payment** is:

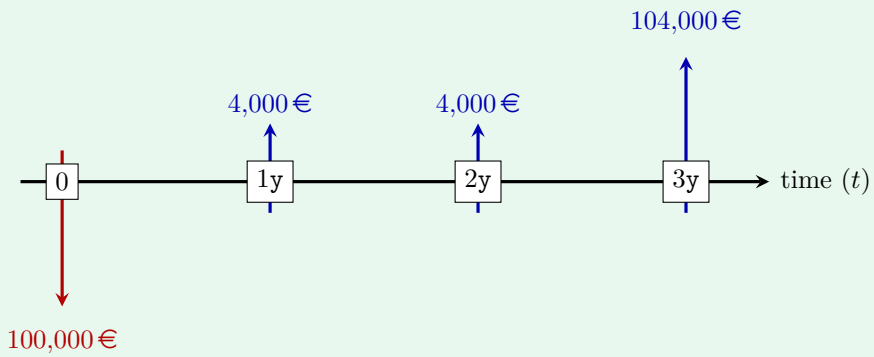
$$C = \frac{c \cdot N}{m}.$$



Example 13: Cash flows of a bond

Consider a bond with face value $N = 100{,}000$ €, annual coupon rate $c = 4\%$, and maturity $T = 3$y. Each year the bondholder receives a coupon of $C = c \cdot N = 0.04 \cdot 100{,}000 = 4{,}000$ €.

Time	Cash flow	Description
$t = 0$	$-100{,}000$ €	Principal paid by holder
$t = 1$y	$+4{,}000$ €	Coupon $= 100{,}000 \cdot 4\%$
$t = 2$y	$+4{,}000$ €	Coupon
$t = 3$y	$+104{,}000$ €	Final coupon (4,000 €) + principal (100,000 €)



The holder pays the principal $N = 100{,}000$ € at $t = 0$ and receives three annual coupons of 4,000 € each, with the principal repaid at maturity.

3.2 Bond pricing

To determine what a bond is worth today, we apply the fundamental principle of present value: discount each promised cash flow back to the present and sum them up.

Definition 21: Bond price

The **fair price** (or **dirty price**) of a bond is the net present value of its promised cash flows. For a bond with cash flows c_i at times t_i ($i = 1, \ldots, n$) and discount factors $\mathrm{DF}(t_i)$:

$$B = \sum_{i=1}^{n} c_i \cdot \mathrm{DF}(t_i).$$

When a single **yield** y (annual interest rate) is used to discount all cash flows, the price formula simplifies to the **price-yield relationship**:

$$B = \sum_{i=1}^{n} \frac{c_i}{(1+y)^{t_i}} \qquad \text{(annual compounding)}.$$

This situation is known as the **flat yield curve assumption**.

Example 14: Pricing a 3-year bond

Consider the bond from Example 13: face value $N = 100{,}000\,€$, coupon rate $c = 4\%$, maturity $T = 3\text{y}$, annual coupons of $4{,}000\,€$. If the market yield is $y = 3\%$, what is the fair price?

Solution. The cash flows are: $4{,}000\,€$ at 1y, $4{,}000\,€$ at 2y, and $104{,}000\,€$ at 3y (final coupon plus principal). Using the flat yield curve assumption with $y = 3\%$:

$$B = \frac{4{,}000}{1.03} + \frac{4{,}000}{1.03^2} + \frac{104{,}000}{1.03^3} = 3{,}883 + 3{,}770 + 95{,}148 = 102{,}801\,€.$$

The bond trades *above par* (above $100{,}000\,€$) because the coupon rate $c = 4\%$ exceeds the market yield $y = 3\%$. Investors are willing to pay a *premium* of $2{,}801\,€$ for the above-market coupon payments.

Example 15: Fairly priced bond

Consider the same bond as in Example 14: $N = 100{,}000\,€$, $c = 4\%$, $T = 3\text{y}$. Now suppose the market yield is $y = 4\%$ (instead of 3%). What is the fair price?

Solution. Using the flat yield curve assumption with $y = 4\%$:

$$B = \frac{4{,}000}{1.04} + \frac{4{,}000}{1.04^2} + \frac{104{,}000}{1.04^3} = 3{,}846 + 3{,}698 + 92{,}456 = 100{,}000\,€.$$

The bond trades *at par* (exactly $100{,}000\,€$) because the coupon rate $c = 4\%$ equals the market yield $y = 4\%$. When the coupon rate equals the yield, the bond is fairly priced at its face value.

3.2.1 Clean price, dirty price, and accrued interest

In practice, bonds are rarely traded exactly on a coupon payment date. When a bond is bought between coupon dates, the buyer must compensate the seller for the portion of the coupon that has already accrued. This leads to a distinction between the quoted price (clean price) and the actual payment (dirty price).

Definition 22: Clean price, dirty price, and accrued interest

- The **dirty price** (or **invoice price**) B_{dirty} is the actual total amount paid by the buyer.
- The **accrued interest** AI is the coupon interest that has accumulated from the

last coupon date to the settlement date:

$$AI = N \cdot c \cdot \frac{\text{days since last coupon}}{\text{days in coupon period}}.$$

- The **clean price** (or **quoted price**) is the dirty price minus accrued interest:

$$B_{\text{clean}} = B_{\text{dirty}} - AI.$$

Why this distinction? The dirty price exhibits a "sawtooth" pattern: it rises steadily as the coupon accrues, then drops suddenly when the coupon is paid. The clean price strips out this mechanical effect, giving a smoother picture of how the market values the bond.

Example 16: Clean vs. dirty price

A bond with $N = 100{,}000$ €, annual coupon $c = 4\%$, and last coupon paid 6m ago has a market clean price of 101,500 €. Compute the accrued interest and the dirty price.

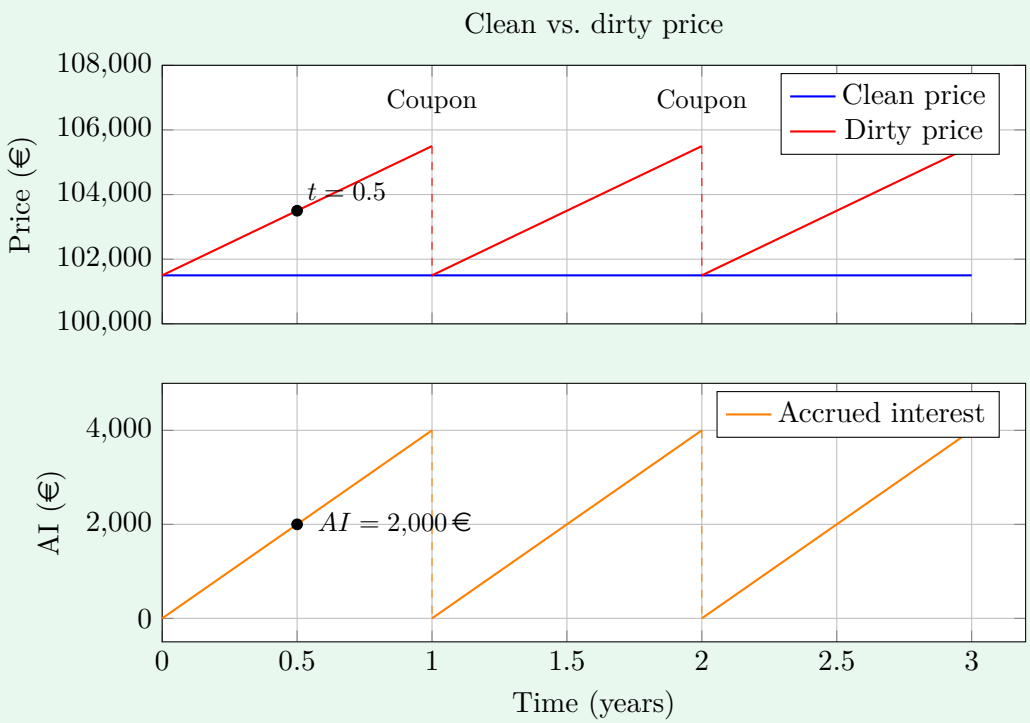
Solution. Accrued interest (30/360 convention; see Definition 12):

$$AI = 100{,}000 \cdot 0.04 \cdot \frac{180}{360} = 2{,}000\,€.$$

Dirty price:

$$B_{\text{dirty}} = 101{,}500 + 2{,}000 = 103{,}500\,€.$$

The buyer pays 103,500 €, of which 2,000 € compensates the seller for the six months of accrued interest.



3.3 Yield to maturity

Definition 23: Yield to maturity (YTM)

The **yield to maturity** y of a bond is the single discount rate that equates the bond's dirty price to the present value of its cash flows:

$$B_{\text{dirty}} = \sum_{i=1}^{n} \frac{c_i}{(1+y)^{t_i}} \qquad \text{(annual compounding)},$$

where c_i is the cash flow at time t_i (coupon for $i < n$, coupon plus principal for $i = n$). If coupons are paid m times per year (e.g. $m = 2$ for semi-annual), the compound convention is:

$$B_{\text{dirty}} = \sum_{i=1}^{n} \frac{c_i}{\left(1 + \frac{y}{m}\right)^{m\,t_i}}.$$

The YTM is a useful *summary statistic* that distils the bond's risk/return profile into a single number, but it has limitations: it assumes a flat yield curve and that coupons are reinvested at the same rate y, which is rarely exact.

Definition 24: Par, premium, and discount

The relationship between a bond's price and its face value N defines three pricing regions:

- A bond trades **at par** when its price equals face value ($B = N$). This occurs when the coupon rate equals the yield to maturity ($c = y$).
- A bond trades **at a premium** when its price exceeds face value ($B > N$). This occurs when the coupon rate exceeds the YTM ($c > y$) — investors pay more than par because the bond's coupons are above market rates.
- A bond trades **at a discount** when its price is below face value ($B < N$). This occurs when the coupon rate is below the YTM ($c < y$) — investors pay less than par because the bond's coupons are below market rates.

Example 17: Computing YTM numerically

A bond has $N = 100{,}000$ €, annual coupon 4%, $T = 3\text{y}$, and trades at $B_{\text{dirty}} = 105{,}000$ €. Find the YTM.

Solution. The cash flows are: $c_1 = 4{,}000$ € at $t_1 = 1\text{y}$, $c_2 = 4{,}000$ € at $t_2 = 2\text{y}$, and $c_3 = 104{,}000$ € at $t_3 = 3\text{y}$. We need to solve:

$$105{,}000 = \frac{4{,}000}{1+y} + \frac{4{,}000}{(1+y)^2} + \frac{104{,}000}{(1+y)^3}.$$

This equation has no closed-form solution; we use trial and error (or Newton's method).

y	**Price**	
3.00%	102,828 €	(too low)
2.00%	105,771 €	(too high)
2.30%	104,874 €	(close)
2.24%	105,041 €	(very close)

By interpolation, $y \approx 2.24\%$. Since the coupon (4%) is well above the YTM (2.24%), the bond trades at a premium.

Definition 25: Par yield

The **par yield** for a given maturity T is the coupon rate r_{par} that would make a bond with that maturity trade at exactly par ($B = N$):

$$N = N\, r_{\text{par}} \sum_{i=1}^{n} \text{DF}(t_i) + N \cdot \text{DF}(T) \qquad \Longrightarrow \qquad r_{\text{par}} = \frac{1 - \text{DF}(T)}{\sum_{i=1}^{n} \text{DF}(t_i)},$$

where $t_n = T$ is the maturity time. Note that the par yield is independent of the face value N.

The term $N \cdot \text{DF}(T)$ accounts for the principal repayment at maturity, while the sum captures the present value of all coupon payments.

Example 18: Computing the par yield

A trader at a fixed-income desk observes the following discount factors, stripped from government bond prices on a given trading day. He wants to determine the fair coupon rate for a newly issued 3y government bond with face value $N = 100$ € that should price at par.

Maturity	$\text{DF}(t_i)$
1y	0.9804
2y	0.9612
3y	0.9423

Solution. Using the par yield formula with $n = 3$ and $T = 3$y:

$$r_{\text{par}} = \frac{1 - \text{DF}(3)}{\text{DF}(1) + \text{DF}(2) + \text{DF}(3)} = \frac{1 - 0.9423}{0.9804 + 0.9612 + 0.9423} = \frac{0.0577}{2.8839} \approx 2.00\%.$$

The bond should carry a coupon of 2.00%. Note that $N = 100$ € does not appear in the calculation: as Definition 25 shows, the par yield depends only on the discount factors, not on the face value. A bond with $N = 1{,}000{,}000$ € and the same maturity would require the same coupon rate of 2.00% to price at par.

3.4 Pull to par

As a bond approaches maturity, its price converges to par regardless of whether it currently trades at a premium or discount. This phenomenon is called pull to par.

Definition 26: Pull to par

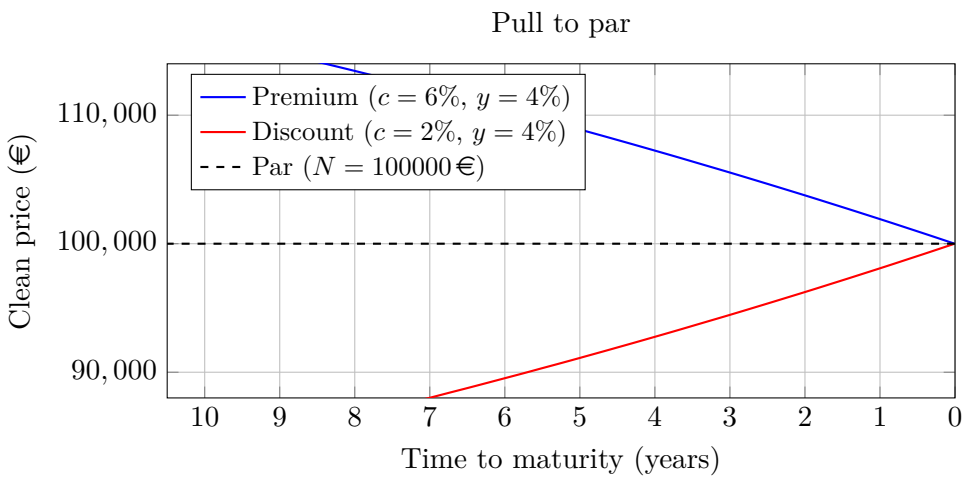
Pull to par is the convergence of a bond's price toward its face value N as time to maturity approaches zero. Formally, for any bond with face value N and no default (Definition 6):

$$\lim_{t \to T} B(t) = N,$$

where $B(t)$ is the bond's dirty price at time t and T is the maturity date.

The economic intuition is straightforward: at maturity, the bond pays exactly N regardless of past coupon rates or yields. As the final payment date approaches, this certain payment dominates the bond's value, causing the price to converge to N. The rate of convergence depends on the bond's current price relative to par:

- **Premium bonds** ($B > N$): price gradually decreases toward N.
- **Discount bonds** ($B < N$): price gradually increases toward N.
- **Par bonds** ($B = N$): price remains stable near N (assuming unchanged yields).



Example 19: Pull to par in action

Consider two bonds, both with $N = 100{,}000$ € and constant YTM $y = 4\%$:

- Bond 1: $c = 6\%$ (premium bond).
- Bond 2: $c = 2\%$ (discount bond).

Compute the price of each bond at 5y, 1y, and 0y to maturity and verify the convergence to par.

Solution. We apply Definition 21 with a flat yield $y = 4\%$. For $T > 1\text{y}$, the coupon sum $\sum_{i=1}^{T}(1+y)^{-i}$ is a finite geometric series, which closes to:

$$\sum_{i=1}^{T} \frac{1}{(1+y)^i} = \frac{1-(1+y)^{-T}}{y},$$

giving the standard annuity factor. The bond price therefore splits into a coupon

annuity and a discounted principal:

$$B(T) = c \cdot N \cdot \frac{1-(1+y)^{-T}}{y} + N \cdot (1+y)^{-T}.$$

Bond 1 (premium): With 5y to maturity:

$$B_1(5\text{y}) = 6{,}000 \cdot \frac{1-1.04^{-5}}{0.04} + 100{,}000 \cdot 1.04^{-5} = 26{,}703 + 82{,}193 = 108{,}896\,€.$$

With 1y to maturity:

$$B_1(1\text{y}) = \frac{6{,}000 + 100{,}000}{1.04} = \frac{106{,}000}{1.04} = 101{,}923\,€.$$

At maturity: $B_1(0) = 100{,}000\,€$.
Bond 2 (discount): With 5y to maturity:

$$B_2(5\text{y}) = 2{,}000 \cdot \frac{1-1.04^{-5}}{0.04} + 100{,}000 \cdot 1.04^{-5} = 8{,}901 + 82{,}193 = 91{,}094\,€.$$

With 1y to maturity:

$$B_2(1\text{y}) = \frac{2{,}000 + 100{,}000}{1.04} = \frac{102{,}000}{1.04} = 98{,}077\,€.$$

At maturity: $B_2(0) = 100{,}000\,€$.
Both bonds converge to par, with the premium bond declining from 108,896 € to 100,000 € and the discount bond rising from 91,094 € to 100,000 €.

Remark (Pull to par and realised returns). Pull to par affects the composition of a bond's total return. For a **premium bond** held to maturity, part of the return comes from coupons and part is a *capital loss* (as price falls to par). For a **discount bond**, there is a *capital gain* in addition to coupon income. However, if yields remain constant at the bond's YTM, the total annualised return equals the YTM in both cases.

3.5 Market liquidity

The pull-to-par phenomenon assumes that bonds can be held to maturity or traded freely at fair prices. In practice, the ease with which bonds can be bought and sold varies considerably across different markets and issuers.

Definition 27: Liquidity

Liquidity refers to the ease with which an asset can be bought or sold in the market without significantly affecting its price. A **liquid** asset can be traded quickly in large volumes with minimal price impact, while an **illiquid** asset requires time to sell or must be sold at a discount to attract buyers.

Market liquidity depends on:
- **Trading volume:** how frequently the asset changes hands.

- **Bid-ask spread:** the difference between the price at which dealers are willing to buy (bid) and sell (ask) the asset; narrower spreads indicate higher liquidity.
- **Market depth:** the ability to execute large transactions without moving the price.

Government bonds are typically highly liquid, while corporate bonds — especially those from smaller issuers — trade less frequently and are therefore less liquid.

Example 20: Liquidity in practice

Consider two bonds, both with face value $N = 1{,}000{,}000$ € and similar maturity:

- **German Bund** (10-year government bond): trades continuously with billions of euros in daily volume. The bid-ask spread is typically 0.01% of face value. An investor wanting to sell 1,000,000 € can execute immediately at the quoted price.
- **Corporate bond** from a mid-sized industrial company: trades infrequently, perhaps once or twice per week. The bid-ask spread may be 0.5% or wider. To sell quickly, the investor might need to accept a price 0.3%–0.5% below the theoretical fair value, representing a **liquidity discount** of 3,000–5,000 €.

The difference in liquidity explains why investors demand higher yields on corporate bonds even when default risk is low: part of the extra yield compensates for the difficulty of selling the bond if needed.

3.6 Credit spread

Not all bonds are equally safe. A corporate bond typically offers a higher yield than a government bond of the same maturity. This extra yield compensates investors for taking on additional risk, and is called the credit spread. Intuitively, if a government bond yields 3% and a corporate bond of the same maturity yields 4.2%, the credit spread is 1.2% = 120 bps. This spread is the price the market charges for bearing the risk that the corporate issuer might not pay in full.

Definition 28: Credit spread

The **credit spread** (or simply **spread**) of a bond is the difference between its yield and the yield of a benchmark risk-free bond of the same maturity:

$$s = y_{\text{risky}} - y_{\text{risk-free}}.$$

The spread compensates the investor for three distinct sources of risk:

1. **Default risk:** the possibility that the issuer fails to make some or all promised payments (coupons or principal).
2. **Liquidity risk:** corporate bonds trade less frequently and in smaller volumes than government bonds, so selling them quickly may require accepting a lower price.
3. **Recovery risk:** even if default occurs, investors may recover a fraction of the face value. The uncertainty about this recovery amount is itself a source of risk.

Credit spreads are quoted in **basis points** (bps), where 1 bps = 0.01% = 0.0001.

Typical spread levels (as of 2024–2025):

- **Investment-grade** corporate bonds: 50–200 bps over government bonds.
- **High-yield** ("junk") bonds: 300–1000+ bps.
- Financial institution bonds: 20–150 bps (depending on creditworthiness).

Spreads are not constant: they **widen** during financial stress (investors demand more compensation) and **narrow** during calm periods as investors accept lower compensation in the so-called "search for yield".

Example 21: Credit spread and bond pricing

A 5y government bond (risk-free benchmark) has YTM $y_{\text{gov}} = 2\%$. A corporate bond with identical structure ($N = 100{,}000$ €, annual coupon $c = 3\%$, $T = 5\text{y}$) trades at a credit spread of 200 bps (2%). Compute the prices of both bonds and interpret the price difference in terms of credit risk.

Solution. The corporate bond's YTM is:

$$y_{\text{corp}} = y_{\text{gov}} + s = 2\% + 2\% = 4\%.$$

Corporate bond price:

$$B_{\text{corp}} = 3{,}000 \cdot \frac{1 - 1.04^{-5}}{0.04} + 100{,}000 \cdot 1.04^{-5} = 13{,}356 + 82{,}193 = 95{,}549\,€.$$

Government bond price (at $y = 2\%$):

$$B_{\text{gov}} = 3{,}000 \cdot \frac{1 - 1.02^{-5}}{0.02} + 100{,}000 \cdot 1.02^{-5} = 14{,}135 + 90{,}573 = 104{,}708\,€.$$

Interpretation: The corporate bond trades at 95,549 € (a discount of 4,451 € below par) while the government bond trades at 104,708 € (a premium of 4,708 € above par). The 9,159 € difference reflects the market's compensation for the corporate bond's additional credit risk. An investor buying the corporate bond accepts a lower price today in exchange for bearing the possibility that the issuer defaults.

Example 22: Spread widening during a crisis

The same corporate bond as in Example 21 initially trades at a spread of 200 bps over the 5y government rate of 2%, giving $y_{\text{corp}} = 4\%$ and price $B = 95{,}549$ €.
Suddenly, concerns about the company's financial health emerge and the credit spread widens to 400 bps, while the government yield remains at 2%. Calculate the new bond price, the loss incurred by the bondholder, and explain what type of risk this illustrates.

Solution. New corporate YTM: $y_{\text{corp}} = 2\% + 4\% = 6\%$.
New corporate bond price:

$$B_{\text{new}} = 3{,}000 \cdot \frac{1 - 1.06^{-5}}{0.06} + 100{,}000 \cdot 1.06^{-5} = 12{,}637 + 74{,}726 = 87{,}363\,€.$$

The bond price dropped from 95,549 € to 87,363 € — a loss of 8,186 € (about 8.6%) — even though the risk-free rate did not change. This loss is purely due to credit spread widening. This illustrates **spread risk**: corporate bondholders are exposed not only to interest rate movements but also to changes in credit perception.

Remark (Credit spread vs. default probability). The credit spread is related to, but not identical to, the probability of default. A simple approximation is:

$$s \approx p \cdot (1 - R),$$

where p is the annual **default probability** and R is the **recovery rate** (fraction of par recovered in default). In words: the spread roughly equals the probability of default multiplied by the **loss-given-default** (the fraction you don't recover).
For example, a 2% spread with 40% recovery suggests $p \approx 0.02/(1-0.4) = 3.3\%$ annual default probability. However, this formula ignores **liquidity premia** (investors demand extra return for hard-to-sell bonds), **risk aversion** (investors dislike uncertainty), and other market frictions, so actual spreads typically exceed this theoretical level.

3.7 Bond portfolios

In practice, investors rarely hold a single bond in isolation. A fixed income investor — whether a pension fund, an insurance company, or a proprietary trading desk — typically holds a collection of bonds simultaneously, managed as a whole. This collection is called a portfolio.

Definition 29: Portfolio

A **portfolio** is a collection of n financial instruments held simultaneously. The **portfolio value** is the sum of the market values of its constituents V_i:

$$V_{\text{port}} = \sum_{i=1}^{n} V_i.$$

Because the portfolio value is a simple sum, any property that is additive at the instrument level — such as present value, risk sensitivity, or cash flows — can be computed for the portfolio by aggregating across its components.

Example 23: A three-bond portfolio

A fixed income investor holds the following three bonds, all with face value $N = 100{,}000$ € and annual coupons:

	Bond	Coupon c	Maturity T	Market yield y
1	Sovereign (safe)	2%	2y	2%
2	Corporate (investment)	5%	2y	4%
3	Corporate (stressed)	3%	2y	6%

Compute the price of each bond, the portfolio value, and discuss how the three positions relate to one another.

Solution.
All three bonds share the same maturity $T = 2\text{y}$, so each has two cash flows: a coupon $C = c \cdot 100{,}000$ at 1y, and a final payment $C + 100{,}000$ at 2y. We apply the bond pricing formula (Definition 21) with a flat yield curve for each bond.

Bond 1 ($c = 2\%$, $y = 2\%$):

$$B_1 = \frac{2{,}000}{1.02} + \frac{102{,}000}{1.02^2} = 1{,}961 + 98{,}039 = 100{,}000\,€.$$

Since $c = y$, Bond 1 trades at par (Definition 24).

Bond 2 ($c = 5\%$, $y = 4\%$):

$$B_2 = \frac{5{,}000}{1.04} + \frac{105{,}000}{1.04^2} = 4{,}808 + 97{,}078 = 101{,}886\,€.$$

Since $c > y$, Bond 2 trades at a premium: investors pay above par for its above-market coupons.

Bond 3 ($c = 3\%$, $y = 6\%$):

$$B_3 = \frac{3{,}000}{1.06} + \frac{103{,}000}{1.06^2} = 2{,}830 + 91{,}670 = 94{,}500\,€.$$

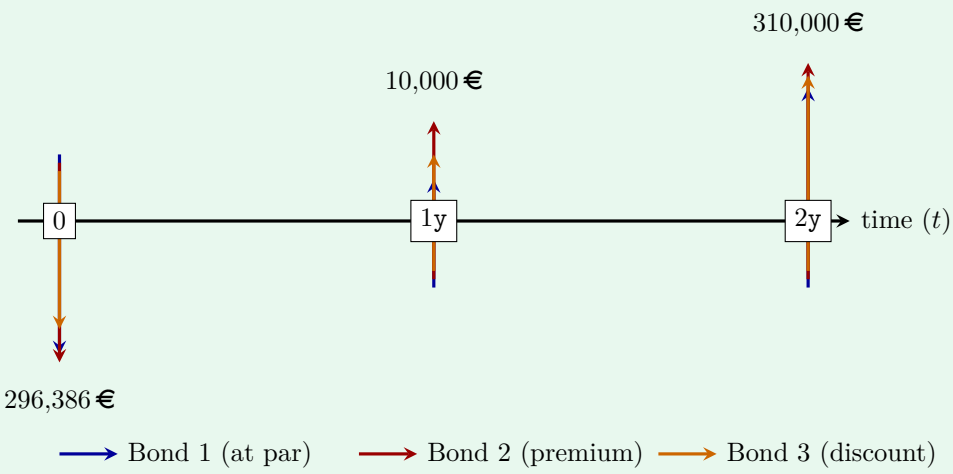
Since $c < y$, Bond 3 trades at a discount: the market demands a higher yield to compensate for the perceived risk of this issuer, and the price falls below par accordingly.

Portfolio value:

$$V_{\text{port}} = B_1 + B_2 + B_3 = 100{,}000 + 101{,}886 + 94{,}500 = 296{,}386\,€.$$

Aggregate cash flows. The portfolio cash flows at each date are obtained by summing across the three bonds:

Date	Bond 1	Bond 2	Bond 3	Portfolio
1y	2,000 €	5,000 €	3,000 €	10,000 €
2y	102,000 €	105,000 €	103,000 €	310,000 €



Discussion. The three bonds illustrate the full spectrum of pricing regions in a single portfolio. Bond 1 (sovereign) trades at par: it is the risk-free benchmark. Bond 2 (investment-grade corporate) trades at a premium because its coupon exceeds the market yield — it was issued when rates were higher and now looks attractive. Bond 3 (stressed corporate) trades at a discount: the market demands a higher yield to compensate for credit risk, driving the price below par.

The portfolio value, 296,386 €, is simply the sum of the three prices. The NPV of holding the portfolio to maturity — computed by discounting the aggregate cash flows at each bond's own yield — equals exactly zero, as required by no-arbitrage: each bond is fairly priced at its market yield. What the portfolio *does* provide is **diversification**: Bond 2's premium offsets part of Bond 3's discount, and the safe income from Bond 1 cushions the portfolio against adverse moves in either corporate issuer.

CHAPTER 4

Zero-Coupon Bonds

4.1 Zero-coupon bonds

In Chapter 3 we studied bonds that make periodic coupon payments. The simplest possible bond eliminates coupons entirely: it makes a single payment of its face value at maturity. This instrument is the fundamental building block of fixed income.

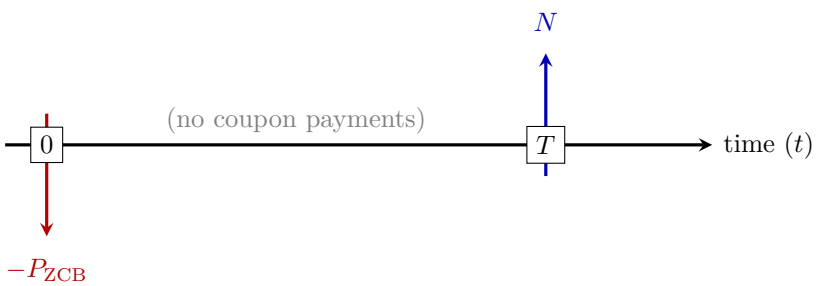
Definition 30: Zero-coupon bond (ZCB)

A **zero-coupon bond** (also called a **discount bond** or **pure discount bond**) is a bond that pays no coupons. It makes a single payment of its face value N at maturity T. Its price today is simply:

$$P_{\text{ZCB}}(T) = N \cdot \text{DF}(T),$$

where $\text{DF}(T)$ is the discount factor for maturity T (Definition 16).

Since there are no intermediate payments, a ZCB always trades at a discount to par (when $r > 0$; see Definition 24). The entire return comes from the **price appreciation**: you buy at $P < N$ and receive N at maturity.



The cash flow structure of a zero-coupon bond: pay P_{ZCB} today, receive face value N at maturity.

Example 24: Pricing a zero-coupon bond

A 3y ZCB with $N = 100{,}000$ € is priced using an annual rate of $r = 3\%$. What is the price today, and what is the investor's total return?

Solution.

$$P = 100{,}000 \cdot \frac{1}{1.03^3} = 100{,}000 \cdot 0.9151 = 91{,}510\,€.$$

You pay 91,510 € today and receive 100,000 € in 3y. The 8,490 € difference is the investor's entire return, consisting purely of price appreciation — there are no intermediate coupon payments.

4.2 Decomposing a coupon bond into ZCBs

Since the price of a ZCB with face value $N = 1$ is exactly $\mathrm{DF}(T)$, zero-coupon bonds are the basic "building blocks" of fixed income. Any coupon bond can be viewed as a portfolio of ZCBs, and the entire discount curve can be extracted from ZCB prices.

Replication by ZCBs

Every fixed-rate coupon bond is equivalent to a portfolio of zero-coupon bonds — one ZCB for each coupon payment and one for the final principal repayment:

$$B = \sum_{i=1}^{n} c_i \cdot \mathrm{DF}(t_i) = \sum_{i=1}^{n} c_i \cdot P_{\mathrm{ZCB}}(t_i; N = 1),$$

where c_i are the bond's cash flows (coupons and principal), $\mathrm{DF}(t_i)$ is the discount factor to time t_i, and $P_{\mathrm{ZCB}}(t_i; N = 1)$ is the price of a unit zero-coupon bond maturing at t_i.

This decomposition follows directly from the bond pricing formula (Definition 21).

Example 25: Bond as a portfolio of ZCBs

Consider a 3y bond with $N = 100{,}000\,€$ and annual coupon rate $c = 4\%$ (the same bond as in Example 13). Given the discount factors $\mathrm{DF}(1) = 0.990$, $\mathrm{DF}(2) = 0.943$, and $\mathrm{DF}(3) = 0.889$, decompose the bond into ZCBs and compute its price.

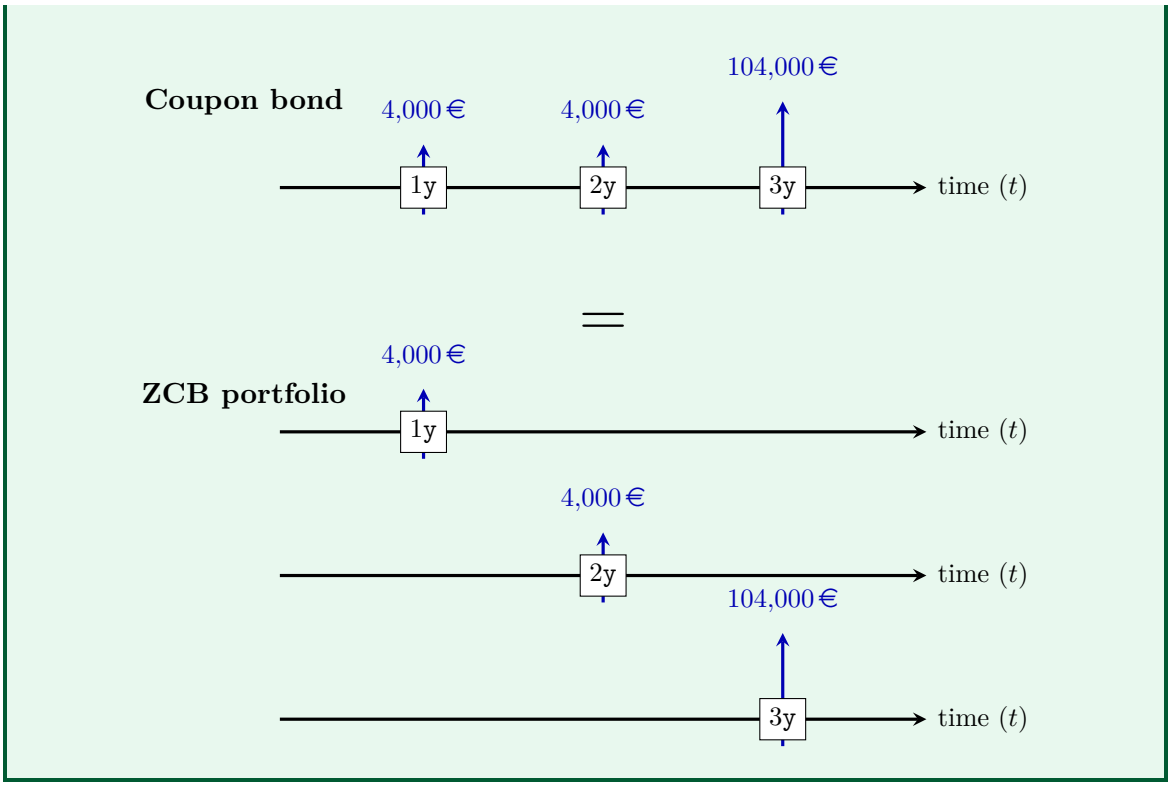
Solution. The bond generates three cash flows, each equivalent to a zero-coupon bond:

Cash flow	Time	Equivalent ZCB	Value
4,000 €	1y	1y ZCB with face value 4,000 €	$4{,}000 \cdot 0.990 = 3{,}960\,€$
4,000 €	2y	2y ZCB with face value 4,000 €	$4{,}000 \cdot 0.943 = 3{,}772\,€$
104,000 €	3y	3y ZCB with face value 104,000 €	$104{,}000 \cdot 0.889 = 92{,}456\,€$

The bond price is the sum of these three present values:

$$B = 3{,}960 + 3{,}772 + 92{,}456 = 100{,}188\,€.$$

This demonstrates that a coupon bond is nothing more than a portfolio of zero-coupon bonds, one for each promised payment.



Remark (Why this decomposition matters). The ZCB decomposition is more than a mathematical curiosity. It has three important practical consequences:

- **Pricing consistency:** If a coupon bond's market price differs from the sum of its ZCB components, an arbitrage opportunity exists. Traders would "strip" the bond (sell the individual cash flows as ZCBs) or "reconstitute" it (buy ZCBs and package them into a coupon bond) until prices align.
- **Curve construction:** The spot curve (which we construct in Chapter 7) is extracted by bootstrapping discount factors from coupon bond prices, treating each bond as a portfolio of ZCBs.
- **Risk management:** Bond's interest rate sensitivity can be broken down maturity by maturity, revealing which part of the yield curve drives the most risk.

Remark (Comparison: coupon bond vs. ZCB). The following table summarises the key differences:

Feature	Coupon bond	Zero-coupon bond
Intermediate payments	Yes (coupons)	None
Reinvestment risk	Yes	None
Interest rate sensitivity	Moderate	Highest for given maturity
Typical trading	Par or near par	Deep discount
Price at maturity	N	N

A ZCB carries no **reinvestment risk** (the risk that interim cash flows must be reinvested at unfavourable rates), because there are no interim cash flows. However, it has the highest **interest rate sensitivity** for its maturity, since all value is concentrated at the final payment.

CHAPTER 5

Spot Rates and Forward Interest Rates

5.1 Spot rates

So far, when computing discount factors and present values, we have used a single interest rate r without specifying its precise nature. In practice, the rate at which a cash flow is discounted depends on *when* it is received. Different maturities generally command different rates. We now formalise this idea.

Definition 31: Spot rate (zero rate)

The **spot rate** $r(T)$ for maturity T is the annualised rate of return on a zero-coupon investment from today ($t = 0$) to time T. It represents the pure time value of money for that specific horizon, with no intermediate cash flows.
The corresponding discount factor is:

$$\mathrm{DF}(T) = \frac{1}{(1 + r(T))^T} \quad \text{(annual)}, \qquad \mathrm{DF}(T) = e^{-r(T)\cdot T} \quad \text{(continuous)}.$$

In general, $r(T)$ varies with T: a 1y investment may earn a different rate than a 5y investment. The collection of spot rates across all maturities forms the **spot curve**, which we will construct in Chapter 7. For now, the key point is that each maturity has its own discount rate.

Example 26: Spot rates in practice

Suppose the market quotes the following spot rates (annually compounded):

Maturity T	**Spot rate** $r(T)$	**Discount factor** $\mathrm{DF}(T)$
1y	3.00%	$1/1.03^1 = 0.9709$
2y	3.50%	$1/1.035^2 = 0.9335$
3y	4.00%	$1/1.04^3 = 0.8890$

The spot curve is upward sloping: longer maturities are discounted more heavily, reflecting the higher return demanded for tying up capital for longer.

A zero-coupon bond paying 100,000 € in three years has present value:

$$\text{PV} = 100{,}000 \cdot \text{DF}(3) = 100{,}000 \cdot 0.8890 = 88{,}900\,€.$$

Using a single flat rate of, say, $r = 3.5\%$ for all maturities would give $\text{PV} = 100{,}000/1.035^3 = 90{,}194\,€$ — a meaningfully different answer. This illustrates why using the correct spot rate for each maturity matters.

Remark (Spot rates vs. the single rate r). In Chapters 1 and 2, we used a single rate r for discounting. This is a special case of the spot rate framework where $r(T) = r$ for all T (a *flat* term structure). The spot rate formalism generalises this to allow different rates for different maturities.

Note that a zero-coupon bond (Definition 30) with face value $N = 1$ and maturity T has price exactly $\text{DF}(T) = 1/(1 + r(T))^T$. This is why spot rates are also called **zero rates**: they are the rates implied by ZCB prices.

5.2 The idea behind forward rates

Spot rates tell us the cost of borrowing from today to some future date. But what if you want to lock in a rate now for borrowing that starts *in the future*? For example, you know you will need a 1y loan starting 2y from now. What rate should you expect?
The answer is the forward rate — and it is determined by today's spot rates through a no-arbitrage argument (Definition 11).

Definition 32: Forward rate

The **forward rate** $f(T_1, T_2)$ is the interest rate, determined today, for borrowing or lending over the future period $[T_1, T_2]$. It is also denoted f_{T_1,T_2} when subscript notation is more convenient.
The forward rate is defined by the no-arbitrage condition that *two investment strategies with the same start and end dates must produce the same return*:

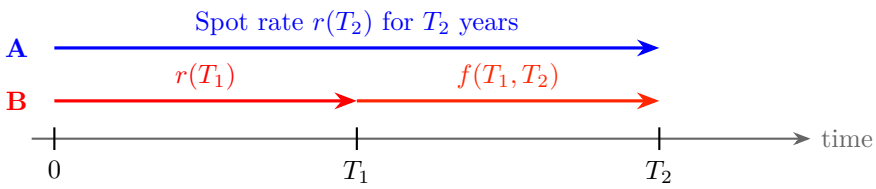
Strategy A: Invest at the spot rate $r(T_2)$ for T_2 years.
Strategy B: Invest at $r(T_1)$ for T_1 years, then roll over at $f(T_1, T_2)$ for the remaining $T_2 - T_1$ years.

No-arbitrage (Definition 11) requires these to give the same terminal value:

$$\underbrace{e^{r(T_2)\cdot T_2}}_{\text{Strategy A}} = \underbrace{e^{r(T_1)\cdot T_1} \cdot e^{f(T_1,T_2)\cdot(T_2-T_1)}}_{\text{Strategy B}} \qquad \text{(continuous compounding)},$$

or equivalently, for annual compounding:

$$\underbrace{(1 + r(T_2))^{T_2}}_{\text{Strategy A}} = \underbrace{(1 + r(T_1))^{T_1} \cdot (1 + f(T_1, T_2))^{T_2 - T_1}}_{\text{Strategy B}}.$$



Solving for $f(T_1, T_2)$ from the no-arbitrage condition:

Forward rate formula

Continuous compounding:

$$f(T_1, T_2) = \frac{r(T_2) \cdot T_2 - r(T_1) \cdot T_1}{T_2 - T_1}.$$

Annual compounding:

$$(1 + f(T_1, T_2))^{T_2 - T_1} = \frac{(1 + r(T_2))^{T_2}}{(1 + r(T_1))^{T_1}} = \frac{\mathrm{DF}(T_1)}{\mathrm{DF}(T_2)}.$$

Remark (Intuition for the forward rate formula). The forward rate formulas have deep economic intuition:

Discount factor interpretation: The ratio $\mathrm{DF}(T_1)/\mathrm{DF}(T_2)$ measures the growth factor over $[T_1, T_2]$. Since $\mathrm{DF}(T_1) > \mathrm{DF}(T_2)$ (for positive rates), this ratio exceeds 1, reflecting the accumulation of interest over $[T_1, T_2]$.

No-arbitrage enforcement: If the forward rate deviated from the no-arbitrage value, investors could profit by borrowing at one rate and lending at another, locking in a risk-free gain. Market forces eliminate such opportunities, pinning forward rates to their unique equilibrium.

Continuous compounding: The continuous formula is particularly clean. The product $r(T) \cdot T$ represents the total continuously compounded return over $[0, T]$, so

$$f(T_1, T_2) = \frac{(\text{total return to } T_2) - (\text{total return to } T_1)}{T_2 - T_1},$$

which is analogous to computing an average rate of change in calculus.

Example 27: Computing a forward rate (annual compounding)

The spot rates are $r(1) = 0.50\%$ and $r(2) = 1.00\%$ (annual compounding). Find the 1y forward rate starting in 1y, i.e. $f(1, 2)$.

Solution. Using the no-arbitrage condition:

$$(1 + r(1))(1 + f(1, 2)) = (1 + r(2))^2.$$

Substituting:

$$(1.005)(1 + f(1, 2)) = (1.01)^2 = 1.0201.$$

Solving for $f(1,2)$:

$$1 + f(1,2) = \frac{1.0201}{1.005} = 1.01503 \quad \Longrightarrow \quad f(1,2) = 1.503\%.$$

Interpretation: The forward rate (1.503%) is *higher* than both spot rates. This always happens when the spot curve is upward-sloping: to make the "average" over $[0, 2\mathtt{y}]$ equal 1% when the first year's rate is only 0.5%, the second year must "compensate" with a higher rate.

Example 28: Computing a forward rate (continuous compounding)

The spot rates are $r(2) = 3.00\%$ and $r(5) = 4.00\%$ (continuously compounded). Find the $3\mathtt{y}$ forward rate starting in $2\mathtt{y}$, i.e. $f(2,5)$.

Solution. Using the continuous compounding formula:

$$f(2,5) = \frac{r(5)\cdot 5 - r(2)\cdot 2}{5-2} = \frac{0.04\cdot 5 - 0.03\cdot 2}{3} = \frac{0.20 - 0.06}{3} = \frac{0.14}{3} = 4.667\%.$$

Interpretation: The forward rate (4.667%) exceeds the $5\mathtt{y}$ spot rate (4%). This reflects the upward-sloping yield curve: rates are expected to be higher in the future period $[2\mathtt{y}, 5\mathtt{y}]$ than the average over $[0, 5\mathtt{y}]$.

Example 29: Building a spot curve from forward rates

Suppose you know: $r(1) = 1.00\%$ and the sequence of $1\mathtt{y}$ forward rates $f(1,2) = 1.50\%$, $f(2,3) = 1.75\%$, $f(3,4) = 2.00\%$ (annual compounding). Find the $4\mathtt{y}$ spot rate $r(4)$.

Solution. Under annual compounding, chaining the rates:

$$(1 + r(4))^4 = (1 + r(1)) \cdot (1 + f(1,2)) \cdot (1 + f(2,3)) \cdot (1 + f(3,4)).$$

Substituting:

$$(1 + r(4))^4 = (1.01)(1.015)(1.0175)(1.02) = 1.06395.$$

Taking the fourth root:

$$r(4) = 1.06395^{1/4} - 1 = 1.01562 - 1 = 1.562\%.$$

Financial interpretation: The spot rate $r(4) = 1.562\%$ is the single constant rate that, if applied for $4\mathtt{y}$, would produce the same terminal value as the sequence of varying forward rates.

Geometric vs. arithmetic mean. The $4\mathtt{y}$ spot rate (1.562%) is close to the simple average of the four $1\mathtt{y}$ rates:

$$\frac{1.00 + 1.50 + 1.75 + 2.00}{4} = 1.5625\%.$$

The small difference arises because the spot rate is a *geometric* mean of $(1+\text{rate})$ factors, while the simple average is arithmetic. By the AM–GM inequality, the geometric mean is always slightly lower than the arithmetic mean for positive rates.

CHAPTER 6

Floating-Rate Bonds

6.1 The logic of floating rates

A fixed-rate bond (Chapter 3) commits to paying the same coupon for its entire life. While this provides certainty, it creates a problem: when market interest rates rise, the bond's fixed coupons become less attractive, causing its market price to fall sharply. Conversely, when rates fall, the bond's price rises. This interest rate sensitivity (Chapter 9) can be undesirable for investors who want stable principal values.

A floating-rate bond solves this problem by periodically adjusting its coupon to track current market rates. When rates rise, the bond's coupons increase, keeping its value near par. When rates fall, coupons decrease, again maintaining value near par. The investor trades away the potential capital gains from falling rates in exchange for protection against capital losses from rising rates.

Definition 33: Floating-rate note (FRN)

A **floating-rate note** (FRN) is a bond whose coupon rate is not fixed but instead resets periodically based on a market reference rate plus a fixed spread. The coupon payment for period i is:

$$C_i = N \cdot (L_i + s) \cdot \Delta t_i,$$

where N is the face value, L_i is the **reference rate** observed at the start of period i (the **reset date** or **fixing date**), s is the **fixed spread** (can be positive, negative, or zero), and Δt_i is the year fraction for period i (using the appropriate day-count convention).

Common reference rates L_i include:

- **EURIBOR** (Euro Interbank Offered Rate) for euro-denominated FRNs.
- **SOFR** (Secured Overnight Financing Rate) for USD FRNs.
- **SONIA** (Sterling Overnight Index Average) for GBP FRNs.

These overnight reference rates are discussed in more detail in Chapter 7, where we introduce the OIS curve and the broader family of interest rate curves.

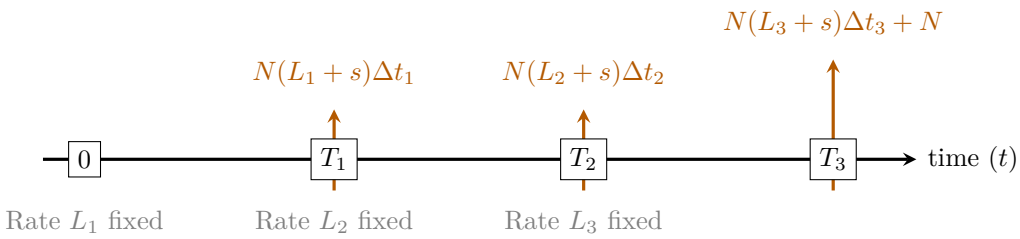
Remark (The role of the spread). The spread s reflects the issuer's **credit quality** and **liquidity**:

- $s = 0$: The issuer can borrow at exactly the reference rate (very rare; implies

near-government quality).
- $s > 0$: The issuer pays a premium above the reference rate (typical for corporate issuers; compare Definition 28).
- $s < 0$: The issuer pays less than the reference rate (extremely rare; might occur for very high-quality issuers when the reference rate itself contains a credit component).

Unlike the reference rate, which changes at each reset date, the spread s is **fixed for the life of the bond**.

To understand how FRNs work, consider the timeline of events:



At each reset date T_{i-1}, the reference rate L_i is observed and locked in for the next period. The corresponding coupon is paid at the end of that period, at time T_i.

Key features of the FRN coupon mechanism:
- **Reset in advance:** The rate L_i for period i is set at the *beginning* of the period (at time T_{i-1}).
- **Payment in arrears:** The coupon is paid at the *end* of the period (at time T_i).
- **Known in advance:** Once L_i is fixed, the next coupon payment is known with certainty.

6.2 Valuation at reset dates

The most important property of floating-rate bonds is that they trade at or very near par on reset dates (assuming zero or small spread).

FRN par pricing

Consider a floating-rate note with zero spread ($s = 0$) valued exactly at a reset date. The bond's dirty price equals its face value:

$$B_{\text{dirty}} = N.$$

This result holds regardless of the current level of interest rates or the bond's remaining maturity.

Remark (Intuition for par pricing). Why does an FRN trade at par on reset dates? Consider what happens at a generic reset date T_{i-1}:
1. The reference rate L_i is observed (say, $L_i = 3\%$).
2. The coupon for the next period is locked in: $C_i = N \cdot L_i \cdot \Delta t_i$.
3. This coupon will be paid at time T_i.
4. After paying C_i, the bond resets again at T_i and repeats the process.

From the perspective of time T_{i-1}, the bond is equivalent to a deposit that pays $N \cdot L_i \cdot \Delta t_i$ at time T_i, plus a new FRN starting at T_i with the same structure. Since

L_i equals the current market rate for period $[T_{i-1}, T_i]$, the present value of the next coupon is:

$$\text{PV(next coupon)} = \frac{N \cdot L_i \cdot \Delta t_i}{1 + L_i \cdot \Delta t_i}.$$

The new FRN starting at T_i will also trade at par when T_i arrives. Discounting this par value back to T_{i-1}:

$$\text{PV(future FRN)} = \frac{N}{1 + L_i \cdot \Delta t_i}.$$

Adding these components:

$$B(T_{i-1}) = \frac{N \cdot L_i \cdot \Delta t_i + N}{1 + L_i \cdot \Delta t_i} = \frac{N(1 + L_i \cdot \Delta t_i)}{1 + L_i \cdot \Delta t_i} = N.$$

By induction, this argument applies at every reset date, so the bond always trades at par (assuming zero spread).

Example 30: FRN at par (zero spread)

Consider a 3y FRN with the following characteristics:

- Face value: $N = 100{,}000\,€$
- Reset frequency: Annual
- Reference rate: 1y EURIBOR
- Spread: $s = 0\,\text{bps}$

Today is a reset date. The current 1y EURIBOR is 2.00%. The spot curve (Definition 31) gives us:

Maturity	**Spot rate**	DF(t)
1y	2.00%	0.980
2y	2.50%	0.952
3y	2.80%	0.921

The implied forward rates (Definition 32) are:

Forward rate	**Value**
$f(0,1)$	2.000%
$f(1,2)$	3.003%
$f(2,3)$	3.366%

What is the fair price of this FRN today?

Solution. Method 1: Direct application of the par pricing result.
Since today is a reset date and the spread is zero, the FRN must trade at par:

$$B = 100{,}000\,€.$$

Method 2: Cash flow valuation (verification).
The coupons, using the forward rates as projections for future reference rates, are:

- Year 1: $L_1 = 2.000\%$ (just fixed today), so $C_1 = 100{,}000 \cdot 0.020 = 2{,}000\,€$.
- Year 2: $L_2 = f(1,2) = 3.003\%$ (expected), so $C_2 \approx 100{,}000 \cdot 0.030 = 3{,}000\,€$.
- Year 3: $L_3 = f(2,3) = 3.366\%$ (expected), so $C_3 = 100{,}000 \cdot 0.03366 = 3{,}366\,€$.

Present value calculation:

$$\begin{aligned} B &= C_1 \cdot \mathrm{DF}(1) + C_2 \cdot \mathrm{DF}(2) + (C_3 + N) \cdot \mathrm{DF}(3) \\ &= 2{,}000 \cdot 0.980 + 3{,}000 \cdot 0.952 + 103{,}366 \cdot 0.921 \\ &= 1{,}960 + 2{,}856 + 95{,}200 \\ &= 100{,}016\,€ \approx 100{,}000\,€. \end{aligned}$$

(The small difference is due to rounding in the forward rate calculations. With exact forward rates derived from the discount factors, we would get exactly 100,000 €.)

Example 31: FRN with positive spread

Consider the same FRN as in Example 30, but now with a spread of $s = 150\,\text{bps}$ (1.50%). What is the fair price?

Solution. The positive spread increases each coupon payment by $100{,}000 \cdot 0.015 = 1{,}500\,€$. The adjusted coupons are:

- Year 1: $C_1 = 100{,}000 \cdot (0.020 + 0.015) = 3{,}500\,€$.
- Year 2: $C_2 = 100{,}000 \cdot (0.030 + 0.015) = 4{,}500\,€$.
- Year 3: $C_3 = 100{,}000 \cdot (0.03366 + 0.015) = 4{,}866\,€$.

Present value:

$$\begin{aligned} B &= 3{,}500 \cdot 0.980 + 4{,}500 \cdot 0.952 + 104{,}866 \cdot 0.921 \\ &= 3{,}430 + 4{,}284 + 96{,}582 \\ &= 104{,}296\,€. \end{aligned}$$

Interpretation: The positive spread causes the FRN to trade at a premium above par. The premium roughly equals the present value of the spread payments:

$$\mathrm{PV}(\text{spread}) = 1{,}500 \cdot (0.980 + 0.952 + 0.921) = 1{,}500 \cdot 2.853 = 4{,}280\,€,$$

which accounts for most of the 4,296 € premium. The remaining difference comes from the interaction between the spread and the forward rates.

6.3 Valuation between reset dates

When valuing an FRN between reset dates, we must account for the accrued interest on the current period's coupon (which has already been fixed) and the fact that at the next reset date, the bond will trade at par (plus any spread adjustment).

Example 32: FRN between reset dates

Consider the zero-spread FRN from Example 30. Suppose we are valuing it `6m` after the last reset date. At that reset date:

- The `1y` EURIBOR was fixed at $L_1 = 2.00\%$.
- This locked in the next annual coupon: $C_1 = 100{,}000 \cdot 0.02 = 2{,}000\,€$ (to be paid `6m` from now).

The current `6m` spot rate is $r(0.5) = 1.80\%$, so $\mathrm{DF}(0.5) = 1/\sqrt{1.018} = 0.991$.
Compute the FRN's dirty price, accrued interest, and clean price.

Solution. Step 1 — Value at next reset date.
In 6m (at the next reset date), the FRN will pay the current period's coupon (2,000 €) and then reset to trade at par (100,000 €). Total value at next reset date: 102,000 €.

Step 2 — Discount to today.

$$B_{\text{dirty}} = 102{,}000 \cdot \text{DF}(0.5) = 102{,}000 \cdot 0.991 = 101{,}082\,€.$$

Step 3 — Calculate clean price.
Accrued interest for 6m on the 2,000 € annual coupon (using 30/360; Definition 12):

$$AI = 2{,}000 \cdot \frac{180}{360} = 1{,}000\,€.$$

Clean price (Definition 22):

$$B_{\text{clean}} = 101{,}082 - 1{,}000 = 100{,}082\,€.$$

Interpretation: The dirty price (101,082 €) exceeds par because it includes accrued interest. The clean price (100,082 €) is very close to par, with the small premium reflecting the time value of the already-fixed coupon payment that will arrive in 6m. As we approach the next reset date, the dirty price will approach 102,000 € (par plus full coupon), while the clean price will approach exactly 100,000 €.

Remark (FRN vs. fixed-rate bond: a comparison). The following table summarises the key differences between the two bond types:

Feature	Fixed-rate	Floating-rate
Coupon	Fixed for life	Resets each period
Price at par	Only when $c = y$	At every reset date ($s = 0$)
Interest rate sensitivity	High (long duration)	Very low (short duration)
Reinvestment risk	Yes	Minimal
Credit spread exposure	Yes	Yes

The FRN's very low **duration** (Chapter 9) is precisely why borrowers and investors use them to manage interest rate risk. However, FRNs still carry **credit risk** (Definition 28): if the issuer's creditworthiness deteriorates, the spread demanded by the market may exceed the fixed spread s, causing the FRN to trade below par.

CHAPTER 7

Interest Rate Curves

In Chapter 5 we introduced spot rates and forward rates for individual maturities. In reality, the rate you can earn (or must pay) depends on *how long* you invest. A 1y deposit does not earn the same rate as a 10y bond. The collection of rates across maturities is called a curve.

7.1 The yield curve

Definition 34: Yield curve

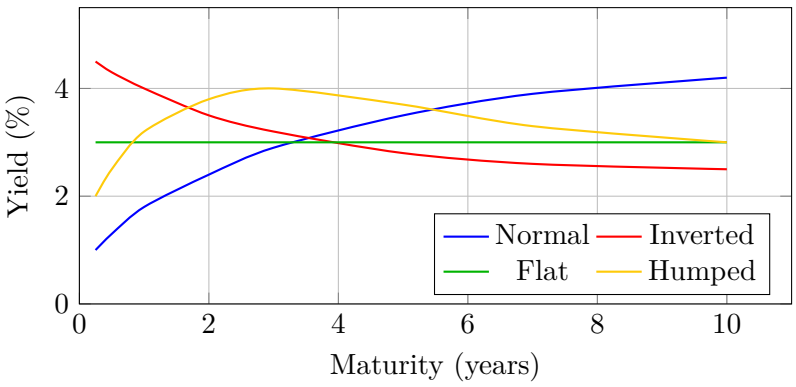
The **yield curve** is the function $y\colon T \mapsto y(T)$ that maps each maturity T to the yield to maturity (Definition 23) of a reference bond (typically a government bond) with that maturity.

The yield curve is one of the most important objects in all of finance. Its shape conveys the market's expectations about future interest rates, inflation, and economic growth.

Typical shapes:

- **Normal (upward-sloping):** Long-term rates are higher than short-term rates. This is the most common shape, reflecting the fact that investors demand a **term premium** for locking up money for longer.
- **Inverted:** Short-term rates exceed long-term rates. Often interpreted as a recession signal: the market expects central banks to cut rates in the future.
- **Flat:** Rates are similar across all maturities.
- **Humped:** Rates peak at an intermediate maturity and then decline.

Stylised yield curve shapes



7.1.1 Constructing the yield curve

The yield curve is constructed directly from market-traded bonds at various maturities. The construction process involves:

1. **Data collection:** Gather prices (or yields) of actively traded government bonds across all available maturities.
2. **YTM calculation:** For each bond, calculate its yield to maturity using the pricing formula (Definition 21).
3. **Plotting:** Plot maturity T on the horizontal axis and YTM $y(T)$ on the vertical axis.
4. **Interpolation:** Since bonds exist only at discrete maturities, use interpolation (linear, cubic spline, or parametric models such as Nelson–Siegel; see Appendix B) to create a continuous curve between observed points.

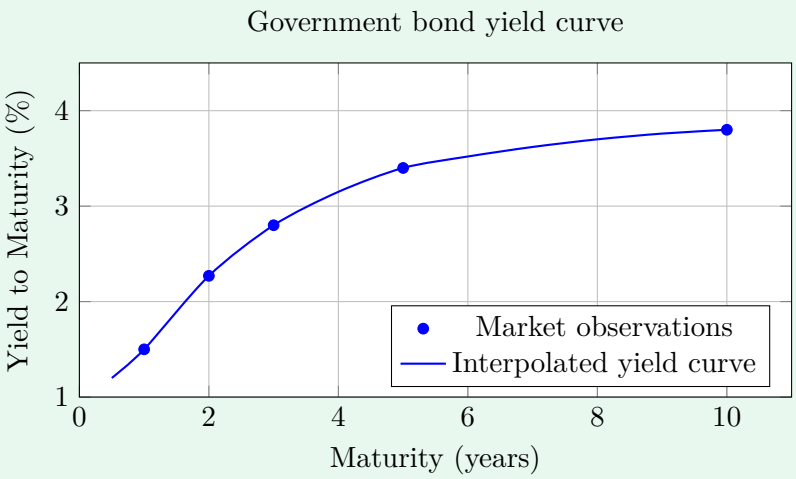
The key insight is that the yield curve uses the YTM of coupon-bearing bonds, which reflects a complex average of spot rates over different periods. This makes it convenient for market participants but less suitable for precise valuation—hence the need for the spot curve.

Example 33: Constructing a simple yield curve

The following government bonds are observed in the market (all with $N = 10{,}000$ € and annual coupons). Construct the yield curve.

Maturity	Coupon rate	Dirty price	YTM
1y	2.0%	10,049 €	1.50%
2y	3.0%	10,143 €	2.27%
3y	2.5%	9,925 €	2.80%
5y	4.0%	10,350 €	3.40%
10y	3.5%	9,680 €	3.80%

Solution. The yield curve is simply the plot of maturity versus YTM:



The curve shows a typical upward-sloping shape with decreasing slope (flattening) at longer maturities, which is common in normal market conditions. The interpolation fills in yields for maturities where no bonds are traded (e.g., 4y, 6y, 7y).

7.2 The spot curve and bootstrapping

The spot curve is the most fundamental interest rate curve in fixed income markets. Unlike the yield curve, which uses yields to maturity of coupon-bearing bonds, the spot curve provides pure discount rates for each maturity without the averaging effect of intermediate coupon payments.

Definition 35: Spot curve (zero-coupon curve)

The **spot curve** (or **zero-coupon curve**) is the function $r\colon T \mapsto r(T)$ that maps each maturity T to its corresponding spot rate $r(T)$ (Definition 31).

The spot curve represents the term structure of risk-free interest rates and is the curve needed for pricing all fixed income securities.

7.2.1 The bootstrapping procedure

In practice, zero-coupon bonds may not be traded for every maturity we need. Instead, we observe market prices of coupon-bearing bonds. Bootstrapping is the iterative procedure that extracts the spot curve from these coupon bond prices.

Definition 36: Bootstrapping

Given market prices of n bonds with maturities $T_1 < T_2 < \cdots < T_n$, **bootstrapping** is the sequential procedure for extracting the spot curve:

1. Use the shortest-maturity bond to find $\mathrm{DF}(T_1)$ directly (this bond typically has one remaining cash flow).
2. Use the next bond and the already-known $\mathrm{DF}(T_1)$ to isolate $\mathrm{DF}(T_2)$.
3. Continue iteratively: each subsequent bond equation contains exactly one unknown discount factor, which can be solved algebraically once all earlier discount factors are known.
4. Convert each discount factor to its corresponding spot rate using:

$$r(T) = \mathrm{DF}(T)^{-1/T} - 1 \quad \text{(annual)}, \qquad r(T) = -\frac{\ln(\mathrm{DF}(T))}{T} \quad \text{(continuous)}.$$

5. If needed, interpolate between the bootstrapped points to obtain a complete spot curve. Common methods include linear interpolation, cubic splines, or parametric models such as Nelson–Siegel (Appendix B).

Remark (Why bootstrapping works). Bootstrapping works because of a simple principle: each successive bond adds exactly one new maturity to the payment schedule. When we price the n-th bond, it has cash flows at times $T_1, T_2, \ldots, T_n$. Since we have already determined $\mathrm{DF}(T_1), \ldots, \mathrm{DF}(T_{n-1})$ from earlier bonds, only $\mathrm{DF}(T_n)$ remains unknown — making it directly solvable. This creates a triangular system where each equation introduces one new unknown, allowing us to solve sequentially by forward substitution.

Example 34: Bootstrapping the spot curve from bond prices

We observe three annual-coupon government bonds, all with face value $N = 10{,}000$ €:

Maturity	Annual coupon	Dirty price
1y	2%	10,049 €
2y	3%	10,143 €
3y	2.5%	9,925 €

Extract the spot curve by computing the discount factors DF(1), DF(2), DF(3) and the corresponding spot rates $r(1)$, $r(2)$, $r(3)$.

Solution. To extract the spot curve, we use the general bond pricing formula:

$$B = \sum_{i=1}^{n} C_i \cdot \mathrm{DF}(T_i) + N \cdot \mathrm{DF}(T_n).$$

Step 1 — Extract DF(1) **from the** 1y **bond.** The 1y bond pays a single cash flow of 10,200 € at $t = 1\mathrm{y}$ (coupon 200 € plus principal 10,000 €). The pricing equation is:

$$10{,}049 = 10{,}200 \cdot \mathrm{DF}(1) \quad \Longrightarrow \quad \mathrm{DF}(1) = \frac{10{,}049}{10{,}200} = 0.98520.$$

Converting to spot rate: $r(1) = 1/0.98520 - 1 = 1.502\%$.

Step 2 — Extract DF(2) **from the** 2y **bond.** The 2y bond pays 300 € at $t = 1\mathrm{y}$ and 10,300 € at $t = 2\mathrm{y}$. We already know DF(1), so the pricing equation has only one unknown:

$$10{,}143 = 300 \cdot \mathrm{DF}(1) + 10{,}300 \cdot \mathrm{DF}(2) = 300 \cdot 0.98520 + 10{,}300 \cdot \mathrm{DF}(2).$$

Isolating DF(2):

$$10{,}143 - 295.56 = 10{,}300 \cdot \mathrm{DF}(2) \quad \Longrightarrow \quad \mathrm{DF}(2) = \frac{9{,}847.44}{10{,}300} = 0.95606.$$

Converting to spot rate: $r(2) = (1/0.95606)^{1/2} - 1 = 2.274\%$.

Step 3 — Extract DF(3) **from the** 3y **bond.** The 3y bond pays 250 € at $t = 1\mathrm{y}$ and $t = 2\mathrm{y}$, and 10,250 € at $t = 3\mathrm{y}$. We know DF(1) and DF(2):

$$9{,}925 = 250 \cdot 0.98520 + 250 \cdot 0.95606 + 10{,}250 \cdot \mathrm{DF}(3).$$

Solving for DF(3):

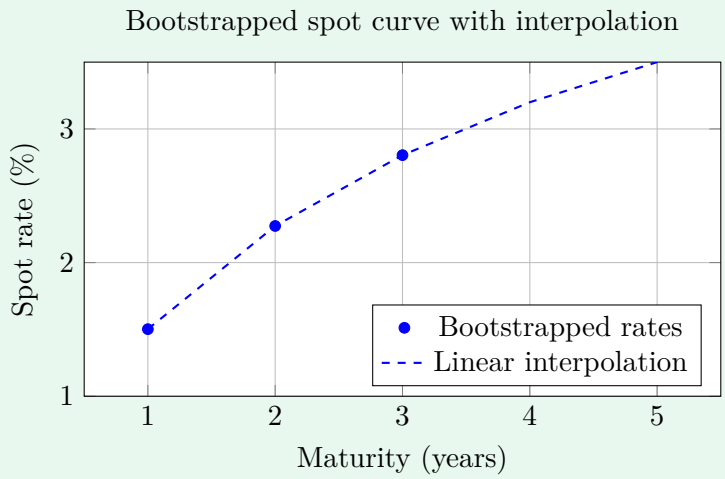
$$\mathrm{DF}(3) = \frac{9{,}925 - 246.30 - 239.02}{10{,}250} = \frac{9{,}439.68}{10{,}250} = 0.92094.$$

Converting to spot rate: $r(3) = (1/0.92094)^{1/3} - 1 = 2.804\%$.

Step 4 — Summary and visualisation.

Maturity	Discount factor	Spot rate
1y	0.985	1.502%
2y	0.956	2.274%
3y	0.921	2.804%

The bootstrapped spot curve is upward-sloping, indicating that longer-term rates are higher than short-term rates — a typical "normal" yield curve shape reflecting a positive term premium.



The dashed line shows a simple linear interpolation extending the curve to 4y and 5y. In practice, more sophisticated methods (cubic splines, Nelson–Siegel; see Appendix B) would be used for smoother interpolation.

Remark (Spot curve vs. yield curve). Key differences and similarities:

- **Yield curve:** plots yields to maturity of coupon-bearing bonds; each yield is an *average* rate across all cash flows.
- **Spot curve:** plots zero-coupon rates; each rate is the *pure* discount rate for a single maturity.
- Both represent the term structure of interest rates but measure it differently.
- For upward-sloping curves, the spot curve typically lies above the yield curve (yields average lower early rates with higher later rates).
- The spot curve is the fundamental pricing tool; it can be extracted from the yield curve via bootstrapping.
- The yield curve is more directly observable (coupon bonds trade actively); the spot curve must be constructed.

7.3 The forward curve

The forward rate $f(T_1, T_2)$ (Definition 32) is the rate for borrowing/lending during $[T_1, T_2]$ locked in today at no cost, as implied by the no-arbitrage condition on today's market.

Definition 37: Forward curve

The **forward curve** is the collection of all forward rates implied by the current spot curve through no-arbitrage relationships.

Discrete compounding: For a given compounding frequency (e.g., annual), the forward curve is the function

$$f\colon (T_1, T_2) \mapsto f(T_1, T_2)$$

that maps each time interval $[T_1, T_2]$ to the forward rate $f(T_1, T_2)$ applicable to that period.
For annual compounding with integer maturities, the forward curve is often represented as the sequence of one-year forward rates $\{f(T-1, T)\}_{T=1,2,3,\ldots}$, where $f(T-1, T)$ is the 1y rate starting at time $T-1$.

Continuous compounding: The forward curve is the function

$$f : T \mapsto f(T)$$

that maps each maturity T to the **instantaneous forward rate** $f(T)$, representing the continuously compounded rate applicable to the infinitesimal interval $[T, T + dT]$ as implied by today's spot curve.

7.3.1 Constructing the forward curve

The forward curve is derived from the spot curve using the no-arbitrage relationship established in Chapter 5. Two equivalent approaches exist:

Method 1: From spot rates (discrete). For annual compounding, the 1y forward rate starting at year $T-1$ is:

$$1 + f(T-1, T) = \frac{(1 + r(T))^T}{(1 + r(T-1))^{T-1}} = \frac{\mathrm{DF}(T-1)}{\mathrm{DF}(T)}.$$

Method 2: Instantaneous forward rate (continuous). For continuous compounding, the instantaneous forward rate at maturity T is:

$$f(T) = \frac{d}{dT}[r(T) \cdot T] = r(T) + T \cdot r'(T).$$

The product $r(T) \cdot T$ represents the total continuously compounded return from time 0 to time T. The forward rate at T asks: "What incremental rate applies to the next infinitesimal period $[T, T + dT]$?" This is precisely the marginal rate of accumulation — the derivative. The formula shows that the instantaneous forward rate equals the spot rate $r(T)$ plus a correction term $T \cdot r'(T)$ that captures how the spot curve's slope affects forward rates. If the spot curve is upward-sloping ($r'(T) > 0$), forward rates exceed spot rates; if downward-sloping, forward rates lie below spot rates.

Example 35: Deriving the forward curve (discrete)

Using the spot rates from Example 34: $r(1) = 1.502\%$, $r(2) = 2.274\%$, $r(3) = 2.804\%$. Compute the 1y forward rates $f(0, 1)$, $f(1, 2)$, and $f(2, 3)$.

Solution. Forward rate $f(0, 1)$: The forward rate from today to year 1 is simply the 1y spot rate:

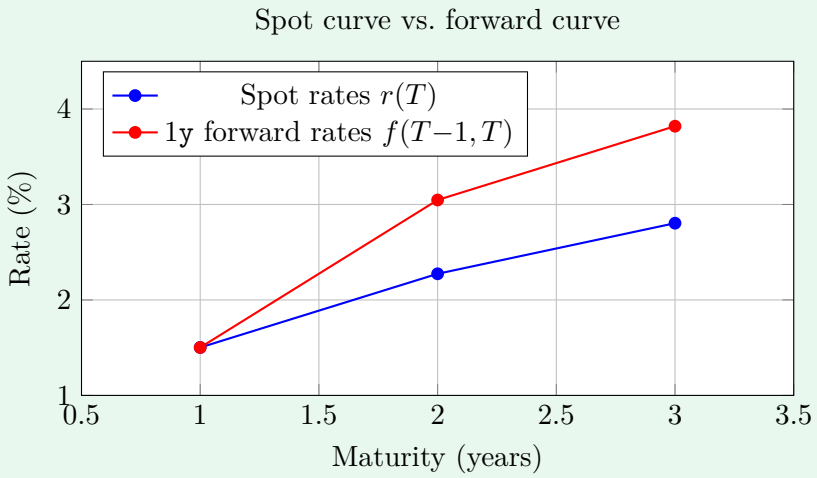
$$f(0, 1) = r(1) = 1.502\%.$$

Forward rate $f(1, 2)$: The 1y rate starting in year 1:

$$1 + f(1, 2) = \frac{(1.02274)^2}{1.01502} = \frac{1.04595}{1.01502} = 1.03047 \quad \Longrightarrow \quad f(1, 2) = 3.047\%.$$

Forward rate $f(2,3)$: The 1y rate starting in year 2:

$$1 + f(2,3) = \frac{(1.02804)^3}{(1.02274)^2} = \frac{1.08590}{1.04595} = 1.03820 \quad \Longrightarrow \quad f(2,3) = 3.820\%.$$



The forward curve lies above the spot curve because the spot curve is upward-sloping. Each forward rate is higher than the previous one, reflecting the market's expectation (or risk premium) for rising rates. The forward rate $f(2,3) = 3.820\%$ is the rate you can lock in today for a 1y loan starting in 2y.

Example 36: Instantaneous forward curve (continuous)

Suppose the continuously compounded spot rate curve is given by the Nelson–Siegel parametric form (Appendix B):

$$r(T) = 2\% + 3\% \cdot e^{-0.5T} - 1\% \cdot T \cdot e^{-0.5T}.$$

Derive the instantaneous forward curve $f(T)$ and evaluate it at $T = 1\text{y}, 2\text{y}, 3\text{y}$.

Solution. We use the formula $f(T) = r(T) + T \cdot r'(T)$. First, compute the spot rates:

$$\begin{aligned}
r(1) &= 2\% + 3\% \cdot e^{-0.5} - 1\% \cdot 1 \cdot e^{-0.5} = 2\% + 1.82\% - 0.61\% = 3.21\%,\\
r(2) &= 2\% + 3\% \cdot e^{-1.0} - 1\% \cdot 2 \cdot e^{-1.0} = 2\% + 1.10\% - 0.74\% = 2.37\%,\\
r(3) &= 2\% + 3\% \cdot e^{-1.5} - 1\% \cdot 3 \cdot e^{-1.5} = 2\% + 0.67\% - 0.67\% = 2.00\%.
\end{aligned}$$

Next, find the derivative:

$$\begin{aligned}
r'(T) &= 3\% \cdot (-0.5)\, e^{-0.5T} - 1\% \cdot e^{-0.5T} - 1\% \cdot T \cdot (-0.5)\, e^{-0.5T}\\
&= e^{-0.5T}\big[-2.5\% + 0.5\% \cdot T\big].
\end{aligned}$$

Now compute the instantaneous forward rates:
At $T = 1\text{y}$:

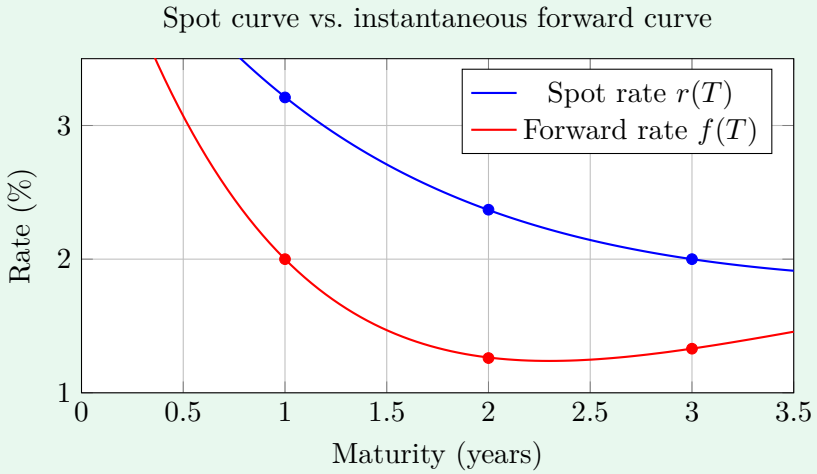
$$\begin{aligned}
r'(1) &= e^{-0.5}[-2.5\% + 0.5\%] = 0.6065 \cdot (-2.0\%) = -1.21\%,\\
f(1) &= 3.21\% + 1 \cdot (-1.21\%) = 2.00\%.
\end{aligned}$$

At $T = 2$y:

$$r'(2) = e^{-1.0}[-2.5\% + 1.0\%] = 0.3679 \cdot (-1.5\%) = -0.55\%,$$
$$f(2) = 2.37\% + 2 \cdot (-0.55\%) = 1.26\%.$$

At $T = 3$y:

$$r'(3) = e^{-1.5}[-2.5\% + 1.5\%] = 0.2231 \cdot (-1.0\%) = -0.22\%,$$
$$f(3) = 2.00\% + 3 \cdot (-0.22\%) = 1.33\%.$$



In this example, the spot curve is downward-sloping (hump-shaped with declining rates), so the instantaneous forward rates lie *below* the spot rates. This reflects the fact that $r'(T) < 0$ for these maturities. The forward curve shows where rates are headed in the instantaneous sense: the market implies that very short-term rates will be lower in the future than current long-term averages suggest.

Remark (Forward rates and expectations). While forward rates are sometimes interpreted as the market's expected future spot rates, this is only true under the **pure expectations hypothesis**. In reality, forward rates include both expectations and **term premia** (risk compensation). An upward-sloping forward curve may reflect rising rate expectations, positive term premia, or both.

7.4 The OIS (Overnight Index Swap) curve

Modern fixed income markets require a consistent framework for discounting future cash flows. Since the 2008 financial crisis, the industry standard has shifted from interbank lending rates to rates based on overnight transactions.

The logic is straightforward: an overnight loan — lent today, repaid tomorrow — leaves almost no time for the borrower to default and no uncertainty about future rates. This makes overnight rates the closest approximation to a truly risk-free rate available in the market. Longer interbank rates (such as LIBOR) embed credit and liquidity risk over weeks or months, and were moreover based on bank surveys rather than actual transactions, leaving them open to manipulation.

Definition 38: Overnight index

An **overnight index** (or **overnight rate**) is a reference interest rate representing the cost of overnight borrowing between banks, published daily by central banks or regulatory authorities based on actual market transactions.

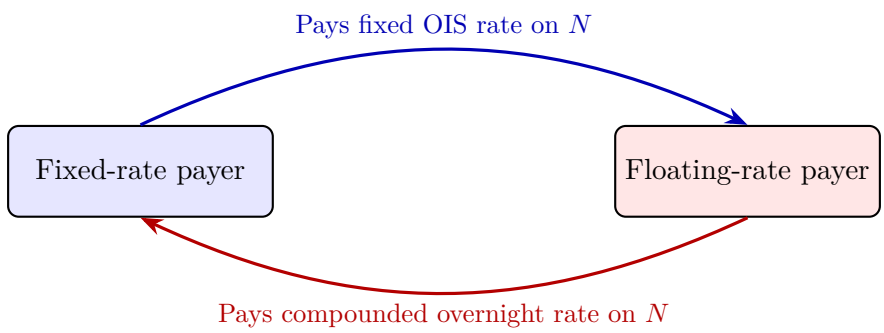
Examples include **ESTR** (Euro Short-Term Rate, euro area), **SOFR** (Secured Overnight Financing Rate, US, based on Treasury repo transactions), and **SONIA** (Sterling Overnight Index Average, UK). These are all transaction-based rates, making them more robust than the survey-based LIBOR they replaced.

Definition 39: Overnight Index Swap (OIS)

An **Overnight Index Swap** is a financial agreement between two parties to exchange interest payments on a notional principal N over a period $[0, T]$:

- **Fixed leg:** One party pays a fixed rate (the **OIS rate**) on N at maturity T.
- **Floating leg:** The other party pays the compounded overnight index rate accumulated over the same period.
- **No principal exchange:** Only the interest payments are exchanged; the notional N never changes hands.

The **OIS rate** for maturity T is the fixed rate that makes the agreement fair at inception — that is, the fixed rate such that the present value of the fixed leg equals the present value of the floating leg.



Mathematically, if r_i denotes the overnight rate on day i and there are n business days in $[0, T]$, the compounded return on the floating leg is:

$$\left(\prod_{i=1}^{n}(1 + r_i \cdot \delta_i)\right) - 1,$$

where δ_i is the day-count fraction for day i (typically 1/360 or 1/365; see Definition 12). For practical purposes, OIS rates can be interpreted as the market's expectation of the average overnight rate over $[0, T]$.

Definition 40: OIS curve

The **OIS curve** is the function $r_{\text{OIS}} \colon T \mapsto r_{\text{OIS}}(T)$ that maps each maturity T to the corresponding OIS rate. Since OIS transactions have minimal credit risk, the OIS curve represents the closest available approximation to a true **risk-free rate curve**.

The OIS curve serves two related but distinct purposes. First, it is a market-observable curve, like the yield curve: we can read off OIS rates for standard maturities and plot them against maturity. Second, and more importantly, it provides the discount factors needed for

present-value calculations. These two objects — rates and discount factors — are equivalent representations of the same information; the discount factor is simply how we convert a rate into a multiplier suitable for discounting:

$$\mathrm{DF}_{\mathrm{OIS}}(T) = \frac{1}{1 + r_{\mathrm{OIS}}(T) \cdot T}.$$

We therefore bootstrap the OIS curve primarily to obtain discount factors $\mathrm{DF}_{\mathrm{OIS}}(T)$, which are then used throughout the valuation of fixed income instruments and derivatives. The plotted curve of OIS rates is simply the most interpretable way to display these discount factors.

The OIS curve has become the industry standard for:

- Discounting derivative cash flows (interest rate swaps, options, etc.).
- Computing present values in collateralised transactions.
- Serving as the baseline for measuring credit and liquidity spreads (Definition 28).

7.4.1 Constructing the OIS curve

The construction follows the same bootstrapping methodology as the spot curve (Section 7.2):

1. **Data collection:** Gather market-quoted OIS rates for standard maturities (`1w`, `1m`, `3m`, `6m`, `1y`, `2y`, `5y`, `10y`, `30y`).
2. **Bootstrapping:** Sequentially solve for discount factors. For a simplified single-payment OIS at maturity T_i:
$$\mathrm{DF}(T_i) = \frac{1}{1 + r_{\mathrm{OIS}}(T_i) \cdot T_i}.$$
(For actual OIS with more frequent or compounded payments, a more detailed bootstrapping is needed.)
3. **Interpolation:** Use interpolation (linear, log-linear, cubic spline, or Nelson–Siegel) to create a smooth curve between observed maturities.

Example 37: Bootstrapping the OIS curve and pricing a bond

The following OIS rates are observed in the euro market.

Maturity	OIS Rate
1y	3.60%
2y	3.65%
3y	3.70%
5y	3.75%

(a) Bootstrap the OIS discount factors.
(b) Use the resulting discount curve to price a 3y government bond with a 4% annual coupon and face value $N = 10\,000$ €.

Solution. Part (a): Bootstrapping the discount factors. Using the simplified single-payment formula $\mathrm{DF}(T) = 1/(1 + r_{\mathrm{OIS}}(T) \cdot T)$:

$$\mathrm{DF}(1) = \frac{1}{1 + 0.0360 \cdot 1} = \frac{1}{1.0360} = 0.96525,$$

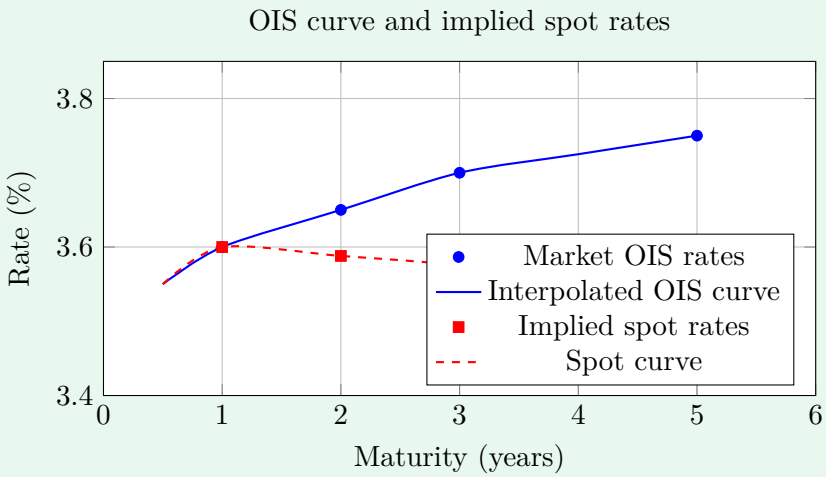
$$\mathrm{DF}(2) = \frac{1}{1 + 0.0365 \cdot 2} = \frac{1}{1.0730} = 0.93197,$$

$$\mathrm{DF}(3) = \frac{1}{1 + 0.0370 \cdot 3} = \frac{1}{1.1110} = 0.90009,$$

$$\mathrm{DF}(5) = \frac{1}{1 + 0.0375 \cdot 5} = \frac{1}{1.1875} = 0.84211.$$

Maturity	OIS Rate	Discount Factor	Implied spot rate
1y	3.60%	0.96525	3.600%
2y	3.65%	0.93197	3.588%
3y	3.70%	0.90009	3.576%
5y	3.75%	0.84211	3.548%

The implied spot rates are computed as $r(T) = \mathrm{DF}(T)^{-1/T} - 1$ and are slightly below the quoted OIS rates — a consequence of the simplified single-payment formula treating the OIS as if all interest accrues only at maturity (linear), while the spot rate applies compound discounting.



Part (b): Pricing the 3y bond. The bond pays coupons of 400 € at $t = 1\mathtt{y}$ and $t = 2\mathtt{y}$, and a final cash flow of 10 400 € at $t = 3\mathtt{y}$. Discounting each cash flow with the bootstrapped OIS discount factors:

$$\begin{aligned} B &= 400 \cdot \mathrm{DF}(1) + 400 \cdot \mathrm{DF}(2) + 10\,400 \cdot \mathrm{DF}(3) \\ &= 400 \cdot 0.96525 + 400 \cdot 0.93197 + 10\,400 \cdot 0.90009 \\ &= 386.10 + 372.79 + 9\,360.94 = 10\,119.83\,€. \end{aligned}$$

The bond prices above par (10 119.83 € vs. face value 10 000 €) because its coupon (4%) exceeds the prevailing discount rates (≈3.7% at the 3y horizon). This illustrates the core purpose of the bootstrapping exercise: the discount factors extracted from OIS market quotes provide a coherent, risk-free benchmark for valuing any stream of fixed cash flows.

Remark (OIS vs. LIBOR/EURIBOR). **LIBOR** (London Interbank Offered Rate) and **EURIBOR** (Euro Interbank Offered Rate) were benchmark rates representing the average rates at which major banks could borrow unsecured funds from each other in the interbank market, published daily for multiple tenors (overnight to 12m) based on panel bank submissions.
Before 2008, these interbank lending rates were widely used for discounting. Post-

crisis analysis revealed they contained **credit risk** (banks might default on unsecured loans), **liquidity risk** (banks might struggle to access funding), and **manipulation risk** (LIBOR was based on surveys, not actual transactions).
OIS rates, being based on actual overnight transactions (often secured by collateral), eliminate most of these issues. The **OIS–LIBOR spread** measures the credit/liquidity premium in the interbank market. During the 2008 crisis, this spread widened from typical levels of 10–20 bps to over 350 bps, signalling severe stress in bank funding markets.
LIBOR has been phased out (USD LIBOR ended June 2023), replaced by overnight risk-free rates like SOFR and SONIA. EURIBOR continues with reforms emphasising transaction data.

7.5 Estimation and discount curves

Valuing any financial instrument requires two logically independent steps: *projecting* future cash flows and *discounting* them to the present. When a single flat rate r is used, both steps collapse into one. In practice, however — and universally so since 2008 — the rate used to *project* cash flows is not the same as the rate used to *discount* them. This gives rise to two separate curves, each with a distinct role.

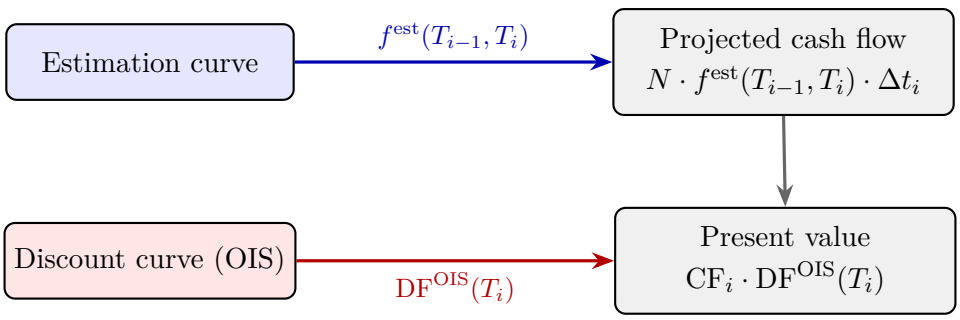
Definition 41: Estimation curve

The **estimation curve** (also called the **forwarding curve** or **projection curve**) is the function $f^{\text{est}} : (T_1, T_2) \mapsto f^{\text{est}}(T_1, T_2)$ whose forward rates serve as estimates for the unknown future rate fixings that determine floating-rate payments. It is constructed from market instruments referencing the same benchmark rate as the cash flows being projected (e.g. EURIBOR swaps for EURIBOR-linked payments, SOFR swaps for SOFR-linked payments).

Definition 42: Discount curve

The **discount curve** is the function $\text{DF}^{\text{disc}} : T \mapsto \text{DF}^{\text{disc}}(T)$ that provides the discount factors used to convert projected future cash flows into present values. Since the 2008 financial crisis, market practice uses the OIS curve (Definition 40) as the standard discount curve, as it represents the closest available approximation to a risk-free rate.

The two curves answer different questions: the estimation curve asks *how much* will be paid at each future date; the discount curve asks *what that payment is worth today*. The diagram below illustrates this separation for a single floating-rate cash flow at time T_i.



7.5.1 The dual-curve framework

Before 2008, the interbank rate (LIBOR or EURIBOR) was considered virtually risk-free, and a single curve served both for projection and discounting. The crisis revealed that interbank rates embed significant credit and liquidity premia — the OIS–LIBOR spread widened from 10–20 bps to over 350 bps at the height of the crisis, meaning the two roles could no longer be performed by the same curve without introducing mispricing.
The industry's response was the **dual-curve framework**: use the benchmark's forward curve (e.g. EURIBOR) for projection, and the OIS curve for discounting.

Dual-curve valuation

The present value of a stream of floating-rate cash flows under the dual-curve framework is:

$$\mathrm{PV} = \sum_{i=1}^{n} N \cdot f^{\mathrm{est}}(T_{i-1}, T_i) \cdot \Delta t_i \cdot \mathrm{DF}^{\mathrm{OIS}}(T_i),$$

where $f^{\mathrm{est}}(T_{i-1}, T_i)$ is the forward rate from the estimation curve and $\mathrm{DF}^{\mathrm{OIS}}(T_i)$ is the discount factor from the OIS curve.

In the **single-curve** approach, the same curve provides both forward rates and discount factors — equivalent to assuming the benchmark rate is risk-free. The dual-curve approach abandons this assumption.

Example 38: Single-curve vs. dual-curve valuation

Consider a 2y floating-rate note with annual payments and notional $N = 100{,}000\,€$. (A fixed-for-floating swap works the same way; we study swaps in detail in Chapter 12.) The market provides the following data, reflecting a situation where EURIBOR trades above OIS due to a credit premium:

Maturity	EURIBOR forward	OIS discount factor	EURIBOR discount factor
1y	3.80%	0.9650	0.9634
2y	4.10%	0.9300	0.9255

Compute the present value of the floating leg under (a) the dual-curve framework and (b) the single-curve approach. Compare the results.

Solution.
Part (a): Dual-curve valuation.
Project each floating payment using EURIBOR forwards, then discount with OIS discount factors:

$$\mathrm{PV}_1 = 100{,}000 \cdot 0.0380 \cdot 0.9650 = 3{,}667\,€,$$

$$\mathrm{PV}_2 = 100{,}000 \cdot 0.0410 \cdot 0.9300 = 3{,}813\,€,$$

$$\mathrm{PV}^{\mathrm{dual}}_{\mathrm{float}} = 3{,}667 + 3{,}813 = 7{,}480\,€.$$

Part (b): Single-curve valuation.
Use EURIBOR forwards for both projection and discounting:

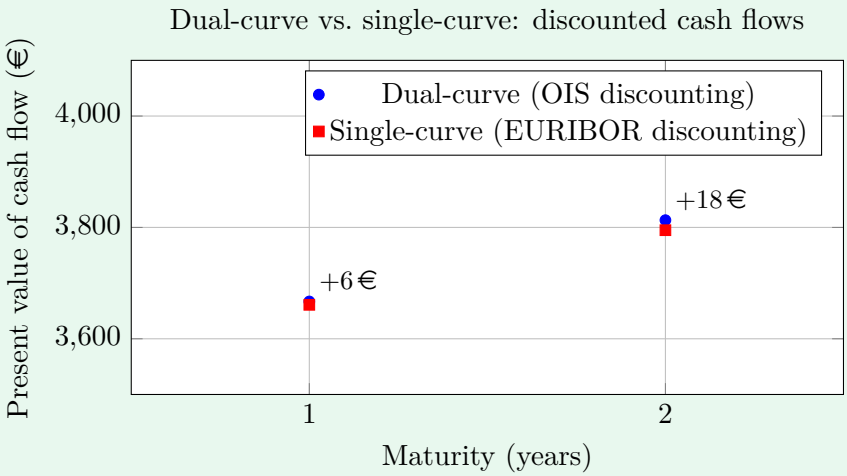
$$\mathrm{PV}_1 = 100{,}000 \cdot 0.0380 \cdot 0.9634 = 3{,}661\,€,$$

$$\mathrm{PV}_2 = 100{,}000 \cdot 0.0410 \cdot 0.9255 = 3{,}795\,€,$$

$$\mathrm{PV}_{\text{float}}^{\text{single}} = 3{,}661 + 3{,}795 = 7{,}456\,€.$$

Comparison.

	Dual-curve	**Single-curve**
PV at 1y	3,667 €	3,661 €
PV at 2y	3,813 €	3,795 €
Total PV	**7,480 €**	**7,456 €**



The dual-curve approach gives a higher present value (7,480 € vs. 7,456 €, a difference of 24 € on a 100,000 € notional). The discrepancy arises because OIS discount factors are *larger* than EURIBOR discount factors at each maturity — OIS rates are lower, so future cash flows are discounted less steeply — and the gap widens at longer maturities. On a 100 M € portfolio this difference scales to approximately 24,000 €, which is economically significant.

Remark (Multi-curve in practice). The term "multi-curve" reflects the fact that modern markets may require more than two curves. A bank pricing a derivatives portfolio might simultaneously use an ESTR-based OIS curve for discounting, a 3m EURIBOR curve for projecting 3m fixings, and a 6m EURIBOR curve for projecting 6m fixings — these two differ because of the **tenor basis**, the premium banks demand for longer unsecured lending. Add separate curves for each currency in cross-currency transactions, and a large institution may maintain dozens of curves. The dual-curve framework described here is the simplest and most important instance of this broader paradigm.

7.6 The Treasury curve

Government bonds serve as benchmarks for pricing other fixed income securities. The Treasury curve represents the yields on these sovereign obligations.

Definition 43: Treasury securities

Treasury securities (or **government bonds**, **sovereign bonds**) are debt obligations issued by a national government. They are considered the safest debt instruments in their respective currencies (assuming the government is creditworthy). Categories include:

- **Treasury bills (T-bills):** Zero-coupon instruments with maturities up to 1y.
- **Treasury notes (T-notes):** Coupon-bearing securities with maturities from 2y to 10y.
- **Treasury bonds (T-bonds):** Coupon-bearing securities with maturities beyond 10y (up to 30y).

Examples include US Treasuries, German Bunds, UK Gilts, and Japanese JGBs (Japanese Government Bonds).

Definition 44: Treasury curve

The **Treasury curve** (or **sovereign yield curve**) is the term structure of interest rates constructed from the yields to maturity of actively traded sovereign bonds across various maturities. Formally, it represents the function $y\colon T \mapsto y(T)$ where $y(T)$ is the yield to maturity of a sovereign zero-coupon bond (or its equivalent) with maturity T.

The Treasury curve serves multiple roles:

- **Risk-free benchmark:** It approximates the risk-free rate term structure for a given currency, as sovereign debt of major economies (US, Germany, UK) carries minimal default risk.
- **Pricing reference:** Corporate bonds are priced as spreads over the Treasury curve: $y_{\text{corporate}}(T) = y_{\text{Treasury}}(T) + s(T)$, where $s(T)$ is the credit spread (Definition 28).
- **Economic indicator:** The curve's shape provides information about market expectations; for instance, an inversion historically precedes recessions.

Example 39: Constructing a Treasury curve

The following US Treasury securities are observed (all with face value 10 000 \$). Construct the Treasury yield curve and interpret its shape.

Security	Maturity	Coupon (annual)	Dirty price	YTM
T-bill	6m	0%	9 780 \$	4.49%
T-note	2y	3.5%	9 920 \$	3.85%
T-note	5y	3.0%	9 650 \$	3.65%
T-note	10y	3.25%	9 730 \$	3.55%
T-bond	30y	3.5%	9 880 \$	3.58%

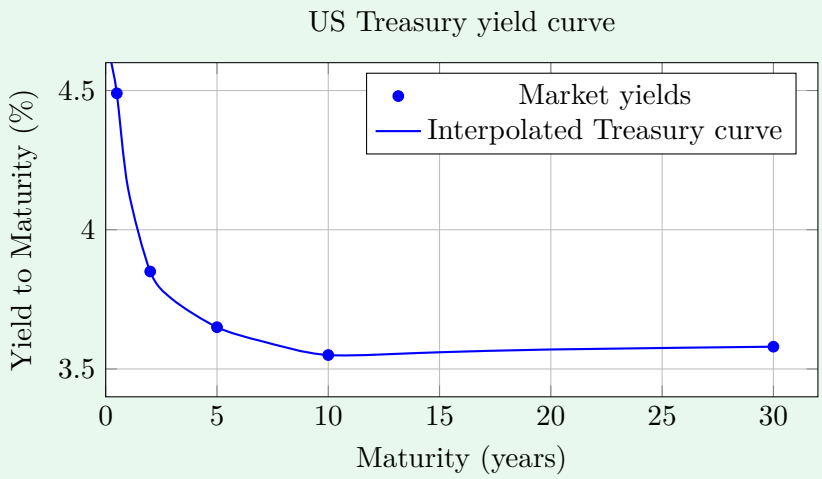
Solution. Step 1 — Spot check for 2y note:
For the 2y note: $N = 10\,000\,\$$, $c = 3.5\%$, $B = 9\,920\,\$$. Cash flows: $350\,\$$ at $t = 1\text{y}$ and $10\,350\,\$$ at $t = 2\text{y}$.

$$9\,920 = \frac{350}{1+y} + \frac{10\,350}{(1+y)^2}.$$

Trying $y = 3.85\%$: $350/1.0385 + 10\,350/1.0385^2 = 337.02 + 9\,596.95 = 9\,933.97 \approx 9\,920$. ✓

(The small discrepancy is due to rounding; actual YTM is close to 3.85%.)

Step 2 — Plot the yield curve:



Interpretation:
The curve shows an **inverted short end**: the 6m T-bill yields 4.49%, significantly higher than the 2y note (3.85%). This inversion typically signals that the Federal Reserve has raised short-term rates aggressively to combat inflation, the market expects the Fed to cut rates in the medium term as the economy slows, and there is an elevated recession risk.
Beyond 2y, the curve flattens considerably, with the 10y yield (3.55%) only slightly below the 2y (3.85%), and the 30y (3.58%) barely higher than the 10y. This flat long end suggests anchored long-term inflation expectations.

7.7 The inflation curve

Inflation erodes the purchasing power of future cash flows. The inflation curve quantifies the market's expectations (or more precisely, compensation) for this erosion.

Definition 45: Inflation

Inflation is the rate of increase in the general price level of goods and services in an economy over a period of time.

Definition 46: Breakeven inflation rate

The **breakeven inflation rate** for maturity T is the difference between the **nominal yield** on a standard government bond and the **real yield** on an inflation-linked government bond of the same maturity:

$$\mathrm{BE}(T) = y_{\text{nominal}}(T) - y_{\text{real}}(T).$$

Nominal bonds pay fixed cash flows in currency units, while inflation-linked bonds adjust their payments based on a price index. The breakeven rate represents the average annual inflation rate over $[0, T]$ at which an investor would be indifferent between the two.

Definition 47: Inflation curve

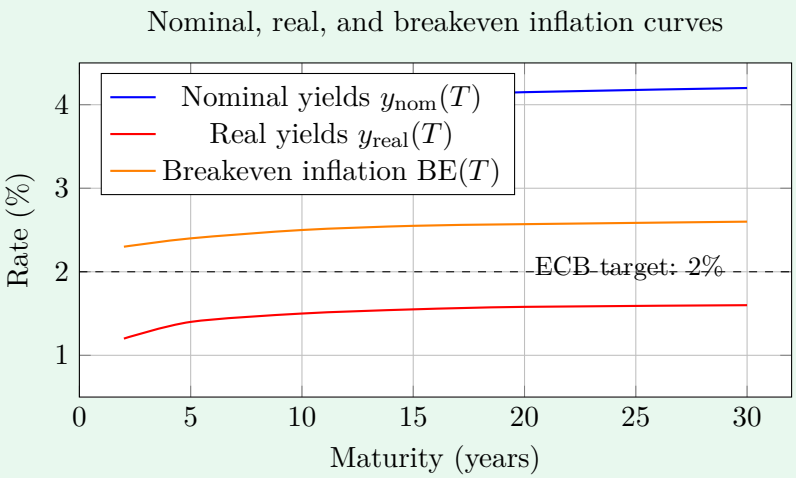
The **inflation curve** is the function $T \mapsto \mathrm{BE}(T)$ that represents the collection of breakeven inflation rates across all maturities T. It represents the market's implied inflation compensation over different time horizons.

Example 40: Constructing the inflation curve from breakeven rates

The following market data are observed for a European government. Construct the inflation curve and interpret it in the context of the ECB's 2% inflation target.

Maturity	Nominal yield	Real yield	Breakeven inflation
2y	3.50%	1.20%	2.30%
5y	3.80%	1.40%	2.40%
10y	4.00%	1.50%	2.50%
30y	4.20%	1.60%	2.60%

Solution. The breakeven inflation rates are computed as $\mathrm{BE}(T) = y_{\mathrm{nom}}(T) - y_{\mathrm{real}}(T)$ and shown in the fourth column.



Interpretation: The breakeven inflation curve is upward-sloping (from 2.30% at 2y to 2.60% at 30y) and lies entirely above the ECB's 2% target, suggesting the market expects inflation to persistently exceed the target or demands a positive inflation risk premium. The widening gap between nominal and real yields at longer maturities reflects greater inflation uncertainty over extended horizons.

Remark (Breakeven inflation $\neq$ expected inflation). The breakeven inflation rate should *not* be interpreted as the market's pure expectation of future inflation. It also includes an **inflation risk premium** (compensation for uncertainty, typically 20–50 bps), a **liquidity premium** (inflation-linked bonds are less liquid, lowering breakeven rates), and technical factors like convexity and indexation lags. The true expected inflation is approximately:

$$\mathbb{E}[\pi] \approx \mathrm{BE} - \text{inflation risk premium} + \text{liquidity premium}.$$

> Separating these components requires econometric models or surveys of professional forecasters.

7.8 The par curve

The par yield represents the fair coupon rate for a newly issued bond. It is independent of the face value N. The par curve addresses a practical question: if a government or corporation wanted to issue a new bond at par for a given maturity, what coupon rate should it carry?

Definition 48: Par curve

The **par curve** (or **par yield curve**) is the function $T \mapsto c_{\text{par}}(T)$ that maps each maturity T to the coupon rate $c_{\text{par}}(T)$ at which a bond with that maturity would trade at par (i.e., at face value N).

7.8.1 Constructing the par curve

The par curve is constructed from the discount function (spot curve) using the par yield formula (Definition 25):

1. **Obtain discount factors:** Bootstrap the spot curve from observed market bond prices (Section 7.2).
2. **Apply par yield formula:** For each maturity T with annual payments:

$$c_{\text{par}}(T) = \frac{1 - \text{DF}(T)}{\sum_{i=1}^{T} \text{DF}(i)}.$$

3. **Plot the curve:** Create the par curve by plotting maturity T against $c_{\text{par}}(T)$.

Example 41: Constructing the par curve from discount factors

Using the discount factors from Example 34: $\text{DF}(1) = 0.985$, $\text{DF}(2) = 0.956$, $\text{DF}(3) = 0.921$. Compute the par yields for 1y, 2y, and 3y maturities, and compare with the spot rates.

Solution. **1y par yield:**

$$c_{\text{par}}(1) = \frac{1 - 0.985}{0.985} = \frac{0.015}{0.985} = 1.502\%.$$

This equals the 1y spot rate $r(1) = 1.502\%$ — which is always true for the first maturity.

2y par yield:

$$c_{\text{par}}(2) = \frac{1 - 0.956}{0.985 + 0.956} = \frac{0.044}{1.941} = 2.263\%.$$

Compare with the 2y spot rate $r(2) = 2.274\%$. The par yield is slightly lower.

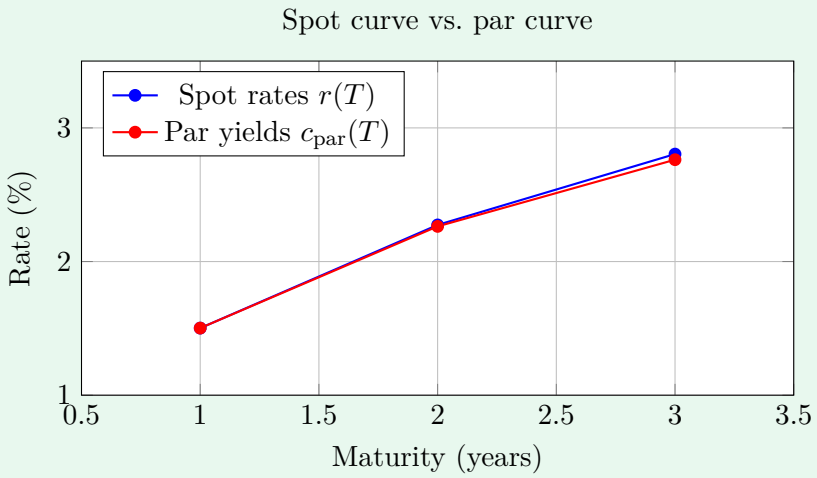
3y par yield:

$$c_{\text{par}}(3) = \frac{1 - 0.921}{0.985 + 0.956 + 0.921} = \frac{0.079}{2.862} = 2.762\%.$$

Compare with the 3y spot rate $r(3) = 2.804\%$. Again, the par yield is slightly lower.

Summary:

Maturity	Par yield	Spot rate
1y	1.502%	1.502%
2y	2.263%	2.274%
3y	2.762%	2.804%



Interpretation: The par curve lies *below* the spot curve for all maturities beyond 1y when the term structure is upward-sloping. This occurs because the par yield represents a weighted average of rates over the bond's life, with earlier payments (at lower rates) weighted more heavily due to higher discount factors. For a downward-sloping term structure, par yields would lie *above* spot rates.

7.9 Relationship between curves

Understanding how different rate curves relate to each other is essential for fixed income analysis and arbitrage detection.

7.9.1 Curve hierarchy for upward-sloping term structures

When the term structure is upward-sloping (normal conditions), the curves typically satisfy:

Hierarchy of curves

For upward-sloping term structures:

$$\text{Forward rates} > \text{Spot rates} > \text{Par yields} \approx \text{Yield curve}.$$

For downward-sloping or inverted term structures, this hierarchy reverses.

Intuition:

- **Forward rates** are highest because they isolate future periods, and in an upward-sloping environment, future rates are expected to be higher than current rates.
- **Spot rates** represent the average return over $[0, T]$, including both low early rates and high later rates, so they lie between forwards and par yields.
- **Par yields** are lowest because they weight early (lower-rate) cash flows more heavily due to higher discount factors.

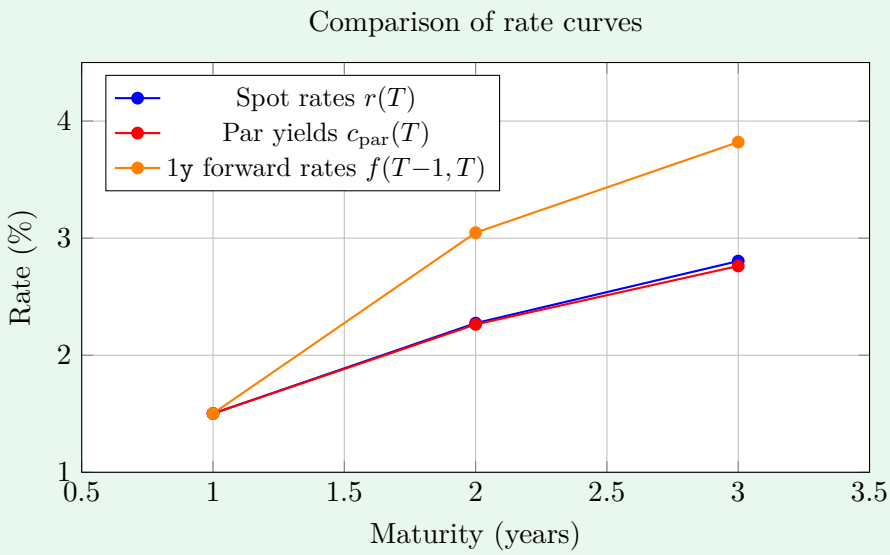
- **Yield curve** (YTM of coupon bonds) lies close to the par curve, typically slightly above due to the averaging effect of coupons at different rates.

Example 42: Comparing all curves

Using our bootstrapped data with spot rates $r(1) = 1.502\%$, $r(2) = 2.274\%$, $r(3) = 2.804\%$, we have computed the par yields (Example 41) and forward rates (Example 35):

Maturity	Spot rate	Par yield	1y forward rate
1y	1.502%	1.502%	1.502%
2y	2.274%	2.263%	3.047%
3y	2.804%	2.762%	3.820%

Solution.



Observations:

1. The hierarchy Forward > Spot > Par holds, consistent with an upward-sloping term structure.
2. The gaps between curves widen as maturity increases: the 3y forward rate (3.820%) is +1.016 percentage points above the spot rate, while the 2y forward is only +0.773 above.
3. At 1y, all curves coincide — as they must, since there is only one period and no averaging.

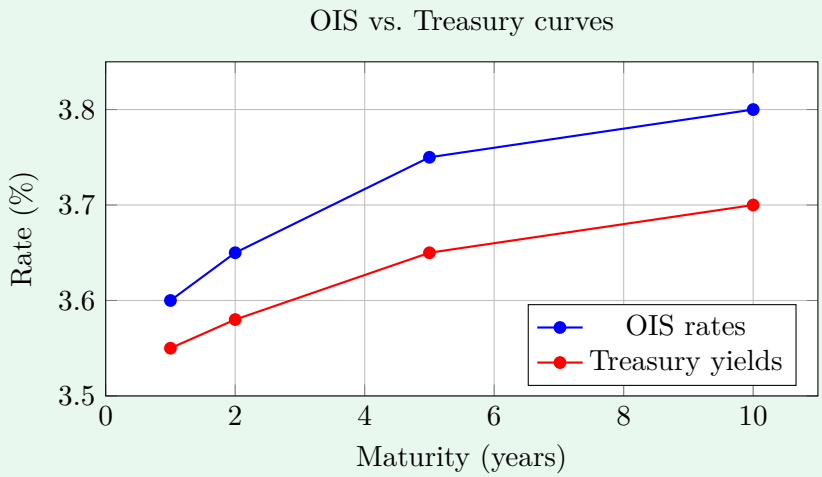
7.9.2 OIS vs. Treasury curves

Example 43: Comparing OIS and Treasury curves

Consider the following market data. Analyse the relationship between the OIS and Treasury curves.

Maturity	OIS rate	Treasury yield
1y	3.60%	3.55%
2y	3.65%	3.58%
5y	3.75%	3.65%
10y	3.80%	3.70%

Solution.



Key observations:
The OIS curve lies *above* the Treasury curve at all maturities, with the spread ranging from 5 bps at 1y to 10 bps at 10y. This pattern seems counterintuitive — OIS is supposed to be "risk-free", so why is it higher than government bonds? Treasuries benefit from a **convenience yield** (they serve as high-quality collateral and satisfy regulatory requirements, creating additional demand that lowers yields), **flight-to-quality** flows (in times of stress, investors flee to Treasuries, pushing prices up and yields down), and **repo specialness** (Treasury securities can be lent in the repo market at favourable rates).
The 5–10 bps spread represents the combined value of these Treasury-specific factors. For derivative valuation, OIS remains the correct discount curve despite trading above Treasuries.

7.9.3 Summary: curve relationships

Curve	Purpose and characteristics
Spot curve	Pure zero-coupon rates; fundamental pricing curve; bootstrapped from bond prices
Forward curve	Future period rates implied by spot curve; highest for upward-sloping term structures
Par curve	Coupon rates for new bonds to trade at par; derived from spot curve; lowest for upward-sloping structures
Yield curve	YTM of coupon bonds; observable directly; close to par curve
OIS curve	Risk-free discount curve for derivatives; based on overnight rates; industry standard post-2008
Estimation curve	Projects unknown future rate fixings; built from instruments referencing the relevant index (e.g. EURIBOR swaps)
Discount curve	Provides discount factors for present value calculations; typically the OIS curve for collateralised transactions
Treasury curve	Government bond benchmark; includes convenience yield; may trade below OIS
Inflation curve	Breakeven inflation rates; difference between nominal and real yields

Remark (Which curve to use?). The choice of curve depends on the application:

- **Projecting floating cash flows:** Use the estimation curve (Definition 41) built from instruments referencing the same index as the cash flows being projected (e.g. EURIBOR swaps for EURIBOR-linked payments).
- **Discounting derivative cash flows:** Use the OIS curve (Definition 40) as the discount curve (Definition 42), as it represents the risk-free rate for collateralised transactions.
- **Government bond analysis:** Use the Treasury (sovereign) curve as the baseline risk-free benchmark for a given currency.
- **Corporate bond pricing:** Price relative to the Treasury curve by adding a credit spread (Definition 28).
- **New bond issuance:** Use the par curve to determine the coupon rate that will price a bond at face value.
- **Inflation analysis:** Use the inflation curve (breakeven rates) to assess market-implied inflation expectations or price inflation-linked bonds.
- **Forward rate analysis:** Use the forward curve to understand market expectations of future interest rates or to price forward-starting contracts (Chapter 11).
- **Arbitrage detection:** Compare relationships between spot, forward, and par curves to identify pricing inconsistencies.

CHAPTER 8

Annuities, Deposits, and Amortised Loans

In Chapters 3 and 6 we studied bonds, where the principal is repaid in a single lump sum at maturity. In practice, however, most real-world loans use a different repayment structure: the borrower makes periodic payments that gradually retire both interest and principal. The mathematical object underlying this structure is the annuity — a stream of payments at regular intervals.

Annuities appear everywhere in finance: mortgage payments, pension payouts, insurance products, lease agreements, and the coupon streams of bonds. This chapter develops the annuity formula, applies it to the simplest fixed income instrument — the term deposit — and then to the most common loan structure — the amortised loan. We close with floating annuities, which connect the annuity framework to the floating-rate mechanics of Chapter 6.

8.1 Fixed annuity

Definition 49: Annuity (ordinary, fixed)

An **ordinary fixed annuity** (or simply **annuity**) is a finite series of m equal payments A, each made at the *end* of successive periods. If the discount rate per period is r_p, the present value of the annuity is:

$$\mathrm{PV} = A \cdot \frac{1-(1+r_p)^{-m}}{r_p}.$$

The factor

$$a_{m\rceil r_p} = \frac{1-(1+r_p)^{-m}}{r_p}$$

is called the **annuity factor** (or **present value interest factor of an annuity**).

8.1.1 Derivation of the annuity formula

The annuity formula is a direct application of the present value concept from Chapter 2. Each payment A occurs at a different time, so we discount each one individually and sum them.

Consider m payments of A at times $1, 2, \ldots, m$ (measured in periods). Each payment is

discounted at rate r_p per period:

$$\mathrm{PV} = \frac{A}{(1+r_p)^1} + \frac{A}{(1+r_p)^2} + \cdots + \frac{A}{(1+r_p)^m} = A \sum_{k=1}^{m} \frac{1}{(1+r_p)^k}.$$

The sum is a finite geometric series with first term $1/(1+r_p)$ and common ratio $q = 1/(1+r_p)$, therefore

$$\sum_{k=1}^{m} \frac{1}{(1+r_p)^k} = \frac{1}{1+r_p} \cdot \frac{1-(1+r_p)^{-m}}{1-(1+r_p)^{-1}} = \frac{1-(1+r_p)^{-m}}{r_p}.$$

Hence $\mathrm{PV} = A \cdot a_{m\rceil r_p}$, as claimed.

Remark (Boundary behaviour of the annuity factor). The annuity factor has two instructive limits:

- As $m \to \infty$ (**perpetuity**): $a_{\infty\rceil r_p} = 1/r_p$. This is the present value of an infinite stream of payments, discussed in Section 8.1.2 below.
- As $r_p \to 0$: $a_{m\rceil r_p} \to m$ (each payment has present value A, so the total is $m \cdot A$). This is consistent with the fact that, at a zero discount rate, there is no time value of money.

Annuity payment for a loan

If a loan of N is to be repaid in m equal periodic instalments at rate r_p per period, the payment is obtained by inverting $N = A \cdot a_{m\rceil r_p}$:

$$A = \frac{N}{a_{m\rceil r_p}} = N \cdot \frac{r_p}{1-(1+r_p)^{-m}}.$$

This is the formula used to compute the constant payment A in an amortised loan, as we will see in Section 8.3.

Example 44: 10-year fixed annuity

A homeowner takes a mortgage of $N = 200{,}000$ € at an annual rate of $r = 5\%$, repaid in monthly instalments over $T = 10$y. Compute the monthly payment, the total amount paid, and the total interest.

Solution. The monthly rate is $r_p = 0.05/12 = 0.4167\%$ and the total number of payments is $m = 12 \cdot 10 = 120$.

Step 1 — Annuity factor:

$$a_{120\rceil 0.004167} = \frac{1-(1.004167)^{-120}}{0.004167} = \frac{1-0.607}{0.004167} = \frac{0.393}{0.004167} = 94.28.$$

Step 2 — Monthly payment:

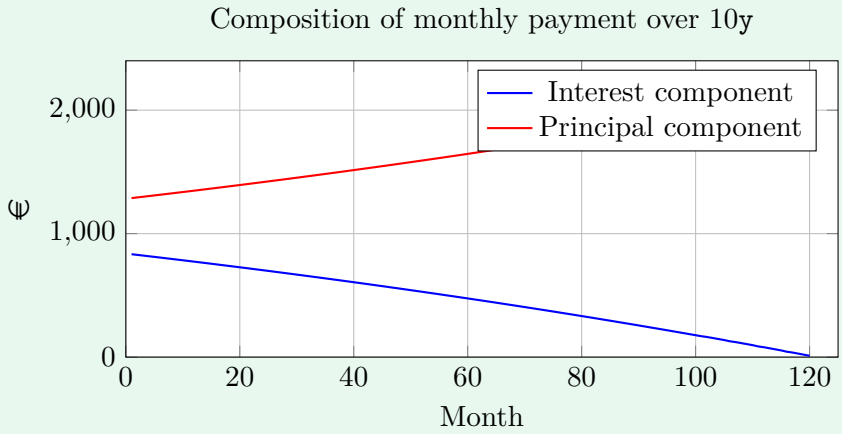
$$A = \frac{200{,}000}{94.28} = 2{,}122\,€/\text{month}.$$

Step 3 — Total paid and total interest:

Total paid over 10y: $2{,}122 \cdot 120 = 254{,}640$ €. Total interest: $254{,}640 - 200{,}000 = 54{,}640$ €.

Step 4 — Composition over time:
Over the life of the loan, the monthly payment stays constant at 2,122 €, but the split between interest and principal shifts dramatically:

- Month 1: interest $= 200{,}000 \cdot 0.004167 = 833$ €, principal $= 2{,}122 - 833 = 1{,}289$ €.
- Month 60 (halfway): outstanding balance $\approx 110{,}260$ €, interest ≈ 460 €, principal $\approx 1{,}662$ €.
- Month 120: interest ≈ 9 €, principal $\approx 2{,}113$ €.



Early in the loan, most of each payment covers interest; by the end, almost all of it retires principal. We will see this pattern again in the amortisation schedule in Example 48.

8.1.2 Perpetuity

Definition 50: Perpetuity

A **perpetuity** is an annuity with infinitely many payments ($m \to \infty$). Its present value is:

$$\mathrm{PV}_{\text{perp.}} = \frac{A}{r_p}.$$

This follows directly from the annuity formula (Definition 49) by taking $m \to \infty$, so that $(1 + r_p)^{-m} \to 0$.

Example 45: Pricing a perpetuity

A university endowment fund promises to pay 50,000 € annually forever to support a research chair. If the discount rate is $r = 4\%$ per year, what is the present value of this perpetuity?

Solution.

$$\mathrm{PV} = \frac{50{,}000}{0.04} = 1{,}250{,}000\,€.$$

The university should set aside 1,250,000 € today to fund the perpetual annual payment of 50,000 €. Note that even though infinitely many payments are received, the present value is finite because distant payments are heavily discounted.

8.1.3 Relationship between annuities and bond pricing

The annuity formula provides a compact expression for bond prices. Recall from Definition 21 that a bond with face value N, coupon rate c, maturity T, and yield y has price:

$$B = \sum_{i=1}^{T} \frac{c\,N}{(1+y)^i} + \frac{N}{(1+y)^T} = c\,N \cdot a_{T\rceil y} + N \cdot (1+y)^{-T}.$$

The first term is the present value of the coupon stream (an annuity with payment $c\,N$), and the second is the present value of the principal repayment (a single future cash flow). This decomposition makes it transparent why a bond's price depends on the coupon rate, the yield, and the maturity in the way it does.

Example 46: Bond price via the annuity formula

Reprice the 3y bond from Example 14 ($N = 100{,}000$ €, $c = 4\%$, $y = 3\%$) using the annuity formula.

Solution. The annuity factor for 3 periods at 3% is:

$$a_{3\rceil 0.03} = \frac{1 - 1.03^{-3}}{0.03} = \frac{1 - 0.915}{0.03} = \frac{0.085}{0.03} = 2.829.$$

Bond price:

$$B = 4{,}000 \cdot 2.829 + 100{,}000 \cdot 1.03^{-3} = 11{,}316 + 91{,}510 = 102{,}826\,€,$$

confirming the result from Chapter 3 (the small difference from 102,801 € is due to rounding in the annuity factor).

8.2 Deposits

The term deposit is the simplest fixed income instrument and the natural starting point before studying more complex repayment structures.

Definition 51: Deposit (term deposit)

A **term deposit** is the simplest fixed income instrument: you place N euros with a bank for a fixed period at a fixed rate r. At maturity, you receive the principal back plus interest:

$$\text{Terminal amount} = N + N \cdot r \cdot t = N(1 + r \cdot t),$$

using simple interest, which is the market convention for maturities $\leq$ 1y.

A deposit is essentially a zero-coupon bond (Definition 30) with a single cash flow at maturity. The difference is institutional: bonds are tradeable securities; deposits are bilateral contracts with a bank.

Example 47: Term deposit

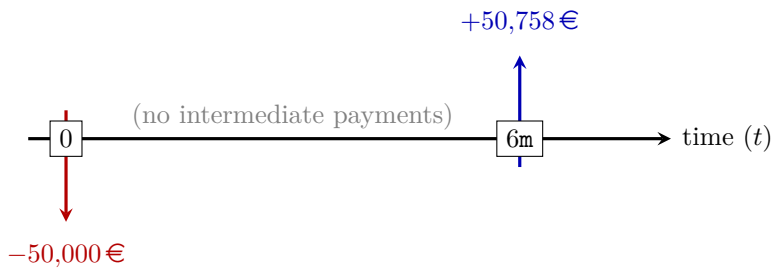
You deposit $N = 50{,}000$ € for 6m at $r = 3\%$ p.a. under the Act/360 day-count convention, with 182 actual days in the period. Compute the interest payment and the total amount received at maturity.

Solution. The year fraction under Act/360 is $t = 182/360$. The interest payment is:

$$I = 50{,}000 \cdot 0.03 \cdot \frac{182}{360} = 758\,€.$$

At maturity you receive:

$$\text{Terminal amount} = 50{,}000 + 758 = 50{,}758\,€.$$



8.3 Amortised loans

Most real-world loans (mortgages, car loans, student loans) are not like bonds — you do not pay just interest and then return the full principal at the end. Instead, you gradually repay both interest and principal over time.

Definition 52: Amortised loan

An **amortised loan** is repaid through a series of periodic payments, each containing:

1. an **interest component**: interest on the outstanding balance, and
2. an **amortisation component** (or **principal repayment**): a partial repayment of the principal.

In a standard **constant-payment amortisation**, the total payment A is *constant* throughout the life of the loan. Over time, the interest portion decreases (because the outstanding balance shrinks) and the amortisation portion increases.

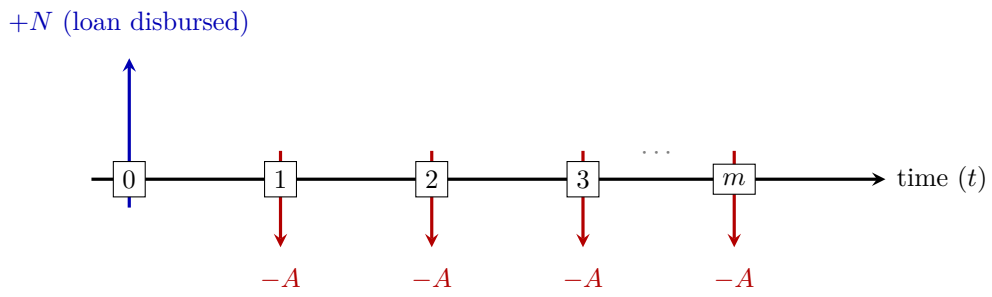
Amortised loan formula

The constant payment A is computed using the annuity formula (Definition 49):

$$A = N \cdot \frac{r_p}{1 - (1 + r_p)^{-m}},$$

where $r_p = r/n$ is the **periodic rate** (r is the annual rate, n is the payment frequency per year), and $m = n \cdot T$ is the total number of payments over the loan's maturity T. For each period i with outstanding balance N_i:

$$\begin{aligned} \text{Interest}_i &= N_i \cdot r_p, \\ \text{Amortisation}_i &= A - \text{Interest}_i, \\ N_{i+1} &= N_i - \text{Amortisation}_i. \end{aligned}$$



Cash flows of an amortised loan from the borrower's perspective: receive N at inception, then make m equal payments of A, each containing both interest and principal.

Example 48: Amortisation schedule

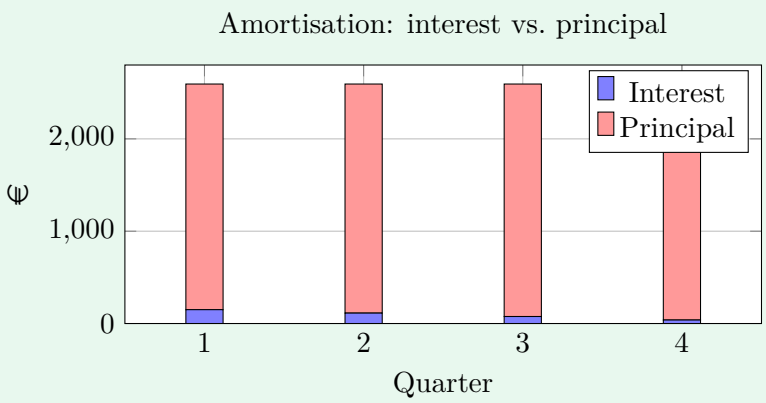
A loan of $N = 10{,}000$ € at an annual rate of $r = 6\%$ with quarterly payments ($n = 4$) and maturity $T = 1\text{y}$. Compute the quarterly payment A and construct the full amortisation schedule.

Solution. The quarterly rate is $r_p = 6\%/4 = 1.5\%$ and the total number of payments is $m = 4$.

$$A = 10{,}000 \cdot \frac{0.015}{1-(1.015)^{-4}} = 10{,}000 \cdot \frac{0.015}{0.0572} = 2{,}594\,€.$$

Quarter	Balance (€)	Interest (€)	Amortisation (€)	New balance (€)
1	10,000	150	2,444	7,556
2	7,556	113	2,481	5,075
3	5,075	76	2,518	2,557
4	2,557	38	2,556	≈ 0
Total		**377**	**10,000**	

Notice that the total payment is always 2,594 €, but the split between interest and amortisation changes. In the first quarter, 150 € goes to interest; by the last quarter, only 38 € does. Over time, as the outstanding balance decreases, the interest component shrinks and the principal repayment component grows — a characteristic feature of amortised loans.



Remark (Amortised loan vs. standard bond). It is instructive to compare the two repayment structures side by side:

Feature	Standard bond	Amortised loan
Principal repayment	Lump sum at maturity	Gradual, each period
Periodic payment	Coupon only	Interest + principal
Outstanding balance	Constant ($= N$)	Decreasing
Total interest paid	Higher	Lower
Interest rate risk	Higher	Lower

A standard bond (also called a **bullet bond**), as in Definition 20, carries higher **interest rate risk** because the full principal remains outstanding until maturity, giving it a longer **duration** (Chapter 9). An amortised loan has lower duration because the principal is repaid gradually, reducing the weighted average time of cash flows.

Not all amortised loans use constant total payments. Alternative structures include:

- **Constant amortisation (linear amortisation):** Each period, the same amount of principal is repaid, so the total payment decreases over time (since the interest component shrinks on a constant amortisation amount).
- **Bullet repayment with amortisation:** A portion of principal is repaid periodically, with the remaining balance due at maturity.
- **Interest-only period:** Some loans have an initial period where only interest is paid, followed by an amortising period. This is common in certain mortgage structures.

The constant-payment structure (Definition 52) is by far the most common for consumer loans and mortgages.

8.4 Floating annuity

Just as floating-rate bonds (Chapter 6) adjust their coupons to track market rates, a floating annuity adjusts its periodic payment when the reference rate changes.

Definition 53: Floating annuity

A **floating annuity** is an amortised loan in which the interest rate resets periodically based on a market reference rate. Between reset dates, the payment is constant (computed using the current rate and outstanding balance). At each reset date, a new rate r'_p is observed, and the payment is recalculated for the remaining term.

Specifically, at a reset date with outstanding balance N', remaining m' periods, and new periodic rate r'_p, the new payment is:

$$A_{\text{new}} = N' \cdot \frac{r'_p}{1 - (1 + r'_p)^{-m'}}.$$

This is identical to the fixed annuity formula, applied to the current balance and rate. Between resets, the loan behaves exactly like a fixed annuity; at each reset, the parameters are updated.

Remark (Fixed vs. floating annuity: risk trade-off). The choice between a fixed and floating annuity reflects a trade-off:

- **Fixed annuity:** Payment is predictable for the entire term, but the borrower bears **opportunity cost** if rates fall (they are locked in at the higher rate).
- **Floating annuity:** Payments adjust to market conditions, but the borrower bears **payment risk** — if rates rise sharply, monthly payments can increase significantly.

This is analogous to the fixed-rate vs. floating-rate bond distinction from Chapter 6, but viewed from the borrower's perspective.

Example 49: Floating annuity with rate reset

A borrower takes a 2y loan of $N = 100{,}000$ € with semi-annual payments. The interest rate resets annually:

- Year 1 rate: $r = 4\%$ p.a. ($r_p = 2\%$ per half-year).
- Year 2 rate (determined at end of year 1): $r = 6\%$ p.a. ($r_p = 3\%$ per half-year).

Compute the payment schedule and total interest paid, and compare with a fixed annuity at 4% throughout.

Solution. Year 1 payments. At inception, there are $m = 4$ semi-annual periods remaining at $r_p = 2\%$:

$$A_1 = 100{,}000 \cdot \frac{0.02}{1 - 1.02^{-4}} = 100{,}000 \cdot \frac{0.02}{0.0776} = 26{,}263\,€.$$

The amortisation schedule for year 1:

Period	Opening (€)	Interest (€)	Principal (€)	Closing (€)
1	100,000	2,000	24,263	75,737
2	75,737	1,515	24,748	50,989

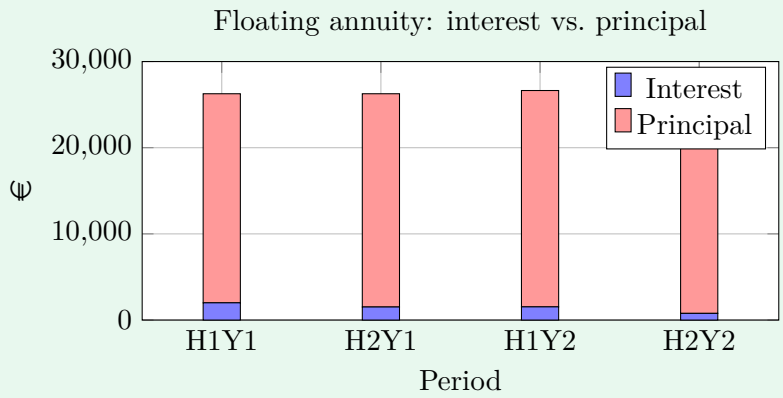
After two payments (end of year 1), the outstanding balance is $N' = 50{,}989$ €.

Year 2 payments. The rate resets to $r_p = 3\%$ with $m' = 2$ periods remaining:

$$A_2 = 50{,}989 \cdot \frac{0.03}{1 - 1.03^{-2}} = 50{,}989 \cdot \frac{0.03}{0.0574} = 26{,}638\,€.$$

Period	Opening (€)	Interest (€)	Principal (€)	Closing (€)
3	50,989	1,530	25,108	25,881
4	25,881	776	25,862	≈ 0

Summary. The payment increased from 26,263 € to 26,638 € when rates rose from 4% to 6%. Total interest paid: $2{,}000 + 1{,}515 + 1{,}530 + 776 = 5{,}821$ €. Compare this with a fixed annuity at 4% throughout, where total interest would be approximately 5,052 €. The rate increase cost the borrower an additional 769 € in interest.



Notice the jump in the interest component from period 2 to period 3, caused by the rate reset. The total bar height also changes (26,263 € to 26,638 €), unlike a fixed annuity where it remains constant.

CHAPTER 9

Duration and Convexity

In Chapter 3 we saw that a bond's price depends on the yield: when yields rise, prices fall, and vice versa. But *how much* does the price change for a given yield movement? This chapter introduces the two key measures of interest rate sensitivity: duration (a first-order measure) and convexity (a second-order correction).

9.1 Macaulay duration

Definition 54: Macaulay duration

The **Macaulay duration** D_{Mac} of a bond is the weighted average time to receipt of its cash flows, where the weight of each cash flow is its present value as a fraction of the bond's total price:

$$D_{\text{Mac}} = \frac{1}{B} \sum_{i=1}^{n} t_i \cdot \frac{c_i}{(1+y)^{t_i}},$$

where B is the bond price, c_i is the cash flow at time t_i, and y is the yield to maturity. Macaulay duration is measured in **years**.

Macaulay duration has a beautiful interpretation: it is the "centre of gravity" of the bond's cash flows in the time dimension, with each cash flow weighted by its economic importance (present value). A bond that pays most of its value early (high coupons, short maturity) has a low duration; a bond that concentrates value at the end (low coupons, long maturity) has a high duration.

Remark (Properties of Macaulay duration).

- For a zero-coupon bond (Definition 30), $D_{\text{Mac}} = T$ (the maturity). All value is concentrated at a single payment.
- For a coupon bond, $D_{\text{Mac}} < T$ (the duration is less than the maturity). Early coupon payments pull the "centre of gravity" toward the present.
- For a perpetuity (Definition 50), $D_{\text{Mac}} = (1+y)/y$. Even though the perpetuity pays forever, its duration is finite.
- Higher coupon $\Rightarrow$ lower duration (more value received early).
- Higher yield $\Rightarrow$ lower duration (distant cash flows are discounted more heavily, reducing their weight).

Example 50: Macaulay duration of a 3-year bond

Compute the Macaulay duration of a bond with $N = 100{,}000$ €, coupon $c = 4\%$ (annual), maturity $T = 3$y, and yield $y = 3\%$.

Solution. First, compute the present value of each cash flow:

t_i	**Cash flow** c_i	$\mathrm{DF}(t_i) = (1.03)^{-t_i}$	$\mathrm{PV}_i = c_i \cdot \mathrm{DF}(t_i)$	$t_i \cdot \mathrm{PV}_i$
1y	4,000 €	0.971	3,883 €	3,883
2y	4,000 €	0.943	3,770 €	7,540
3y	104,000 €	0.915	95,177 €	285,531
			$B = 102{,}830$ €	$\Sigma = 296{,}954$

$$D_{\mathrm{Mac}} = \frac{296{,}954}{102{,}830} = 2.888\text{y}.$$

The Macaulay duration is 2.888 years, slightly less than the 3y maturity. The coupons at 1y and 2y pull the duration below the maturity.
For comparison, a 3y zero-coupon bond at the same yield would have $D_{\mathrm{Mac}} = 3.000$y (exactly equal to maturity).

9.2 Modified duration

While Macaulay duration measures the time profile of cash flows, practitioners need a direct measure of *price sensitivity*. This leads to modified duration.

Definition 55: Modified duration

The **modified duration** D_{mod} measures the percentage price change per unit change in yield:

$$D_{\mathrm{mod}} = \frac{D_{\mathrm{Mac}}}{1 + y/m},$$

where m is the compounding frequency ($m = 1$ for annual, $m = 2$ for semi-annual). For continuous compounding, $D_{\mathrm{mod}} = D_{\mathrm{Mac}}$. Duration is measured in **years**.

Remark (Derivation: modified duration as a derivative). Assume annual compounding ($m = 1$). Modified duration arises from a Taylor expansion of the bond price function $B(y)$ in Definition 21. Taking the first derivative:

$$\frac{dB}{dy} = -\frac{1}{1+y} \sum_{i=1}^{n} t_i \cdot \frac{c_i}{(1+y)^{t_i}} = -\frac{D_{\mathrm{Mac}}}{1+y} \cdot B.$$

Dividing both sides by B:

$$\frac{1}{B}\frac{dB}{dy} = -\frac{D_{\mathrm{Mac}}}{1+y} = -D_{\mathrm{mod}}.$$

Hence the first-order approximation for the percentage price change is:

$$\frac{\Delta B}{B} \approx -D_{\mathrm{mod}} \cdot \Delta y.$$

Duration approximation

For a small parallel shift Δy in the yield:

$$\Delta B \approx -D_{\text{mod}} \cdot B \cdot \Delta y.$$

The negative sign reflects the inverse relationship between bond prices and yields.

Example 51: Using modified duration

The bond from Example 50 has $D_{\text{Mac}} = 2.888\text{y}$, $y_0 = 3\%$, and $B_0 = 102{,}830\,€$. Estimate the new price after a $+50$ bps yield rise and compare with the exact value.

Solution. Modified duration:

$$D_{\text{mod}} = \frac{2.888}{1.03} = 2.804.$$

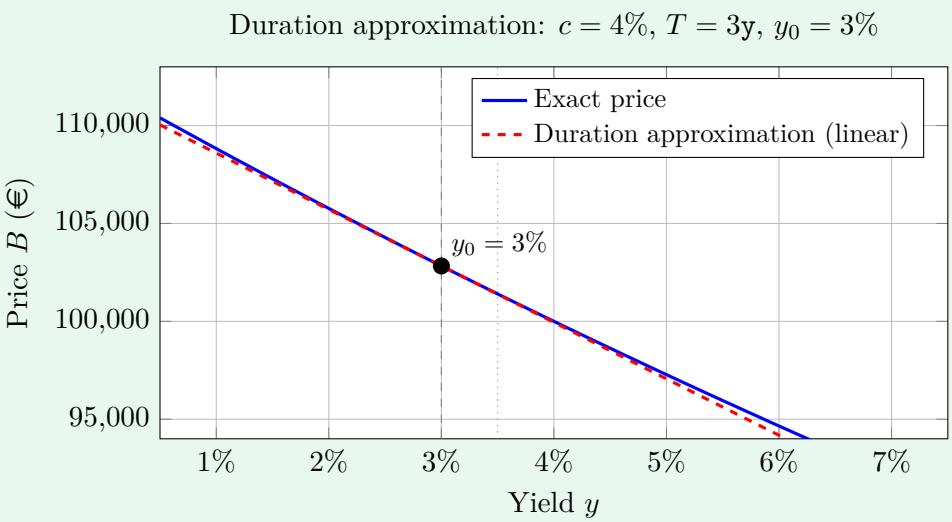
Estimated price change and new price:

$$\Delta B \approx -2.804 \cdot 102{,}830 \cdot 0.005 = -1{,}442\,€, \qquad B_{\text{new}} \approx 102{,}830 - 1{,}442 = 101{,}388\,€.$$

Exact value:

$$B_{\text{exact}} = \frac{4{,}000}{1.035} + \frac{4{,}000}{1.035^2} + \frac{104{,}000}{1.035^3} = 101{,}401\,€.$$

The duration estimate is accurate to within 11 € for this small move. The residual error arises from the curvature of the price–yield curve, which the linear approximation ignores.



The red line is tangent to the blue curve at $y_0 = 3\%$. For a small move of $+50$ bps the two are nearly indistinguishable; the gap widens for larger moves.

9.3 Convexity

Duration provides a linear approximation to the price–yield relationship. But bond prices are *convex* functions of yields (the price–yield curve bends), so the linear approximation

becomes less accurate for larger yield changes. Convexity measures this curvature.

Definition 56: Convexity

The **convexity** $\mathcal{C}$ of a bond is the second derivative of the price–yield relationship (Definition 21), normalised by the price:

$$\mathcal{C} = \frac{1}{B}\frac{d^2B}{dy^2} = \frac{1}{B}\sum_{i=1}^{n}\frac{t_i(t_i+1)\cdot c_i}{(1+y)^{t_i+2}}.$$

Convexity is measured in **years**^{2}.

Duration + convexity approximation

The second-order Taylor expansion of the percentage price change is:

$$\frac{\Delta B}{B} \approx -D_{\text{mod}}\cdot\Delta y + \tfrac{1}{2}\mathcal{C}\cdot(\Delta y)^2.$$

The convexity term is always *positive* for standard bonds: the bond gains more than duration predicts when yields fall, and loses less when yields rise.

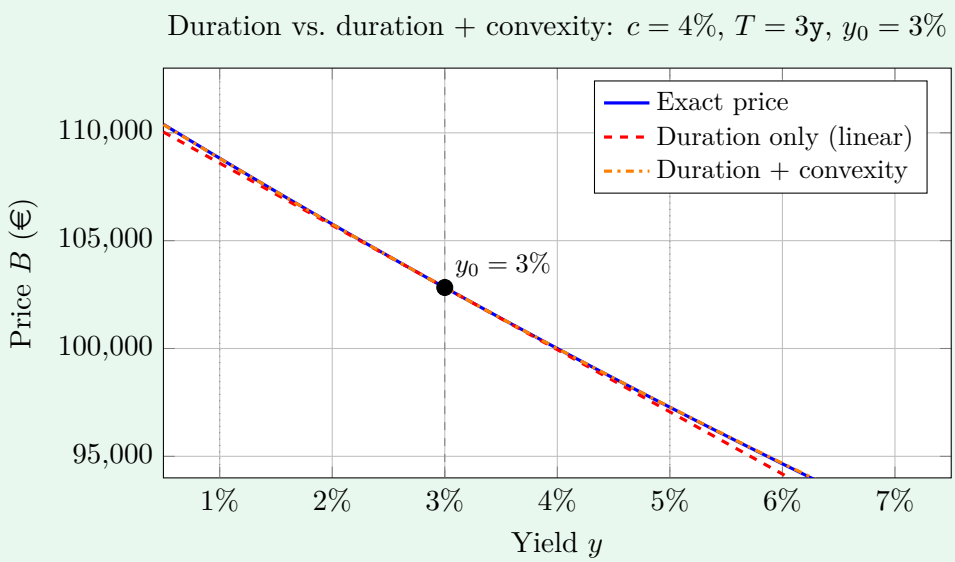
Example 52: Convexity adjustment

The bond from Example 51 has $B_0 = 102{,}830\,€$, $D_{\text{mod}} = 2.804$, and $\mathcal{C} = 10.75$. Compare the duration-only estimate, the duration + convexity estimate, and the exact price for yield moves of ± 200 bps.

Solution. Using the approximation formulas with $B_0 = 102{,}830\,€$:

Yield move	Duration only	Dur + convexity	Exact
-200 bps ($y = 1\%$)	108,601 €	108,822 €	108,822 €
$+200$ bps ($y = 5\%$)	97,059 €	97,280 €	97,281 €

For a 200 bps move, duration alone misses by 221 €; adding the convexity term reduces the error to under 1 €. The duration approximation always underestimates the true price (in both directions) because it ignores the upward bend of the price–yield curve. Convexity corrects for that bend.



The red dashed line (duration) is tangent at y_0 but diverges visibly at ± 200 bps. The orange dash-dotted curve (duration + convexity) lies virtually on top of the exact blue curve throughout the range shown.

Remark (Why convexity is desirable). Convexity is a desirable property for bondholders. Two bonds with the same duration but different convexities will behave differently:

- **Higher convexity:** the bond gains *more* when yields fall and loses *less* when yields rise — it outperforms in volatile environments.
- **Lower convexity:** the bond's price response is closer to the linear duration prediction.

All else being equal, investors prefer higher convexity and will accept a lower yield for it.

Example 53: High vs. low convexity

Consider two bonds, both with face value $N = 100{,}000$ €, coupon $c = 5\%$, and initial yield $y_0 = 5\%$ (so both trade at par). They differ only in maturity:

- **Bond 1:** $T = 5$y, $D_{\text{mod}} = 4.33$, $\mathcal{C} = 23.9$.
- **Bond 2:** $T = 15$y, $D_{\text{mod}} = 10.38$, $\mathcal{C} = 140.3$.

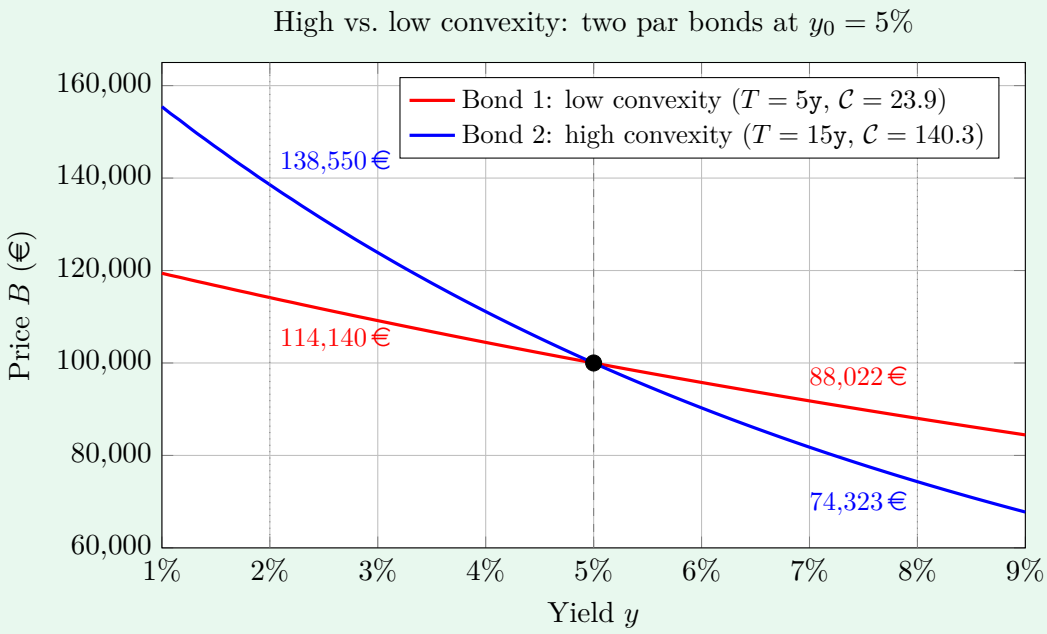
Compute the price of each bond at $y = 2\%$ and $y = 8\%$.

Solution. Applying Definition 21:

	$y = 2\%$	$y = 5\%$	$y = 8\%$
Bond 1 ($T = 5$y)	114,140 €	100,000 €	88,022 €
Bond 2 ($T = 15$y)	138,550 €	100,000 €	74,323 €

When yields fall to 2%, Bond 2 gains 38,550 € vs. 14,140 € for Bond 1. When yields

rise to 8%, Bond 2 loses 25,677 € vs. 11,978 € for Bond 1. Higher convexity amplifies gains and cushions losses — the asymmetry is always favourable.



The two curves cross at $y_0 = 5\%$ and diverge symmetrically. Bond 2 (blue) is visibly more bowed: steeper when yields fall, shallower when yields rise. Bond 1 (red) is flatter and tracks yield moves more linearly.

9.4 Duration and convexity of a portfolio

We extend the concepts of duration and convexity from individual bonds to portfolios. A portfolio (Definition 29) of bonds has a combined interest rate sensitivity that follows naturally from the additive structure of portfolio value. The duration and convexity measures can be aggregated across holdings to summarize the portfolio's first-order and second-order price sensitivity to yield changes.

Definition 57: Portfolio duration and convexity

The **duration of a portfolio** is the market-value-weighted average of the individual bond durations:

$$D_{\text{portfolio}} = \sum_{i=1}^{m} w_i \cdot D_i, \qquad w_i = \frac{B_i}{\sum_{j=1}^{m} B_j},$$

where B_i is the market value of bond i and D_i is its modified duration.

The **convexity of a portfolio** is similarly defined as the market-value-weighted average of the individual bond convexities:

$$C_{\text{portfolio}} = \sum_{i=1}^{m} w_i \cdot C_i,$$

where C_i is the convexity of bond i and w_i are the same market-value weights.

Both duration and convexity are additive in the market-value-weighted sense, making them convenient for risk management. Duration alone provides a linear approximation of portfolio

price sensitivity, accurate only for small yield changes. The combined effect of duration and convexity on portfolio value is captured by the second-order Taylor expansion:

$$\Delta V \approx -D_{\text{portfolio}} \cdot V \cdot \Delta y + \frac{1}{2} C_{\text{portfolio}} \cdot V \cdot (\Delta y)^2,$$

where Δy is the change in yield. Since bonds have positive convexity, the convexity term is always positive, meaning the actual price change is less negative for yield increases and more positive for yield decreases compared to the duration-only estimate. As yield moves become larger, the convexity adjustment becomes increasingly important.

Example 54: Portfolio duration and convexity

A portfolio consists of two bonds with the following characteristics:

Bond	Market value	Mod. duration	Convexity	Weight
A (2y, $c = 3\%$)	50,000 €	1.91	4.2	50%
B (10y, $c = 4\%$)	50,000 €	7.85	98.5	50%
Portfolio	100,000 €			

Compute the portfolio duration and convexity, and estimate the portfolio's price change if yields rise by 25 bps using both the duration-only and duration-plus-convexity approximations.

Solution. Portfolio metrics:

$$D_{\text{portfolio}} = 0.50 \cdot 1.91 + 0.50 \cdot 7.85 = 0.955 + 3.925 = 4.88,$$

$$C_{\text{portfolio}} = 0.50 \cdot 4.2 + 0.50 \cdot 98.5 = 2.10 + 49.25 = 51.35.$$

Duration-only estimate for a 25 bps (0.0025) yield increase:

$$\Delta V_{\text{duration}} \approx -4.88 \cdot 100{,}000 \cdot 0.0025 = -1{,}220\,€.$$

Convexity adjustment:

$$\Delta V_{\text{convexity}} \approx \tfrac{1}{2} \cdot 51.35 \cdot 100{,}000 \cdot (0.0025)^2 = 16\,€.$$

Combined estimate:

$$\Delta V \approx -1{,}220 + 16 = -1{,}204\,€.$$

For a 25 bps move, the convexity adjustment is modest (about 1.3% of the duration effect), but it partially offsets the loss. Note that Bond B, with both higher duration and significantly higher convexity, contributes the vast majority of both the interest rate risk and the convexity benefit. For larger yield moves, this adjustment would become proportionally more significant.

Portfolio duration and convexity assume a *parallel* shift in yields — all maturities move by the same amount. In reality, yield curves can twist, steepen, or flatten. **Key-rate duration** (KRD) measures the sensitivity of a portfolio to yield changes at specific maturity points (e.g. 2y, 5y, 10y, 30y). This allows decomposition of interest rate risk by maturity bucket and is essential for managing portfolios with cash flows across many maturities. KRD analysis relies on the ZCB decomposition from Chapter 4: each cash flow's sensitivity to its own maturity's rate is measured independently.

Remark (Negative convexity and callable bonds). Not all bonds have positive convexity. **Callable bonds** — bonds where the issuer has the right to repay early — exhibit *negative convexity* when interest rates fall sufficiently. This is because the issuer exercises the call option when refinancing becomes attractive, capping the bond's upside price appreciation. When managing portfolios with callable bonds, convexity becomes a critical risk metric alongside duration, as it can create asymmetric price-yield profiles that complicate risk measurement.

CHAPTER 10

Financial Derivatives

The instruments studied so far — bonds (Chapter 3), deposits and annuities (Chapter 8) — are *primary* instruments: they represent direct claims on cash flows generated by a borrower or issuer. In the preceding chapter, we saw how to measure the sensitivity of a bond's price to yield changes using duration and convexity. This chapter introduces a fundamentally different class of instruments whose value is *derived from* the value of something else, and develops a general sensitivity framework — the Greeks — that extends the ideas of duration and convexity to any risk factor.

10.1 What is a derivative?

Definition 58: Financial derivative

A **financial derivative** (or simply **derivative**) is a contract whose value depends on — is *derived from* — the value of one or more variables. These variables, called **underlyings**, can include interest rates, bond prices, exchange rates, stock prices, commodity prices, or credit events.

A derivative does not represent ownership of the underlying; it is a contract whose value is based on that underlying.

Definition 59: Notional principal

The **notional principal** (or **notional amount**) of a derivative is the reference amount on which the contract's payments are calculated. It is typically *not* exchanged between the parties; it serves purely as a scaling factor for computing cash flows.

Remark (Why would parties enter a derivative?). Derivatives serve several key economic purposes, even though they involve no initial exchange of capital:

1. **Hedging (risk reduction).** A company with a floating-rate loan faces uncertainty: if rates rise, its interest expense increases unpredictably. By entering a derivative where it *pays fixed* and *receives floating*, the floating receipts offset the loan payments, effectively locking in a known cost. The derivative transforms uncertain cash flows into certain ones.

2. **Speculation (taking a view).** An investor who believes interest rates will fall can enter a derivative to *receive fixed* and *pay floating*. If rates decline, the fixed receipts exceed the floating payments, generating profit. Derivatives allow taking positions on market views without needing to own the underlying asset.
3. **Arbitrage (exploiting mispricings).** If the same risk can be bought cheaply in one market and sold expensively in another, derivatives enable traders to capture the difference with minimal capital outlay.
4. **Access and efficiency.** Derivatives can provide exposure to markets or instruments that are otherwise difficult or expensive to access directly. A pension fund wanting exposure to a foreign interest rate need not open foreign accounts or deal with currency conversion — a derivative achieves the same economic effect more simply.

The notional principal amplifies these effects: a *small* difference in rates (say 0.20%) becomes a *large* cash flow (20,000 € on a 10,000,000 € notional). This leverage is precisely why derivatives are powerful tools for risk management and why they require careful understanding.

Example 55: Why parties pay money on a derivative

Two companies, A and B, enter into a 1y interest rate derivative contract on a notional of 10,000,000 €:

- Company A agrees to **pay** a fixed rate of 3% per annum.
- Company B agrees to **pay** a floating rate, currently at 2.80%.

The notional of 10,000,000 € is *never* transferred. It serves only to compute the payments:

$$\text{A pays:} \quad 3\% \cdot 10{,}000{,}000 = 300{,}000\,€, \qquad \text{B pays:} \quad 2.80\% \cdot 10{,}000{,}000 = 280{,}000\,€.$$

In practice, only the **net difference** changes hands. Company A pays Company B:

$$300{,}000 - 280{,}000 = 20{,}000\,€.$$

Why would either party agree to this? Because the contract **transfers risk**:

- Company A may have a floating-rate loan and wants predictable costs. By paying fixed and receiving floating, the floating payments it receives offset its loan payments, effectively converting its exposure to a fixed rate.
- Company B may expect rates to fall, or may have the opposite hedging need.

If, at the next payment date, the floating rate has risen to 3.40%, the net payment reverses: B now pays A the difference of 40,000 €. The notional remains untouched throughout — it is merely the *ruler* that scales small rate differences into concrete euro amounts.

Remark (Notional vs. market value). It is essential to distinguish between the **notional** and the **market value** of a derivative. The notional is the contractual reference amount — often very large. The **market value** (also called **mark-to-market value** or **fair value**) is what the contract is actually worth today, which is typically a small fraction of the notional.

Consider a contract with notional 100,000,000 € that obliges one party to pay a fixed rate of 3% and receive a floating rate currently at 3.05%. The net annual cash flow is

> $0.05\% \cdot 100{,}000{,}000 = 50{,}000$ € — a tiny fraction of the notional. The market value of the entire contract might be a few hundred thousand euros, not 100,000,000 €. This distinction matters enormously for risk reporting: the total notional outstanding in global derivatives markets exceeds hundreds of trillions of euros, but the actual economic exposure — the aggregate market value — is orders of magnitude smaller.

10.2 Classification of derivatives

Derivatives are classified along two main dimensions: the *type of contract* and the *trading venue*. We begin with two foundational concepts: what a derivative contract is written on, and where it can be traded.

Definition 60: Asset

An **asset** is anything of economic value that can be owned, traded, or used as the basis of a financial contract. In the context of derivatives, the term is used broadly: it may refer to a physical commodity (oil, gold), a financial instrument (a bond, a share of stock), a currency, or — by extension — a market variable such as an interest rate or a price index.

Definition 61: Volatility

The **volatility** of an asset or market variable quantifies the magnitude of its fluctuations over time. Informally, high volatility corresponds to large and less predictable price movements, whereas low volatility indicates more stable dynamics.
In quantitative finance, volatility is typically defined as the annualised standard deviation σ of the asset's returns.

For example, a volatility of $\sigma = 20\%$ on a stock means that, roughly speaking, the stock's annual return is expected to deviate from its mean by about 20 percentage points.

Definition 62: Exchange

An **exchange** is an organised, regulated marketplace where standardised financial instruments are traded. Exchanges provide transparent pricing, enforce uniform contract specifications, and typically operate a **central counterparty** (CCP) — a clearing house that interposes itself between buyer and seller, guaranteeing performance and virtually eliminating the risk that one party defaults on its obligations.

Major derivatives exchanges include the CME Group (Chicago), Eurex (Frankfurt), and ICE (London/Atlanta).
The four fundamental types of derivative contracts are:

Definition 63: Forward contract

A **forward contract** is an agreement between two parties to buy or sell an asset at a specified future date T (the **delivery date** or **maturity**) for a price K (the **delivery price** or **forward price**) agreed upon today.
No money changes hands at inception. At maturity, the **buyer** receives the asset and pays K; the **seller** delivers the asset and receives K.

The Forward Rate Agreement introduced in the next chapter is a specific type of forward

contract where the "asset" is an interest rate rather than a physical commodity or security.

Definition 64: Futures contract

A **futures contract** has the same economic purpose as a forward contract — locking in a future price — but differs in its institutional structure:

- **Standardisation:** Futures have standardised contract sizes, delivery dates, and quality specifications, set by the exchange.
- **Exchange-traded:** Futures are traded on organised exchanges (e.g. Eurex, CME), not bilaterally.
- **Central counterparty (CCP):** The exchange's clearing house stands between buyer and seller, virtually eliminating counterparty default risk.
- **Daily settlement (marking to market):** Gains and losses are settled *daily* through **margin accounts**, rather than accumulated until maturity. Each day, the losing party's margin account is debited and the gaining party's is credited by the change in the futures price.
- **Margin requirements:** Both parties must post an **initial margin** (a deposit guaranteeing performance) and maintain a **maintenance margin**. If losses erode the margin below this level, a **margin call** requires additional funds.

Feature	Forward	Futures
Trading venue	Bilateral (OTC)	Exchange
Standardisation	Customisable	Standardised
Counterparty risk	Present	Virtually eliminated (CCP)
Settlement	At maturity	Daily (mark-to-market)
Liquidity	Variable	Generally high

Definition 65: Swap

A **swap** is a derivative contract in which two parties agree to exchange sequences of cash flows over a specified period. Typically, one stream of payments is determined by a fixed rate and the other by a floating (variable) market rate.
The simplest and most important example is the **interest rate swap**, where one party pays a fixed interest rate and receives a floating rate (or vice versa) on a notional principal.

Definition 66: Option

An **option** is a derivative contract that gives the holder the **right, but not the obligation**, to buy or sell an underlying asset (or to enter into a financial transaction) at a predetermined price on or before a specified date.
Because the option holder can choose whether to exercise, the payoff is *asymmetric*: the holder's potential loss is limited to the upfront cost of the option (the **premium**), while the potential gain is theoretically unlimited (for a call) or substantial (for a put).

- A **call option** gives the right to *buy* (or to benefit from a rise in the underlying).
- A **put option** gives the right to *sell* (or to benefit from a fall in the underlying).

Remark (Obligations vs. rights). Forwards, futures, and swaps are **symmetric** contracts: both parties have obligations, and the profit or loss can be positive or neg-

ative for either side. Options are **asymmetric**: the buyer pays the premium upfront and then holds only rights; the seller receives the premium but bears obligations. This asymmetry is why options require an upfront payment while forwards and swaps (typically) do not.

Definition 67: OTC and exchange-traded derivatives

Derivatives are traded in two ways:

- **Over-the-counter (OTC):** Contracts are negotiated bilaterally between two parties (e.g. a bank and a corporate client). Terms are fully customisable. OTC contracts carry **counterparty risk** — the risk that one party fails to honour its obligations.
- **Exchange-traded:** Contracts are standardised and traded on an organised exchange with a central counterparty. Counterparty risk is largely eliminated through daily margining and collateral requirements.

Most interest rate derivatives are OTC instruments, though post-2008 regulation has increasingly required central clearing of standardised OTC contracts, blurring the historical distinction.

Example 56: Four contracts on the same underlying

A European energy company expects to purchase 100,000 barrels of oil in six months. Today's market price is 80 € per barrel. The company considers four derivative strategies to manage its price risk exposure. In all four cases, the notional principal — the reference amount on which payments are calculated — equals 100,000 barrels, equivalent to a notional value of 8,000,000 € at current prices. This notional amount is *never* exchanged; it serves purely as a scaling factor to determine cash flows.

1. Forward contract. The company enters a forward contract to buy 100,000 barrels in 6m at a delivery price of $K = 80$ € per barrel. No money is exchanged today: the contract has zero value at inception. In six months, at maturity (delivery date):

- If oil trades at 90 €: the company buys at 80 €, saving $(90 - 80) \cdot 100{,}000 = 1{,}000{,}000$ € relative to the market price. The contract has generated a profit.
- If oil trades at 70 €: the company must still buy at 80 €, losing $(80 - 70) \cdot 100{,}000 = 1{,}000{,}000$ € relative to the market. The contract has generated a loss.

The forward is a symmetric contract: the company has an obligation (not a right); the payoff can be positive or negative. The company gains if the price rises and loses if it falls (in this case, the company is said to hold a *long position* on oil).

2. Futures contract. Economically similar to the forward, but traded on an exchange — for example, Eurex. The futures contract is standardised: contract size, delivery dates, and quality specifications are fixed by the exchange. The company posts an initial margin and its account is adjusted daily through marking to market. If oil rises by 2 € on a given day, 200,000 € is credited to the company's margin account; if it falls by 1 €, 100,000 € is debited. The final economic result over six months is approximately the same as the forward, but gains and losses are realised *incrementally* day by day rather than concentrated at maturity. The exchange clearing house acts as a central counterparty (CCP), virtually eliminating counterparty risk. The futures is also a symmetric contract: the company has obligations in both directions.

3. Swap. Instead of a single delivery, the company enters a 6m swap with monthly settlements. Each month, the company pays a fixed price of 80 € per barrel and

receives the prevailing market price, on a notional principal of 100,000 barrels. In a month where the market price is 83 €:

$$\text{Net cash flow} = (83 - 80) \cdot 100{,}000 = +300{,}000\,€ \quad \text{(company receives)}.$$

If instead the market quotes 77 €, the company pays the difference: $(80-77)\cdot 100{,}000 = 300{,}000\,€$. The swap is also a symmetric contract: it spreads the price exposure across multiple settlement dates rather than concentrating it at a single maturity. The notional principal (8,000,000 € at current prices) is never exchanged; it serves only to compute the monthly flows. The company maintains a long position on oil: it benefits from price increases and suffers from decreases.

4. Call option. The company buys a call option with *strike price* $K = 80\,€$ per barrel, paying a premium of 4 € per barrel. The initial cost is $4 \cdot 100{,}000 = 400{,}000\,€$, paid at purchase. This is the only certain cash flow: unlike the previous contracts, the option requires an upfront payment.

- If oil trades at 90 €: the company exercises its right to buy at 80 €. The option payoff is $\max(90 - 80, 0) = 10\,€$ per barrel, total 1,000,000 €. Net gain after the premium: $(90 - 80 - 4) \cdot 100{,}000 = 600{,}000\,€$.
- If oil trades at 70 €: the company does not exercise (the payoff would be $\max(70 - 80, 0) = 0$) and buys at the market price. The loss is limited to the premium paid: 400,000 €. This is the maximum the company can lose.

The option is an asymmetric contract: the company has a *right* (not an obligation) and the downside is limited to the premium, while the upside is potentially unlimited. This protection has a cost — the premium of 400,000 € — which the first three contracts do not require. The premium represents the value of optionality: the possibility that the oil price may move favourably before expiry.

10.3 Mark-to-market and profit & loss

Derivatives are living contracts: their value changes continuously as market conditions evolve. Tracking these changes is fundamental to risk management.

Definition 68: Mark-to-market (MtM)

The **mark-to-market** value V of a financial instrument is its current market value — the price at which it could be closed out (sold or unwound) in an arm's-length transaction today. For a derivative, this is typically the present value of all expected future net cash flows, computed using current market rates and curves.

At inception, most derivatives (forwards, swaps) are structured so that neither party needs to make an upfront payment: the contract has zero mark-to-market value. As market conditions change, the contract acquires positive value for one party and negative value for the other.

Definition 69: Profit and loss (P&L)

The **profit and loss** (P&L) of a position over a period $[t, t + \Delta t]$ is the change in its mark-to-market value plus any cash flows received (minus any paid) during the period:

$$\text{P\&L} = [V(t + \Delta t) - V(t)] + \text{net cash flows received during } [t, t + \Delta t],$$

where $V(t)$ denotes the mark-to-market value at time t.

For instance, in a forward contract, the profit and loss for the buyer at maturity is:

$$\text{P\&L}_{\text{buyer}} = S_T - K,$$

where S_T is the market price of the asset at time T. The seller's P&L is $K - S_T$.

Example 57: Mark-to-market of a forward contract

A company enters a 1y forward contract to buy 1,000,000 € worth of goods at a fixed price of $K = 50$ € per unit (i.e. 20,000 units). At inception the forward has zero value. After 3m, the market forward price for the same delivery date has risen to 52 €. Compute the mark-to-market value for the buyer.

Solution. The buyer is committed to paying 50 € per unit but can now lock in a sale at 52 € per unit for the same delivery date. The unrealised gain per unit is $52 - 50 = 2$ €. The total mark-to-market value (undiscounted, for simplicity) is:

$$V \approx 20{,}000 \cdot (52 - 50) = 40{,}000\,€.$$

More precisely, this gain would be discounted over the remaining 9m to delivery. If the 9m discount factor is $\text{DF}(0.75) = 0.977$:

$$V = 20{,}000 \cdot 2 \cdot 0.977 = 39{,}080\,€.$$

The contract has positive mark-to-market for the buyer (long position) and equal negative value for the seller (short position).

Definition 70: Long and short positions

A **long position** in a financial instrument or market variable is one that *profits when the value of that instrument or variable rises* and loses when it falls.
A **short position** is one that *profits when the value falls* and loses when it rises.

For primary instruments, the meaning is intuitive: buying a bond is a long position (you profit if its price rises); selling a bond you own is closing a long position. In derivative markets, however, both parties can establish positions without owning the underlying. The terms "long" and "short" then refer to the *direction of exposure* rather than to physical ownership.

Remark (Long and short in the context of interest rates). Interest rate instruments require care with the direction terminology because bond prices and interest rates move inversely (Definition 24):

- A bondholder is **long the bond** but effectively **short interest rates**: the bondholder profits when rates *fall* (causing the bond price to rise).
- A borrower at a fixed rate is **short the bond** (equivalent to having issued it) and effectively **long interest rates**: the borrower benefits when rates *rise* (the fixed obligation becomes relatively cheaper).

Throughout this booklet, when we say a portfolio is "long delta" or "short delta" with

respect to a risk factor, we mean the portfolio gains or loses value, respectively, when that factor increases. We will formalise this notion in Section 10.4.

10.4 Sensitivity measures: the Greeks

The mark-to-market value of a derivative V depends on one or more risk factors — market variables such as interest rates, exchange rates, or volatilities. Risk management requires measuring how sensitive V is to changes in each risk factor. These sensitivities are collectively known as the Greeks, named after the Greek letters traditionally used to denote them.

Definition 71: Risk factor

A **risk factor** is any market variable whose movements cause changes in the mark-to-market value of a financial instrument or portfolio. Common risk factors in fixed income include interest rates (at various maturities), FX rates, credit spreads, and volatilities. A hypothetical or real change in a risk factor is called a **shock**.

10.4.1 Delta

Definition 72: Delta (Δ)

The **delta** of a derivative is the first partial derivative of its value V with respect to the underlying risk factor x:

$$\Delta = \frac{\partial V}{\partial x}.$$

Delta measures the **linear sensitivity** of the instrument's value to a small change in the risk factor.

A delta of $-500{,}000$ means that if the risk factor increases by one unit, the instrument's value decreases by approximately 500,000 €.

Delta is the single most important sensitivity measure. It determines the *direction* and *magnitude* of the instrument's exposure, connecting directly to the long and short positions introduced in Definition 70:

- $\Delta > 0$: the position is **long** the risk factor — it gains when the factor rises.
- $\Delta < 0$: the position is **short** the risk factor — it gains when the factor falls.
- $\Delta = 0$: the position is **delta-neutral** — insensitive (to first order) to small changes in the factor.

Remark (Delta and duration). For a bond, the risk factor is the yield y, and the bond's price sensitivity is captured by modified duration (Definition 55):

$$\frac{\Delta B}{\Delta y} \approx -D_{\text{mod}} \cdot B.$$

Duration is therefore the bond-specific version of delta. The negative sign reflects the inverse price–yield relationship (Definition 24): when yields rise, bond prices fall. In derivative markets, the broader concept of delta generalises this idea to *any* risk factor, not just yields.

10.4.2 Gamma

Definition 73: Gamma (Γ)

The **gamma** of a derivative is the second partial derivative of its value with respect to the underlying risk factor:

$$\Gamma = \frac{\partial^2 V}{\partial x^2}.$$

Gamma measures the **rate of change of delta** itself — how quickly the linear sensitivity shifts as the risk factor moves. A large gamma means that delta changes rapidly, making the instrument's behaviour increasingly non-linear for larger moves.

Gamma is particularly important for two reasons:

- **Accuracy of linear approximations:** If gamma is large, the linear (delta-only) approximation to price changes becomes inaccurate for anything beyond very small moves.
- **Stability of risk-neutral positions:** A portfolio constructed to have zero delta will quickly develop a non-zero delta as the risk factor moves if gamma is large, requiring frequent adjustment to maintain neutrality.

Remark (Gamma and convexity). Just as delta generalises duration, gamma generalises **convexity** — the second-order sensitivity of a bond's price to yield changes. For a bond:

$$\Gamma_{\text{bond}} = \frac{\partial^2 B}{\partial y^2} = \mathcal{C} \cdot B,$$

where $\mathcal{C}$ is the convexity. The conceptual parallel is exact: convexity measures how the bond's delta (duration) changes as yields move.

10.4.3 Other Greeks

While delta and gamma are the most important Greeks for fixed income instruments, three others appear frequently:

- **Vega** ($\mathcal{V}$): Sensitivity to volatility (Definition 61): $\mathcal{V} = \partial V / \partial \sigma$. Important for options, where the price depends critically on expected future volatility.
- **Theta** (Θ): Sensitivity to the passage of time: $\Theta = \partial V / \partial t$. Measures **time decay** — how much value the instrument loses as one day passes, all else equal.
- **Rho** (ρ): Sensitivity to interest rates: $\rho = \partial V / \partial r$. Relevant when the discount rate itself is a risk factor distinct from the underlying.

10.5 Taylor expansion for portfolio P&L

The Greeks allow us to *approximate* the change in value of an instrument or portfolio without fully revaluing it. The mathematical basis is the Taylor expansion.

Taylor expansion for P&L

If the value of an instrument depends on a single risk factor x, and the risk factor experiences a shock of size Δx, the resulting change in value is approximately:

$$\Delta V \approx \underbrace{\Delta \cdot \Delta x}_{\text{linear (delta) effect}} + \underbrace{\tfrac{1}{2}\Gamma \cdot (\Delta x)^2}_{\text{curvature (gamma) effect}}.$$

This is the second-order Taylor expansion of $V(x + \Delta x)$ around x.

The linear term $\Delta \cdot \Delta x$ dominates for small shocks. The quadratic term $\frac{1}{2}\Gamma \cdot (\Delta x)^2$ becomes important for larger moves, especially for instruments with significant curvature (high gamma). For standard fixed income instruments and shocks up to a few percentage points, the second-order approximation is typically accurate to within fractions of a percent.

Since the Greeks are partial derivatives, and differentiation is linear, the Greeks of a portfolio are simply the sums of the individual instruments' Greeks:

Additivity of Greeks

The delta and gamma of a portfolio are the sums of the individual instruments' Greeks:

$$\Delta_{\text{port}} = \sum_{j=1}^{n} \Delta_j, \qquad \Gamma_{\text{port}} = \sum_{j=1}^{n} \Gamma_j.$$

This additivity enables efficient risk management: we can compute the risk of an entire portfolio by aggregating the risks of its components.
The Taylor expansion then applies at the portfolio level: if the risk factor shifts by Δx, the approximate portfolio P&L is:

$$\Delta V_{\text{port}} \approx \Delta_{\text{port}} \cdot \Delta x + \tfrac{1}{2}\,\Gamma_{\text{port}} \cdot (\Delta x)^2.$$

Remark (Multiple risk factors). When a portfolio is exposed to several risk factors $x_1, x_2, \ldots, x_k$ (e.g. rates in different currencies or at different maturities), the P&L approximation generalises to:

$$\Delta V_{\text{port}} \approx \sum_{i=1}^{k} \Delta_i \cdot \Delta x_i + \tfrac{1}{2} \sum_{i=1}^{k} \Gamma_i \cdot (\Delta x_i)^2 + \sum_{i<j} \frac{\partial^2 V}{\partial x_i\, \partial x_j} \cdot \Delta x_i \cdot \Delta x_j,$$

where Δ_i and Γ_i are the delta and gamma with respect to risk factor x_i, and the cross-partial derivatives capture the interaction between different risk factors. In practice, the cross terms are often small and may be neglected for a first approximation.

Remark (Connection to duration and convexity). For a bond with value B, risk factor y (yield), delta $= -D_{\text{mod}} \cdot B$, and gamma $= \mathcal{C} \cdot B$, the Taylor expansion becomes:

$$\Delta B \approx -D_{\text{mod}} \cdot B \cdot \Delta y + \tfrac{1}{2}\,\mathcal{C} \cdot B \cdot (\Delta y)^2.$$

This is exactly the duration-plus-convexity approximation derived in Chapter 9. The general Taylor framework presented here unifies this bond-specific result with the analogous formulas for swaps, options, and portfolios of mixed instruments.

Example 58: Approximating P&L with delta and gamma

An interest rate derivative has the following characteristics:

Quantity	Value	Interpretation
Current value V	50,000 €	Mark-to-market today
Delta Δ	−800,000	Loses 800,000 € per unit increase in factor
Gamma Γ	+120,000,000	Delta becomes more negative as factor rises

The underlying risk factor decreases by 200 bps ($\Delta x = -0.02$). Estimate the P&L using (a) delta only and (b) delta plus gamma.

Solution. **(a) Delta-only approximation:**

$$\Delta V \approx \Delta \cdot \Delta x = (-800{,}000) \cdot (-0.02) = +16{,}000\,€.$$

The negative delta and negative shock produce a gain, consistent with a short position in the risk factor.

(b) Delta + gamma approximation:

$$\Delta V \approx (-800{,}000)(-0.02) + \tfrac{1}{2}(120{,}000{,}000)(-0.02)^2 = 16{,}000 + 24{,}000 = +40{,}000\,€.$$

The gamma term adds 24,000 € to the estimate. Note that the gamma contribution is always *positive* regardless of the direction of the shock (since $(\Delta x)^2 > 0$ and $\Gamma > 0$). This instrument benefits from large moves in either direction — a desirable property.

The estimated new value is $V + \Delta V = 50{,}000 + 40{,}000 = 90{,}000\,€$.

10.5.1 Numerical computation of Greeks

In many practical situations, the value function $V(x)$ is not available in closed form — it may come from a complex pricing model, a Monte Carlo simulation, or simply from observed market data. In such cases, the Greeks are computed numerically using finite difference methods.

Definition 74: Numerical Greeks (finite differences)

Given the value of an instrument at the current risk factor level x and at shifted levels $x \pm \varepsilon$ (where ε is a small perturbation), the Greeks can be approximated as:

Delta (central difference):

$$\Delta \approx \frac{V(x+\varepsilon) - V(x-\varepsilon)}{2\varepsilon}.$$

Gamma (second-order central difference):

$$\Gamma \approx \frac{V(x+\varepsilon) - 2\,V(x) + V(x-\varepsilon)}{\varepsilon^2}.$$

The central difference formula for delta uses values on *both sides* of x, giving a more accurate estimate than a one-sided difference $[V(x+\varepsilon) - V(x)]/\varepsilon$. A typical choice for the perturbation in interest rate applications is $\varepsilon = 1\,\text{bps} = 0.0001$.

Example 59: Computing Greeks numerically

A contract's value is observed at five levels of the underlying risk factor x:

Risk factor level	Contract value
$x - 2\varepsilon$	106,408.91 €
$x - \varepsilon$	106,583.09 €
x	106,533.42 €
$x + \varepsilon$	106,475.37 €
$x + 2\varepsilon$	106,408.91 €

where $\varepsilon = 0.0001$ (1 bps). Compute delta and gamma, and verify the approximation quality by using the computed Greeks to predict $V(x + 2\varepsilon)$.

Solution. **Delta** (central difference):

$$\begin{aligned}\Delta &= \frac{V(x+\varepsilon) - V(x-\varepsilon)}{2\varepsilon} \\ &= \frac{106{,}475.37 - 106{,}583.09}{2 \cdot 0.0001} = \frac{-107.72}{0.0002} = -538{,}600.\end{aligned}$$

The negative delta means the contract *loses* value when the risk factor rises — a short position.

Gamma (second-order central difference):

$$\begin{aligned}\Gamma &= \frac{V(x+\varepsilon) - 2V(x) + V(x-\varepsilon)}{\varepsilon^2} \\ &= \frac{106{,}475.37 - 213{,}066.84 + 106{,}583.09}{(0.0001)^2} \\ &= \frac{-8.38}{10^{-8}} = -838{,}000{,}000.\end{aligned}$$

Verification. Use the Taylor expansion to predict $V(x+2\varepsilon)$ with $\Delta x = 2\varepsilon = 0.0002$:

$$\begin{aligned}V(x+2\varepsilon) &\approx V(x) + \Delta \cdot (2\varepsilon) + \tfrac{1}{2}\Gamma \cdot (2\varepsilon)^2 \\ &= 106{,}533.42 + (-538{,}600)(0.0002) \\ &\quad + \tfrac{1}{2}(-838{,}000{,}000)(0.0002)^2 \\ &= 106{,}533.42 - 107.72 - 16.76 \\ &= 106{,}408.94\,€.\end{aligned}$$

The actual value is 106,408.91 €, so the prediction error is

$$\frac{|106{,}408.94 - 106{,}408.91|}{106{,}408.91} = 0.00003\%,$$

an excellent approximation, confirming the accuracy of the second-order Taylor expansion.

When Greeks are computed numerically, it is good practice to *validate* them by repeating the calculation with a different perturbation size (e.g. 2ε) and checking that the results are consistent. Large discrepancies may indicate numerical instability or data quality issues in the underlying valuations. This cross-validation is particularly important when working with

real portfolio data, where some instruments may exhibit irregular pricing behaviour.

10.6 Hedging

Measuring risk through the Greeks naturally leads to the question of how to *reduce* it.

Definition 75: Hedging

Hedging is the practice of taking an **offsetting position** — a position whose value moves in the opposite direction to an existing exposure — in order to reduce or eliminate risk. When the risk factor moves adversely for the original position, the offsetting position gains value, compensating the loss. A **perfect hedge** eliminates risk entirely; in practice, most hedges are *partial*, reducing but not eliminating exposure.

Definition 76: Delta hedging

Delta hedging is the strategy of making a portfolio **delta-neutral** — i.e. reducing its aggregate delta to zero — by taking an offsetting position whose delta cancels the existing exposure.
If a portfolio has delta Δ_{port} with respect to a risk factor, and a hedging instrument has delta Δ_{hedge} per unit, then the number of **hedging units** required is:

$$n = -\frac{\Delta_{\text{port}}}{\Delta_{\text{hedge}}}.$$

Example 60: Delta hedging a portfolio

A portfolio of interest rate contracts has an aggregate delta of $\Delta_{\text{port}} = +2{,}000{,}000$ with respect to the 5y interest rate, meaning the portfolio gains 2,000,000 € for each 1% increase in the 5y rate. A 5y bond with face value 100,000 € has delta $\Delta_{\text{bond}} = -4{,}500$ per bond (it loses 4,500 € per 1% rate increase). How many bonds must be bought or sold to delta-hedge the portfolio?

Solution.

$$n = -\frac{\Delta_{\text{port}}}{\Delta_{\text{bond}}} = -\frac{+2{,}000{,}000}{-4{,}500} = +444.4.$$

The portfolio requires purchasing approximately 444 bonds. Buying the bonds introduces a negative delta ($444 \cdot (-4{,}500) = -1{,}998{,}000$), which offsets the portfolio's positive delta, resulting in an approximately delta-neutral combined position.
Note that this hedge is effective only for *small* rate moves. For larger moves, the gamma of both the portfolio and the bonds will cause the deltas to change, requiring periodic rebalancing — a process known as **dynamic hedging**.

Remark (Limitations of delta hedging). Delta hedging neutralises first-order risk but leaves the portfolio exposed to:

- **Gamma risk:** Large moves cause the delta to shift, making the hedge imperfect. A **gamma hedge** (using instruments with offsetting gamma, typically options) can address this.

- **Other risk factors:** Delta hedging against one factor (e.g. the 5y rate) does not protect against moves in other factors (e.g. the 2y rate, FX rates, or credit spreads).
- **Model risk:** Computed Greeks may be inaccurate due to model limitations or data quality issues.
- **Discrete rebalancing:** In practice, hedges are adjusted periodically, not continuously. Between adjustments, the portfolio is exposed to risk from gamma and other higher-order effects.

CHAPTER 11

Forward Rate Agreements

In Chapter 5 we derived forward rates from the spot curve using no-arbitrage arguments. Forward rates are not merely theoretical constructs — they are the basis for traded instruments that allow market participants to lock in future borrowing or lending rates. The simplest such instrument is the Forward Rate Agreement.

11.1 What is an FRA?

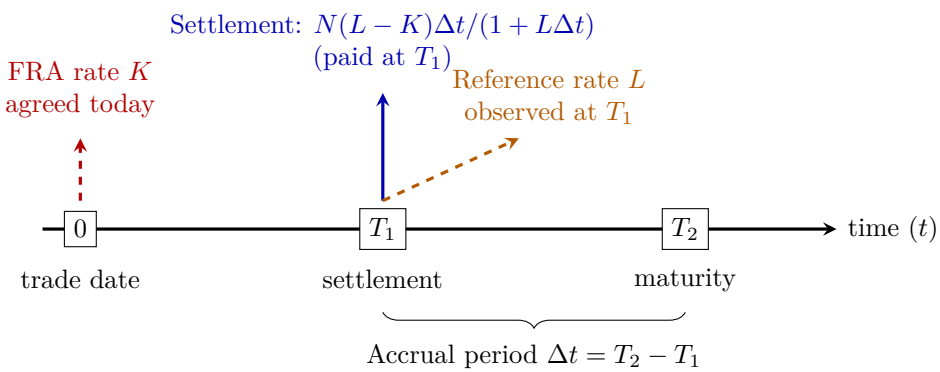
Definition 77: Forward Rate Agreement (FRA)

A **Forward Rate Agreement** (FRA) is an over-the-counter derivative contract in which two parties agree today on an interest rate (the **FRA rate** K) to be applied to a notional principal N over a *future* period $[T_1, T_2]$.
At the **settlement date** T_1, the FRA is settled by comparing the agreed FRA rate K with the prevailing market reference rate L (e.g. EURIBOR; see Definition 33) for the period $[T_1, T_2]$. The settlement amount is:

$$\text{Settlement} = N \cdot (L - K) \cdot \Delta t \cdot \frac{1}{1 + L \cdot \Delta t},$$

where $\Delta t = T_2 - T_1$ is the year fraction and the factor $1/(1 + L \cdot \Delta t)$ discounts the payment from T_2 to T_1. If $L > K$, the seller pays the buyer; if $L < K$, the buyer pays the seller.

The **buyer** of the FRA profits when the reference rate L exceeds the agreed rate K, and loses when $L < K$. The **seller** has the opposite exposure. No principal is exchanged; only the interest rate difference is settled.



11.1.1 FRA notation

Definition 78: FRA notation

FRAs are commonly designated as "$a \times b$" (read "a by b"), where a is the number of months to the settlement date T_1 and b is the number of months to the maturity date T_2.

For example:
- A 3×6 FRA settles in 3m and covers the 3m period from 3m to 6m.
- A 6×12 FRA settles in 6m and covers the 6m period from 6m to 12m.
- A 12×24 FRA settles in 12m and covers the 12m period from 1y to 2y.

11.2 FRA pricing

The fair FRA rate K is determined by no-arbitrage: it must equal the forward rate $f(T_1, T_2)$ for the same period. If it were different, an arbitrage opportunity would exist.

Fair FRA rate

The fair (no-arbitrage) FRA rate for the period $[T_1, T_2]$ equals the forward rate implied by the current spot curve:

$$K = f(T_1, T_2).$$

For simple interest (the convention used for FRAs with maturities $\leq$ 1y):

$$K = \frac{1}{\Delta t}\left(\frac{\mathrm{DF}(T_1)}{\mathrm{DF}(T_2)} - 1\right),$$

where $\Delta t = T_2 - T_1$ is the year fraction and $\mathrm{DF}(\cdot)$ denotes the discount factor.

The FRA rate is derived from a replication argument. Consider two strategies:

Strategy A: Invest 1 at the spot rate $r(T_2)$ for T_2 years, receiving $1/\mathrm{DF}(T_2)$ at T_2.
Strategy B: Invest 1 at the spot rate $r(T_1)$ for T_1 years, receiving $1/\mathrm{DF}(T_1)$ at T_1, then reinvest at the FRA rate K for the period $[T_1, T_2]$, receiving $[1/\mathrm{DF}(T_1)] \cdot (1 + K \cdot \Delta t)$ at T_2.

No-arbitrage requires the terminal values to be equal:

$$\frac{1}{\mathrm{DF}(T_2)} = \frac{1}{\mathrm{DF}(T_1)} \cdot (1 + K \cdot \Delta t).$$

Solving for K:

$$K = \frac{1}{\Delta t}\left(\frac{\mathrm{DF}(T_1)}{\mathrm{DF}(T_2)} - 1\right).$$

This is precisely the forward rate $f(T_1, T_2)$ under simple compounding — confirming that the fair FRA rate equals the forward rate.

Example 61: Pricing a 3×6 FRA

A bank quotes a 3×6 FRA on a notional of $N = 1{,}000{,}000$ €. The current discount factors are $\mathrm{DF}(0.25) = 0.991$ and $\mathrm{DF}(0.50) = 0.982$. Compute the fair FRA rate.

Solution. The accrual period is $\Delta t = 0.50 - 0.25 = 0.25\mathtt{y}$ ($3\mathtt{m}$). Using the formula:

$$K = \frac{1}{0.25}\left(\frac{0.991}{0.982} - 1\right) = 4 \cdot (1.00917 - 1) = 4 \cdot 0.00917 = 3.666\%.$$

The fair 3×6 FRA rate is 3.666%. This is the rate at which the FRA has zero value at inception — neither party has an advantage.

11.3 FRA settlement

At settlement date T_1, the reference rate L is observed and the FRA is settled.

Example 62: FRA settlement

Continuing Example 61: the 3×6 FRA has $N = 1{,}000{,}000$ €, FRA rate $K = 3.666\%$, and $\Delta t = 0.25\mathtt{y}$. At settlement ($T_1 = 3\mathtt{m}$), the $3\mathtt{m}$ EURIBOR fixes at $L = 4.00\%$. Compute the settlement amount and determine who pays whom.

Solution. The settlement amount (paid at T_1) is:

$$\begin{aligned}\text{Settlement} &= N \cdot (L - K) \cdot \Delta t \cdot \frac{1}{1 + L \cdot \Delta t} \\ &= 1{,}000{,}000 \cdot (0.04 - 0.03666) \cdot 0.25 \cdot \frac{1}{1 + 0.04 \cdot 0.25}.\end{aligned}$$

Step by step:

$$\begin{aligned}L - K &= 0.04 - 0.03666 = 0.00334, \\ N \cdot (L - K) \cdot \Delta t &= 1{,}000{,}000 \cdot 0.00334 \cdot 0.25 = 835.00\,€, \\ \text{Discount factor} &= \frac{1}{1 + 0.04 \cdot 0.25} = \frac{1}{1.01} = 0.990, \\ \text{Settlement} &= 835.00 \cdot 0.990 = 826.65\,€.\end{aligned}$$

Since $L > K$ (the market rate exceeded the agreed rate), the **seller pays the buyer** 826.65 €. The FRA buyer locked in borrowing at 3.666% but the market rate turned out to be 4%; the settlement compensates them for the difference.

Remark (Equivalence: FRA settlement vs. forward deposit). The FRA settlement amount is economically equivalent to the present value (at T_1) of the interest rate differential over the accrual period. If the settlement were made at T_2 instead of T_1, the amount would be simply $N \cdot (L - K) \cdot \Delta t$ (without the discounting factor). The convention of settling at T_1 with discounting is called **settlement in advance**; it is standard market practice for FRAs.

11.4 Uses of FRAs

FRAs serve three main purposes:

- **Hedging:** A borrower expecting to take a loan in the future can buy an FRA to lock in

the borrowing rate, eliminating exposure to rising rates.

- **Speculation:** A trader who believes rates will rise can buy an FRA (profiting if $L > K$); one who believes rates will fall can sell an FRA (profiting if $L < K$).
- **Arbitrage:** If FRA rates deviate from forward rates implied by the spot curve, arbitrageurs can lock in riskless profit by trading the FRA against spot instruments.

Example 63: Hedging with an FRA

A company knows it will need to borrow $N = 10{,}000{,}000$ € for 6m starting in 3m. The current 3×9 FRA rate is $K = 3.50\%$. The company buys the FRA to hedge.
Three months later, the 6m EURIBOR fixes at $L = 4.20\%$. What is the effective borrowing cost?

Solution. Step 1 — FRA settlement.
The company bought the FRA (it is the "buyer" or "fixed-rate payer"). Since $L > K$, the company receives a settlement payment:

$$\begin{aligned}\text{Settlement} &= 10{,}000{,}000 \cdot (0.042 - 0.035) \cdot 0.5 \cdot \frac{1}{1 + 0.042 \cdot 0.5} \\ &= 10{,}000{,}000 \cdot 0.0035 \cdot 0.9794 = 34{,}278\,€.\end{aligned}$$

Step 2 — Actual borrowing.
The company borrows 10,000,000 € at the market rate $L = 4.20\%$ for 6m. The interest cost is:

$$I = 10{,}000{,}000 \cdot 0.042 \cdot 0.5 = 210{,}000\,€.$$

Step 3 — Net cost.
The FRA settlement offsets part of the borrowing cost. The present value of the interest saving at T_1 is 34,278 €, which is equivalent to 35,000 € at T_2. Net interest cost:

$$\text{Net interest} \approx 210{,}000 - 35{,}000 = 175{,}000\,€,$$

corresponding to an effective rate of:

$$r_{\text{eff}} = \frac{175{,}000}{10{,}000{,}000 \cdot 0.5} = 3.50\%.$$

The FRA locked in the borrowing cost at exactly 3.50%, regardless of where rates moved. The company is protected against rising rates but forgoes the benefit if rates had fallen below 3.50%.

Remark (FRAs vs. interest rate futures). FRAs and **interest rate futures** (such as Euribor futures or SOFR futures) serve similar economic purposes — both allow locking in future interest rates. Key differences include:

- **Trading venue:** FRAs are OTC (customisable terms); futures are exchange-traded (standardised contracts).
- **Settlement:** FRAs settle once at T_1; futures are marked to market daily with daily margin payments.
- **Credit risk:** FRAs carry bilateral counterparty risk; futures mitigate this through the clearinghouse.

- **Convexity adjustment:** Because of daily margining, futures rates differ slightly from forward rates by a **convexity adjustment** (typically a few basis points), which accounts for the correlation between rate changes and the cost of funding margin payments.

CHAPTER 12

Interest Rate Swaps

In Chapter 11 we introduced the Forward Rate Agreement, which allows two parties to lock in a future interest rate for a single period. An interest rate swap generalises this idea to *multiple* periods: it is essentially a portfolio of FRAs covering consecutive future periods, packaged into a single contract.

12.1 What is an interest rate swap?

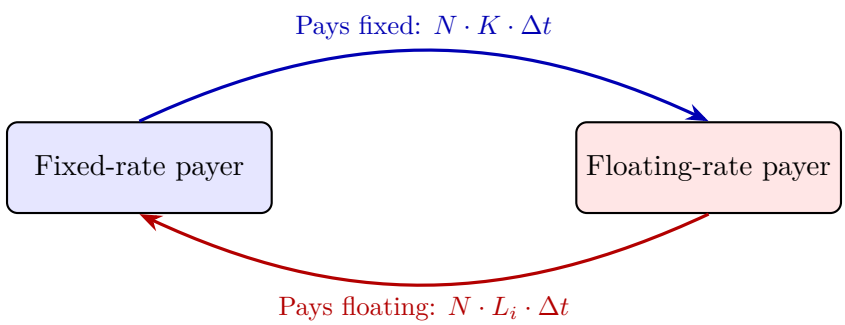
Definition 79: Interest rate swap (IRS)

An **interest rate swap** (IRS) is an OTC derivative contract in which two parties agree to exchange periodic interest payments on a notional principal N for a specified term T. One party pays a **fixed rate** K (the **swap rate**); the other pays a **floating rate** tied to a market reference rate L_i (e.g. EURIBOR, SOFR; see Definition 33). No principal is exchanged — only the net interest difference at each payment date.

The two payment streams are called **legs**:

- **Fixed leg:** Pays $N \cdot K \cdot \Delta t_i$ at each payment date T_i ($i = 1, \ldots, n$).
- **Floating leg:** Pays $N \cdot L_i \cdot \Delta t_i$ at each payment date T_i, where L_i is the reference rate observed at the start of period i (reset in advance, paid in arrears — the same mechanism as for FRNs).

In practice, only the **net** amount is exchanged at each date: if the fixed payment exceeds the floating payment, the fixed-rate payer pays the difference to the floating-rate payer, and vice versa.



Remark (Terminology conventions).

- The **fixed-rate payer** (also called the **swap buyer**) pays fixed and receives floating. This party benefits when rates rise (receiving higher floating payments).
- The **floating-rate payer** (also called the **swap seller**) pays floating and receives fixed. This party benefits when rates fall.
- The **swap rate** K is the fixed rate that makes the swap have zero value at inception.
- The **tenor** is the total length of the swap (e.g. 5y, 10y).
- The **payment frequency** is how often payments are exchanged (e.g. quarterly, semi-annually, annually).

12.2 Swap valuation

To price a swap, we observe that its cash flows can be decomposed into two familiar instruments.

12.2.1 Swap as a portfolio of bonds

Swap = fixed-rate bond − floating-rate bond

From the perspective of the fixed-rate payer, an IRS is equivalent to:

- **Short** a fixed-rate bond (paying fixed coupons $K \cdot N$ and returning N at maturity), and
- **Long** a floating-rate note (receiving floating coupons $L_i \cdot N$ and returning N at maturity).

Since both bonds have the same notional N and the same maturity T, the principal payments cancel:

$$V_{\text{swap}} = V_{\text{float}} - V_{\text{fixed}} = \underbrace{N}_{\text{FRN at par}} - \underbrace{\left(K \cdot N \sum_{i=1}^{n} \text{DF}(T_i) + N \cdot \text{DF}(T_n) \right)}_{\text{fixed-rate bond price}}.$$

On a reset date, the FRN trades at par (see the Key Formula in Chapter 6), so $V_{\text{float}} = N$. The fixed leg is priced as a standard coupon bond (Definition 21).

12.2.2 Swap as a portfolio of FRAs

Alternatively, each net payment $(L_i - K) \cdot N \cdot \Delta t_i$ at time T_i is exactly the payoff of an FRA (Definition 77) with rate K for the period $[T_{i-1}, T_i]$. Therefore:

Swap = portfolio of FRAs

An interest rate swap is equivalent to a portfolio of n Forward Rate Agreements, one for each payment period:

$$V_{\text{swap}} = \sum_{i=1}^{n} N \cdot \left(f(T_{i-1}, T_i) - K \right) \cdot \Delta t_i \cdot \text{DF}(T_i),$$

where $f(T_{i-1}, T_i)$ is the forward rate for each period (Definition 32).

12.3 The swap rate

Definition 80: Swap rate (par swap rate)

The **swap rate** K is the fixed rate that gives the swap zero value at inception. Setting $V_{\text{swap}} = 0$ and solving:

$$K = \frac{1 - \text{DF}(T_n)}{\sum_{i=1}^{n} \Delta t_i \cdot \text{DF}(T_i)}.$$

Remark (Swap rate equals the par yield). Comparing with the par yield formula (Definition 25):

$$r_{\text{par}} = \frac{1 - \text{DF}(T)}{\sum_{i=1}^{n} \text{DF}(t_i)},$$

we see that the swap rate and the par yield are *identical* when the payment dates coincide and $\Delta t_i = 1$ (annual payments). This is no coincidence: a par bond has zero NPV (it trades at face value), and a fair swap also has zero NPV. Both conditions lead to the same equation.

Example 64: Pricing a 3-year swap

Compute the 3y annual swap rate given the following discount factors (bootstrapped from the OIS curve; see Section 7.2):

Maturity	$\text{DF}(T_i)$
1y	0.965
2y	0.932
3y	0.900

Solution. With annual payments ($\Delta t_i = 1$ for all i):

$$K = \frac{1 - \text{DF}(3)}{\text{DF}(1) + \text{DF}(2) + \text{DF}(3)} = \frac{1 - 0.900}{0.965 + 0.932 + 0.900} = \frac{0.100}{2.797} = 3.575\%.$$

Verification via the FRA decomposition. The implied 1y forward rates are:

$$f(0,1) = \frac{1}{\text{DF}(1)} - 1 = \frac{1}{0.965} - 1 = 3.627\%,$$
$$f(1,2) = \frac{\text{DF}(1)}{\text{DF}(2)} - 1 = \frac{0.965}{0.932} - 1 = 3.541\%,$$
$$f(2,3) = \frac{\text{DF}(2)}{\text{DF}(3)} - 1 = \frac{0.932}{0.900} - 1 = 3.556\%.$$

The present value of each floating payment:

$$f(0,1) \cdot \text{DF}(1) = 0.03627 \cdot 0.965 = 0.03500,$$
$$f(1,2) \cdot \text{DF}(2) = 0.03541 \cdot 0.932 = 0.03300,$$
$$f(2,3) \cdot \text{DF}(3) = 0.03556 \cdot 0.900 = 0.03200.$$

Total PV of floating leg (excluding principal): $0.035 + 0.033 + 0.032 = 0.100$.
Total PV of fixed leg per unit rate: $\mathrm{DF}(1) + \mathrm{DF}(2) + \mathrm{DF}(3) = 2.797$.
Hence $K = 0.100/2.797 = 3.575\%$, confirming the result. At this rate, both legs have equal present value.

Example 65: Valuing an existing swap

A 3y swap was entered 1y ago with swap rate $K = 3.00\%$ and notional $N = 100{,}000\,€$. The fixed-rate payer has 2y remaining. The current discount factors are:

Maturity	$\mathrm{DF}(T_i)$
1y	0.971
2y	0.943

What is the mark-to-market value of the swap for the fixed-rate payer?

Solution. **Value of the floating leg:** On a reset date, the FRN trades at par: $V_{\text{float}} = 100{,}000\,€$.

Value of the fixed leg: The remaining fixed payments are 3,000 € at 1y and 103,000 € (coupon plus principal) at 2y:

$$V_{\text{fixed}} = 3{,}000 \cdot 0.971 + 103{,}000 \cdot 0.943 = 2{,}913 + 97{,}129 = 100{,}042\,€.$$

Swap value (fixed-rate payer):

$$V_{\text{swap}} = V_{\text{float}} - V_{\text{fixed}} = 100{,}000 - 100{,}042 = -42\,€.$$

The swap has a small negative value for the fixed-rate payer. Since rates have risen (the current 2y swap rate implied by the discount factors is $\approx 2.98\%$), paying 3.00% fixed is slightly disadvantageous. The fixed-rate payer is "locked in" at a rate that is now marginally above the market.

12.4 Uses of interest rate swaps

Interest rate swaps are among the most widely traded derivatives in the world (with hundreds of trillions of dollars in notional outstanding). Their primary uses include:

- **Hedging:** A company with floating-rate debt can enter a swap (paying fixed, receiving floating) to convert its exposure to a fixed rate, eliminating uncertainty about future interest payments.
- **Asset–liability management:** Banks use swaps to match the interest rate profile of their assets (e.g. fixed-rate mortgages) with their liabilities (e.g. floating-rate deposits).
- **Speculation:** Traders who expect rates to rise can enter as fixed-rate payers (profiting as the swap's mark-to-market value increases); those expecting rates to fall can enter as floating-rate payers.
- **Curve trading:** Swaps at different maturities can be combined to express views on the *shape* of the yield curve (e.g. a **steepener** trade profits from a widening spread between long and short rates).

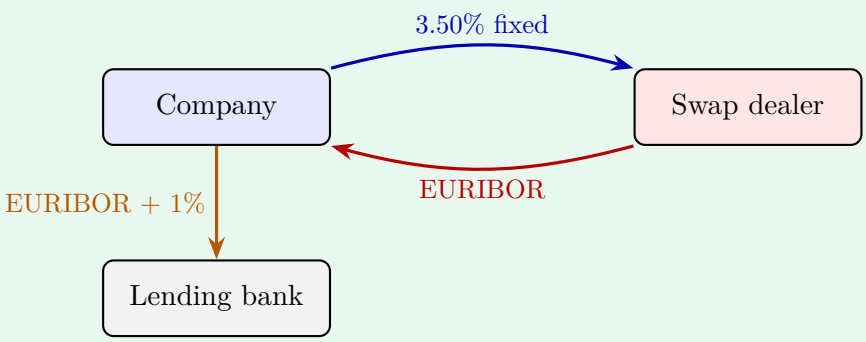
Example 66: Hedging floating-rate debt

A company has a 5y loan of $N = 10{,}000{,}000$ € at EURIBOR + 100 bps, with annual payments. To eliminate interest rate risk, the company enters a 5y interest rate swap paying a fixed rate of $K = 3.50\%$ and receiving EURIBOR.
Determine the company's net fixed borrowing cost and explain how the hedge works.

Solution. At each annual payment date, the company's cash flows are:

Cash flow	**Amount**
Loan interest (pay)	$-N \cdot (L_i + 1.00\%)$
Swap fixed leg (pay)	$-N \cdot 3.50\%$
Swap floating leg (receive)	$+N \cdot L_i$
Net payment	$-N \cdot (3.50\% + 1.00\%) = -N \cdot 4.50\%$

The EURIBOR components cancel: the company receives L_i from the swap and pays $L_i + 1.00\%$ on the loan. The net result is a **fixed** annual cost of 4.50% on 10,000,000 €, regardless of future EURIBOR movements.



The swap transforms the company's floating-rate obligation into a predictable fixed-rate cost. The trade-off: if EURIBOR falls below 3.50%, the company would have been better off without the swap, but it has purchased certainty.

Definition 81: Swap curve

The **swap curve** is the collection of par swap rates (Definition 80) across standard maturities (1y, 2y, 3y, 5y, 7y, 10y, 15y, 20y, 30y). Formally, it is the function $T \mapsto K(T)$ that maps each maturity T to the swap rate of an interest rate swap with that tenor.

The swap curve is one of the most important benchmark term structures in fixed income markets, for several reasons:

- **Liquidity:** Interest rate swaps are among the most actively traded financial instruments in the world, with notional amounts outstanding in the hundreds of trillions of euros. This deep liquidity ensures that swap rates are reliable and up-to-date reflections of market conditions.
- **Maturity coverage:** Swaps are quoted for a wide range of maturities, from 1y to 30y (and beyond), providing a complete term structure. Government bond markets, by contrast, may have limited liquidity at certain maturities.
- **Curve construction:** The swap curve serves as a primary input for bootstrapping the

spot curve and forward curve (Section 7.2). For maturities beyond 2y, swap rates are often the most reliable market quotes available.

Remark (Swap curve vs. OIS curve). The swap curve and the OIS curve (Definition 40) are closely related but conceptually distinct. In the dual-curve framework (Section 7.5), the OIS curve serves as the *discount curve*, while the swap curve — built from swaps referencing a term rate such as EURIBOR — reflects the *estimation curve* for that index. The spread between swap rates and OIS rates at the same maturity captures the credit and liquidity premium embedded in interbank lending rates relative to overnight risk-free rates.

CHAPTER 13

FX Forwards and Cross-Currency Swaps

So far, all instruments in this booklet have been denominated in a single currency. In practice, global capital markets involve multiple currencies. This chapter introduces two instruments that bridge different currency markets: the FX forward and the cross-currency swap. Both are priced using no-arbitrage arguments (Definition 11) that relate interest rates across currencies.

13.1 FX spot and forward rates

Definition 82: FX spot rate

The **FX spot rate** S is the current market price of one unit of a **base currency** expressed in units of a **quote currency**.

For example, if EUR/USD = 1.10, one euro costs USD 1.10.

Definition 83: FX forward rate

The **FX forward rate** $F(T)$ is the exchange rate agreed today for a currency transaction that will take place at a future date T. It is not a forecast of where the spot rate will be at T; rather, it is the unique rate that prevents arbitrage between the two currencies' interest rates.

13.2 Covered interest rate parity (CIP)

The relationship between the FX forward rate and the two currencies' interest rates is one of the most fundamental results in international finance.

Covered interest rate parity

The FX forward rate $F(T)$ is determined by:

$$F(T) = S \cdot \frac{\mathrm{DF}_{\text{quote}}(T)}{\mathrm{DF}_{\text{base}}(T)},$$

where S is the spot rate, $\mathrm{DF}_{\text{base}}(T)$ is the discount factor in the base currency, and $\mathrm{DF}_{\text{quote}}(T)$ is the discount factor in the quote currency.
Equivalently, with simple interest rates r_b (base) and r_q (quote):

$$F(T) = S \cdot \frac{1 + r_q \cdot T}{1 + r_b \cdot T}.$$

CIP follows from a no-arbitrage argument. Consider two strategies for converting 1 unit of the base currency into the quote currency at time T:

Strategy A (direct):

1. Invest 1 base-currency unit at rate r_b for T years.
2. At time T, receive $1 + r_b \cdot T$ in the base currency.
3. Convert at the forward rate $F(T)$: receive $(1 + r_b \cdot T) \cdot F(T)$ in the quote currency.

Strategy B (synthetic):

1. Convert 1 base-currency unit to S quote-currency units at the spot rate today.
2. Invest S at rate r_q for T years.
3. At time T, receive $S \cdot (1 + r_q \cdot T)$ in the quote currency.

No-arbitrage requires both strategies to yield the same terminal amount:

$$(1 + r_b \cdot T) \cdot F(T) = S \cdot (1 + r_q \cdot T).$$

Solving for $F(T)$:

$$F(T) = S \cdot \frac{1 + r_q \cdot T}{1 + r_b \cdot T}.$$

Using discount factors, $1 + r \cdot T = 1/\mathrm{DF}(T)$ (for simple interest), which gives the equivalent form $F(T) = S \cdot \mathrm{DF}_{\text{quote}}(T)/\mathrm{DF}_{\text{base}}(T)$ after inversion.

Definition 84: Forward premium and discount

The **forward premium** (or **forward discount**) is the difference between the forward rate and the spot rate:

$$\text{Forward points} = F(T) - S.$$

- If $F(T) > S$ (**forward premium**): the base currency is *more expensive* in the forward market. This occurs when $r_q > r_b$ (the quote currency has higher interest rates — investors demand compensation for holding the lower-yielding base currency).
- If $F(T) < S$ (**forward discount**): the base currency is *cheaper* in the forward market. This occurs when $r_q < r_b$.

Example 67: Computing an FX forward rate

The EUR/USD spot rate is $S = 1.1000$. The 1y euro discount factor is $\mathrm{DF}_{\text{EUR}}(1) = 0.965$ and the 1y dollar discount factor is $\mathrm{DF}_{\text{USD}}(1) = 0.952$. Compute the 1y EUR/USD forward rate and interpret the result.

Solution. Using CIP with EUR as the base currency and USD as the quote currency:

$$F(1) = S \cdot \frac{\mathrm{DF}_{\text{USD}}(1)}{\mathrm{DF}_{\text{EUR}}(1)} = 1.1000 \cdot \frac{0.952}{0.965} = 1.1000 \cdot 0.9865 = 1.0852.$$

Interpretation: The 1y EUR/USD forward rate (1.0852) is below the spot rate

(1.1000), meaning the euro trades at a forward discount of $1.1000 - 1.0852 = 0.0148$ (or 148 forward points).
This makes economic sense: since USD interest rates (5.04%) exceed EUR rates (3.63%), the euro must trade cheaper in the forward market to compensate investors for holding the lower-yielding currency.

Example 68: FX forward arbitrage

Using the data from Example 67, suppose a dealer quotes a 1y EUR/USD forward at $F^{\text{mkt}} = 1.0900$ (above the fair value of 1.0852). Construct an arbitrage strategy.

Solution. The market forward $F^{\text{mkt}} = 1.0900$ is above the fair forward $F(1) = 1.0852$. This means EUR is overpriced in the forward market: we can buy EUR forward at 1.0900 when it should cost only 1.0852. The arbitrage is to buy EUR forward and replicate the same position synthetically at a lower cost.

Arbitrage strategy (start with zero capital):

At $t = 0$:

1. Borrow $\text{DF}_{\text{EUR}}(1) = 0.965$ € today. In 1y, repay 1 €.
2. Convert 0.965 € to $0.965 \cdot 1.10 = 1.0615$ \$ at the spot rate.
3. Invest 1.0615 \$ at the USD rate. In 1y, receive $1.0615/0.952 = 1.1150$ \$.
4. Buy 1 € forward at $F^{\text{mkt}} = 1.0900$. In 1y, pay 1.0900 \$ and receive 1 €.

At $t = 1$y:

- Receive 1.1150 \$ from the USD investment.
- Pay 1.0900 \$ and receive 1 € under the forward contract.
- Use the 1 € to repay the EUR loan.
- Net profit: $1.1150 - 1.0900 = 0.0250$ \$.

Result: Riskless profit of 0.0250 \$ per euro of notional, with zero initial investment. Scaling to $N = 1{,}000{,}000$ € of notional yields a profit of 25,000 \$.

Economic intuition: The synthetic cost of obtaining 1 € in 1y is 1.0852 \$ (borrow EUR, convert to USD, invest). The market allows us to lock in 1 € for 1.0900 \$. By doing the synthetic strategy and buying EUR forward at the inflated rate, we profit from the difference: $1.0900 - 1.0852 = 0.0048$ \$ per euro of forward notional, or equivalently 0.0250 \$ per euro of spot notional after accounting for interest rates.

13.3 Cross-currency swaps

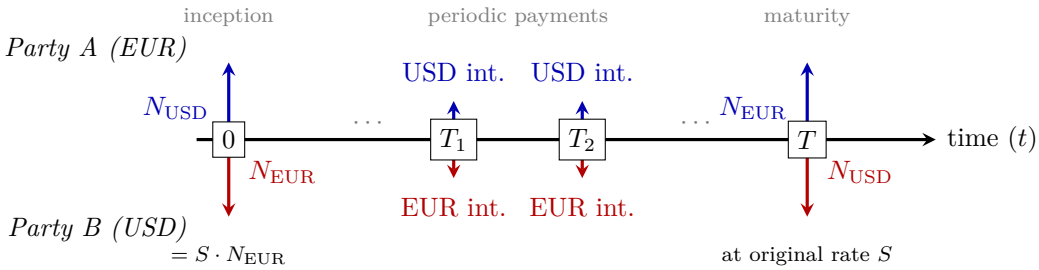
Definition 85: Cross-currency swap (XCCY swap)

A **cross-currency swap** (XCCY swap, also called a **cross-currency basis swap**) is a derivative contract in which two parties exchange:

1. **Principal amounts** in two different currencies at inception and at maturity (at the initial spot rate S).
2. **Periodic interest payments**: one party pays interest in one currency; the other pays interest in the other currency.

Unlike a single-currency IRS (Definition 79), where the notional is purely notional (never exchanged), a cross-currency swap involves **actual exchange of principal** at inception and

maturity. This principal exchange is essential because the two currencies may appreciate or depreciate against each other during the swap's life.



13.3.1 Types of cross-currency swaps

- **Fixed–fixed:** One party pays a fixed rate in currency A; the other pays a fixed rate in currency B.
- **Fixed–floating:** One party pays fixed in one currency; the other pays floating (reference rate) in the other.
- **Floating–floating (basis swap):** Both parties pay floating rates in their respective currencies. The **cross-currency basis spread** b is added to one leg to reflect supply–demand imbalances for funding in different currencies.

13.3.2 Pricing a cross-currency swap

A cross-currency swap is fair (zero NPV at inception) when the present value of cash flows in each currency, converted at the spot rate, is equal. Using CIP (Covered Interest Rate Parity):

Cross-currency swap pricing

For a fixed–fixed XCCY swap with notional N_b in the base currency and $N_q = S \cdot N_b$ in the quote currency, annual payments, and swap rates K_b (base) and K_q (quote), the no-arbitrage condition is:

$$N_b \left(K_b \sum_{i=1}^{n} \mathrm{DF}_b(T_i) + \mathrm{DF}_b(T_n) \right) = S^{-1} \cdot N_q \left(K_q \sum_{i=1}^{n} \mathrm{DF}_q(T_i) + \mathrm{DF}_q(T_n) \right),$$

where DF_b and DF_q are the discount factors in the base and quote currencies, respectively.

Example 69: Pricing a EUR/USD cross-currency swap

A 2y fixed–fixed cross-currency swap has the following terms:

- EUR notional: $N_{\text{EUR}} = 10{,}000{,}000$ €.
- USD notional: $N_{\text{USD}} = S \cdot N_{\text{EUR}} = 1.10 \cdot 10{,}000{,}000 = 11{,}000{,}000$ \$.
- Annual payments in both currencies.

The discount factors are:

Maturity	$\mathrm{DF}_{\mathbf{EUR}}$	$\mathrm{DF}_{\mathbf{USD}}$
1y	0.965	0.952
2y	0.932	0.907

Compute the fair EUR and USD swap rates.

Solution. **EUR swap rate** (using the swap rate formula, Definition 80):

$$K_{\text{EUR}} = \frac{1 - \text{DF}_{\text{EUR}}(2)}{\text{DF}_{\text{EUR}}(1) + \text{DF}_{\text{EUR}}(2)} = \frac{1 - 0.932}{0.965 + 0.932} = \frac{0.068}{1.897} = 3.585\%.$$

USD swap rate:

$$K_{\text{USD}} = \frac{1 - \text{DF}_{\text{USD}}(2)}{\text{DF}_{\text{USD}}(1) + \text{DF}_{\text{USD}}(2)} = \frac{1 - 0.907}{0.952 + 0.907} = \frac{0.093}{1.859} = 5.003\%.$$

Verification. EUR leg present value:

$$\begin{aligned} &10{,}000{,}000 \cdot 3.585\% \cdot (0.965 + 0.932) + 10{,}000{,}000 \cdot 0.932 \\ &\quad = 680{,}055 + 9{,}320{,}000 = 10{,}000{,}055\,€ \approx 10{,}000{,}000\,€. \end{aligned}$$

USD leg present value:

$$\begin{aligned} &11{,}000{,}000 \cdot 5.003\% \cdot (0.952 + 0.907) + 11{,}000{,}000 \cdot 0.907 \\ &\quad = 1{,}022{,}863 + 9{,}977{,}000 = 10{,}999{,}863\,\$ \approx 11{,}000{,}000\,\$. \end{aligned}$$

Converting USD to EUR at spot: $11{,}000{,}000/1.10 = 10{,}000{,}000\,€$. Both legs have the same value when converted to a common currency, confirming the swap is fairly priced.

13.4 The cross-currency basis spread

In a floating–floating cross-currency swap, CIP implies that both parties should simply exchange their respective reference rates (e.g. EURIBOR vs. SOFR) with no additional spread. In practice, however, a cross-currency basis spread b is added to one leg.

Definition 86: Cross-currency basis spread

The **cross-currency basis spread** b is the additional spread applied to one leg of a floating–floating cross-currency swap to make it trade at par. It reflects:

- **Demand imbalances:** If many institutions need USD funding but have EUR, they bid up the cost of swapping EUR for USD, creating a negative basis (the EUR leg pays EURIBOR $- b$, where $b > 0$).
- **Credit risk differences:** Counterparty risk may differ across currencies.
- **Regulatory constraints:** Capital requirements and liquidity rules may make certain currencies more expensive to hold.

In practice, CIP is violated due to balance sheet costs, credit risk, regulatory constraints, and market segmentation. These frictions prevent perfect arbitrage, making the basis spread a measure of the cost of obtaining foreign currency funding through swaps. During the 2008 financial crisis and the 2020 COVID crisis, the EUR/USD basis spread widened dramatically (reaching -80 bps or more), reflecting severe dollar funding stress. Japanese banks routinely face a negative USD/JPY basis due to structural dollar demand.

CHAPTER 14

Interest Rate Options

All the instruments studied so far — bonds, FRAs, swaps — involve *obligations*: both parties must honour the contract regardless of market movements. An option introduces *choice*: the holder has the right, but not the obligation, to enter into a transaction at a predetermined rate. This asymmetry is valuable and must be paid for upfront (the premium).
This chapter develops the theory and practice of interest rate options. We begin by establishing the fundamental payoff structures that distinguish options from symmetric contracts, then study the main interest rate option products — caps, floors, and swaptions — before turning to their valuation. We examine three standard pricing models (Black, Black–Scholes, and Bachelier), discuss the market reality of non-constant volatility through the volatility surface, and conclude with embedded options in mortgage products.

14.1 Option payoffs

Options were introduced in Definition 66 as contracts that grant a right without imposing an obligation. This asymmetry gives options a distinctive payoff structure that sets them apart from forwards, futures, and swaps. Before studying specific interest rate option products, we must understand the general mechanics of option payoffs.

Definition 87: Strike price and payoff

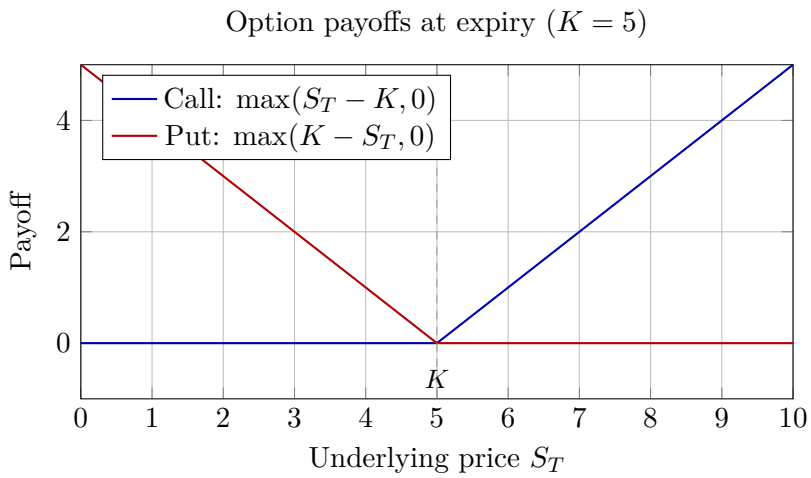
The **strike price** K (or **strike rate**, for interest rate options) is the predetermined price at which the option holder has the right to buy (call option) or sell (put option) the underlying asset if they choose to exercise the option.
At maturity T, let S_T denote the market value of the underlying asset. The option holder will compare S_T with the strike price K to decide whether to exercise.

- A **call option** gives the right to *buy* the underlying at price K. The holder exercises only if $S_T > K$ (the market price exceeds the strike), earning a profit of $S_T - K$. If $S_T \leq K$, the option expires worthless.
- A **put option** gives the right to *sell* the underlying at price K. The holder exercises only if $S_T < K$ (the strike exceeds the market price), earning a profit of $K - S_T$. If $S_T \geq K$, the option expires worthless.

The payoffs of the two basic option types are:

$$\textbf{Call payoff} = \max(S_T - K,\ 0),$$
$$\textbf{Put payoff} = \max(K - S_T,\ 0).$$

The $\max(\cdot, 0)$ operator captures the fact that the holder will never exercise if doing so would result in a loss, since exercise is a right, not an obligation. The payoff diagrams below illustrate the characteristic "hockey stick" shape of option payoffs. The call payoff is zero for all values of S_T below the strike and then rises linearly; the put payoff mirrors this on the downside.



Definition 88: Moneyness

An option's **moneyness** describes the relationship between the current underlying price S and the strike K:

- **In the money (ITM):** The option would have positive payoff if exercised now. For a call: $S > K$; for a put: $S < K$.
- **At the money (ATM):** $S \approx K$. The option is on the boundary.
- **Out of the money (OTM):** The option would have zero payoff if exercised now. For a call: $S < K$; for a put: $S > K$.

Definition 89: Intrinsic value and time value

The price (premium) of an option can be decomposed into two components:

- **Intrinsic value:** The payoff the option would have if exercised immediately: $\max(S - K, 0)$ for a call, $\max(K - S, 0)$ for a put.
- **Time value:** The excess of the option premium over intrinsic value. It reflects the possibility that the underlying may move favourably before expiry. Time value is always non-negative and decreases as expiry approaches (**time decay**).

$$\text{Option premium} = \text{Intrinsic value} + \text{Time value}.$$

The five main factors that determine the price of a European option are:

- **Underlying price S:** Higher S increases call values and decreases put values.
- **Strike price K:** Higher K decreases call values and increases put values.
- **Time to expiry T:** More time generally increases option values (more opportunity for favourable moves).
- **Volatility σ:** Higher volatility increases both call and put values (greater chance of large moves). Volatility (Definition 61) measures how much the underlying fluctuates over time.
- **Interest rate r:** Higher rates increase call values and decrease put values (through the effect on the present value of the strike).

We will quantify these sensitivities precisely when we study option pricing models in Section 14.4.

14.2 Caps, floors, and collars

Having established the general payoff structure of options, we now turn to the most widely traded interest rate option products. An interest rate cap is conceptually a series of call options on an interest rate; a floor is a series of puts.

14.2.1 Interest rate caps

Definition 90: Interest rate cap

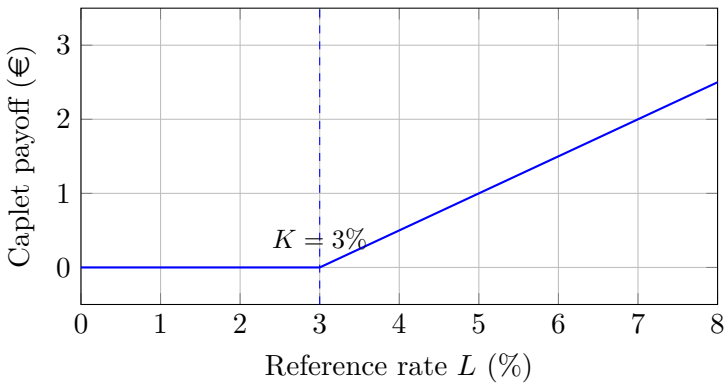
An **interest rate cap** is a series of European call options (**caplets**) on a reference interest rate (e.g. EURIBOR). The cap provides protection against rising interest rates. For each caplet covering the period $[T_{i-1}, T_i]$, the payoff at T_i is:

$$\text{Caplet payoff}_i = N \cdot \max(L_i - K, 0) \cdot \Delta t_i,$$

where N is the notional, L_i is the reference rate observed at T_{i-1}, K is the **strike rate**, and Δt_i is the year fraction.

The cap holder receives a payment whenever the reference rate exceeds the strike. It acts as "insurance" against high rates. The payoff structure $\max(L_i - K, 0)$ is precisely the call option payoff applied to interest rates rather than asset prices. When $L_i > K$, the option is in the money and pays out; when $L_i \leq K$, it is out of the money and expires worthless.

Caplet payoff ($K = 3\%$, $\Delta t = 0.5$, $N = 100$)



14.2.2 Interest rate floors

Definition 91: Interest rate floor

An **interest rate floor** is a series of European put options (**floorlets**) on a reference rate. It provides protection against falling interest rates.
For each floorlet covering the period $[T_{i-1}, T_i]$, the payoff at T_i is:

$$\text{Floorlet payoff}_i = N \cdot \max(K - L_i, 0) \cdot \Delta t_i.$$

A floor holder receives a payment whenever the reference rate falls below the strike K. This is the put option payoff on interest rates: when $L_i < K$, the floor is in the money; when $L_i \geq K$, it is out of the money.

14.2.3 Collars

Definition 92: Interest rate collar

An **interest rate collar** is the combination of a long cap (strike K_{cap}) and a short floor (strike K_{floor}), with $K_{\text{floor}} < K_{\text{cap}}$. It limits the effective borrowing rate to the band $[K_{\text{floor}}, K_{\text{cap}}]$.

A collar is often structured as a **zero-cost collar**: the premium received from selling the floor offsets the premium paid for the cap, so no upfront payment is required. The trade-off is that the borrower gives up the benefit of rates falling below K_{floor}.

Example 70: Cap payoff

A borrower holds a 1y cap with strike $K = 3.00\%$, notional $N = 1{,}000{,}000$ €, and quarterly caplets ($\Delta t = 0.25$). The realised 3m EURIBOR rates are:

Period	EURIBOR L_i	Caplet payoff
Q1	2.50%	$\max(2.50 - 3.00, 0) \cdot 0.25 \cdot 1{,}000{,}000 = 0$ €
Q2	3.20%	$\max(3.20 - 3.00, 0) \cdot 0.25 \cdot 1{,}000{,}000 = 500$ €
Q3	3.80%	$\max(3.80 - 3.00, 0) \cdot 0.25 \cdot 1{,}000{,}000 = 2{,}000$ €
Q4	2.90%	$\max(2.90 - 3.00, 0) \cdot 0.25 \cdot 1{,}000{,}000 = 0$ €
	Total	2,500 €

The cap paid out 2,500 € over the year, compensating the borrower for the two quarters where EURIBOR exceeded 3%. In Q1 and Q4, the caplets were out of the money and had zero intrinsic value. In Q2 and Q3, the caplets were in the money and delivered positive payoffs. This payment offsets the higher borrowing cost, effectively capping the borrower's rate at 3% (plus the upfront premium paid for the cap).

14.3 Swaptions

Definition 93: Swaption

A **swaption** (swap option) gives the holder the right, but not the obligation, to enter into an interest rate swap (Definition 79) at a predetermined swap rate K on a future date.

- A **payer swaption** gives the right to *pay* fixed and receive floating.
- A **receiver swaption** gives the right to *receive* fixed and pay floating.

A swaption is specified by its **expiry** T_0 (when the option expires) and the **tenor** T of the underlying swap (e.g. a "1y into 5y" swaption expires in 1y and gives the right to enter a 5y swap).

Payoff at expiry T_0:
For a **payer swaption** with strike K, the holder exercises if the prevailing swap rate $S(T_0)$ exceeds K (the option is in the money):

$$\text{Payer payoff} = N \cdot \max(S(T_0) - K, 0) \cdot A(T_0),$$

where $A(T_0) = \sum_{i=1}^{n} \Delta t_i \cdot \text{DF}(T_i)$ is the **annuity factor** evaluated at expiry (the present value of receiving 1 per period over the swap's life).

For a **receiver swaption**:

$$\text{Receiver payoff} = N \cdot \max\big(K - S(T_0), 0\big) \cdot A(T_0).$$

The payoff structure of a payer swaption is that of a call option: it benefits from the swap rate rising above the strike. The receiver swaption is a put option: it benefits from the swap rate falling below the strike.

Remark (Put–call parity for swaptions). A payer swaption and a receiver swaption with the same strike K, expiry, and tenor satisfy:

$$\text{Payer} - \text{Receiver} = \text{PV}(\text{forward swap at rate } K).$$

If K equals the current forward swap rate, the payer and receiver swaptions have equal value (analogous to at-the-money put–call parity for equity options).

14.4 Option pricing models

To price caps, floors, and swaptions, we need a model that links observable market quantities (forward rates, discount factors) to option values. The standard approach is to assume the underlying rate follows a stochastic process and derive the option price under the no-arbitrage principle. We examine three models, each with different assumptions and use cases.

14.4.1 The Black–Scholes framework

Before studying the specific models used for interest rate options, we establish the conceptual foundation. The Black–Scholes framework, developed by Fischer Black, Myron Scholes, and Robert Merton in 1973, revolutionised derivatives pricing by showing that options can be valued without needing to predict the future direction of the underlying asset.

Remark (Key assumptions of Black–Scholes). The model relies on several simplifying assumptions.

- **Geometric Brownian motion:** The underlying price S follows a geometric Brownian motion.
- **Log-normal prices:** The underlying price S can never become negative, and its logarithm $\ln S$ is normally distributed.
- **Constant volatility:** σ is known and constant over the life of the option. In reality, volatility varies with both strike and maturity, as we discuss in Section 14.5.
- **Frictionless markets:** No transaction costs, no bid–ask spreads, continuous trading is possible, and unlimited borrowing and lending at the risk-free rate.
- **No arbitrage:** The no-arbitrage principle is the economic engine of the model — it forces the option price to equal the cost of replicating its payoff through continuous delta hedging.

Black–Scholes model

The **Black–Scholes model** provides a closed-form formula for the call and put price of a European option on an asset whose price satisfies the above assumptions:

$$C = S_0 \cdot \Phi(d_1) - K \cdot e^{-rT} \cdot \Phi(d_2),$$
$$P = K \cdot e^{-rT} \cdot \Phi(-d_2) - S_0 \cdot \Phi(-d_1),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

S_0 is the current asset price, K is the strike price, r is the continuously compounded risk-free rate, T is the time to expiry, and Φ is the cumulative standard normal distribution.

Example 71: Black–Scholes option pricing

A European call option has the following parameters: current underlying price $S_0 = 10\,000$ €, strike $K = 10\,500$ €, risk-free rate $r = 2\%$ (continuous compounding), time to expiry $T = 1\text{y}$, volatility $\sigma = 25\%$. Compute the call premium.

Solution. Step 1 — Compute d_1 and d_2:

$$\begin{aligned} d_1 &= \frac{\ln(10\,000/10\,500) + (0.02 + \frac{1}{2} \cdot 0.0625) \cdot 1}{0.25 \cdot 1} \\ &= \frac{-0.04879 + 0.05125}{0.25} = \frac{0.00246}{0.25} = 0.0098, \\ d_2 &= 0.0098 - 0.25 = -0.2402. \end{aligned}$$

Step 2 — Cumulative normal values:

$$\Phi(0.0098) = 0.5039, \qquad \Phi(-0.2402) = 0.4051.$$

Step 3 — Call price:

$$\begin{aligned} C &= 10\,000 \cdot 0.5039 - 10\,500 \cdot e^{-0.02} \cdot 0.4051 \\ &= 5\,039 - 10\,500 \cdot 0.9802 \cdot 0.4051 \\ &= 5\,039 - 4\,169 = 870\,€. \end{aligned}$$

The call costs 870 €. Since the option is out of the money ($S_0 < K$), the intrinsic value is zero; the entire premium is time value, reflecting the possibility that S may exceed 10 500 € before expiry.

14.4.2 The Black model for interest rate options

The Black–Scholes model prices options on assets (stocks, commodities). For interest rate options, we need to adapt the framework to work with forward rates and swap rates. This adaptation is known as the **Black model** or **Black-76**.

The key insight is that under the appropriate forward measure, a forward rate (or swap rate) behaves like an asset price: it is a martingale. This allows us to apply the Black–Scholes machinery directly to the forward rate, replacing the spot price S_0 with the forward rate F.

Black model for caplets

Under the Black model, the price of a caplet on the forward rate $F = f(T_{i-1}, T_i)$ with strike K, notional N, and year fraction Δt is:

$$\text{Caplet} = N \cdot \Delta t \cdot \text{DF}(T_i) \cdot \big[F \cdot \Phi(d_1) - K \cdot \Phi(d_2)\big],$$

where

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T_{i-1}}{\sigma\sqrt{T_{i-1}}}, \qquad d_2 = d_1 - \sigma\sqrt{T_{i-1}},$$

σ is the implied volatility of the forward rate, Φ is the cumulative standard normal distribution, and T_{i-1} is the time to the caplet's fixing date.

Black model for swaptions

Under the Black model, the price of a payer swaption with strike K, notional N, expiry T_0, and underlying swap annuity factor A is:

$$\text{Payer swaption} = N \cdot A \cdot \big[S_0 \cdot \Phi(d_1) - K \cdot \Phi(d_2)\big],$$

where S_0 is the current forward swap rate, $A = \sum_{i=1}^{n} \Delta t_i \cdot \text{DF}(T_i)$ is the annuity factor, and

$$d_1 = \frac{\ln(S_0/K) + \frac{1}{2}\sigma^2 T_0}{\sigma\sqrt{T_0}}, \qquad d_2 = d_1 - \sigma\sqrt{T_0},$$

with σ being the swaption's implied volatility.

Remark (Relation to Black–Scholes). The Black model is obtained from Black–Scholes by the substitution $S_0 e^{rT} \to F$ (replacing the spot price growing at the risk-free rate with the forward price) and factoring out the discount factor $e^{-rT} = \text{DF}(T)$. Specifically:

$$C_{\text{BS}} = S_0 \cdot \Phi(d_1) - K \cdot e^{-rT} \cdot \Phi(d_2) \quad \Rightarrow \quad C_{\text{Black}} = e^{-rT}\big[F \cdot \Phi(d_1) - K \cdot \Phi(d_2)\big].$$

This is why Black-76 is the natural choice for interest rate options: it works directly with forward rates, which are the natural underlying for caps, floors, and swaptions.

Remark (Assumptions of the Black model). The Black model assumes:

- The forward rate (or swap rate) follows a **geometric Brownian motion** under the appropriate forward measure (the T_i-forward measure for caplets, the annuity measure for swaptions).
- Volatility σ is **constant**. In practice, implied volatility varies with strike (smile/skew) and maturity (term structure); see Section 14.5.
- Interest rates are **positive**. The log-normal assumption means rates cannot go negative, which became problematic during the negative-rate environment post-2014.

Example 72: Pricing a caplet with the Black model

Price a caplet with the following parameters:

- Notional: $N = 1{,}000{,}000\,€$.
- Forward rate: $F = 3.50\%$.
- Strike: $K = 3.00\%$.
- Accrual period: $\Delta t = 0.25$ (quarterly).
- Time to fixing: $T = 0.5\mathtt{y}$.
- Implied volatility: $\sigma = 20\%$.
- Discount factor to payment date: $\mathrm{DF}(0.75) = 0.974$.

Solution. Step 1 — Compute d_1 and d_2:

$$\begin{aligned} d_1 &= \frac{\ln(0.035/0.03) + \frac{1}{2} \cdot 0.04 \cdot 0.5}{0.20 \cdot \sqrt{0.5}} \\ &= \frac{\ln(1.1667) + 0.01}{0.1414} = \frac{0.1542 + 0.01}{0.1414} \\ &= \frac{0.1642}{0.1414} = 1.161, \\ d_2 &= 1.161 - 0.1414 = 1.020. \end{aligned}$$

Step 2 — Look up cumulative normal values:

$$\Phi(1.161) = 0.877, \qquad \Phi(1.020) = 0.846.$$

Step 3 — Caplet price:

$$\begin{aligned} \text{Caplet} &= 1{,}000{,}000 \cdot 0.25 \cdot 0.974 \cdot \big[0.035 \cdot 0.877 - 0.03 \cdot 0.846\big] \\ &= 243{,}500 \cdot \big[0.03070 - 0.02538\big] \\ &= 243{,}500 \cdot 0.00532 \\ &= 1{,}295\,€. \end{aligned}$$

The caplet costs 1,295 € upfront. This is the premium for the right to receive a payment if the 3m EURIBOR exceeds 3% in six months. Since the forward rate $F = 3.50\%$ exceeds the strike $K = 3.00\%$, the caplet is currently in the money. Its intrinsic value is approximately $(3.50\% - 3.00\%) \cdot 0.25 \cdot 1{,}000{,}000 \cdot 0.974 = 1{,}218\,€$; the remaining 77 € is time value, reflecting the possibility of further favourable rate movements before expiry.

14.4.3 The Bachelier (Normal) model

The Black model's log-normal assumption implies that interest rates can never become negative. When rates turned negative in the eurozone and Japan (starting around 2014), this assumption became untenable. The Bachelier model provides an alternative that handles negative rates naturally.

Bachelier model (Normal model)

The **Bachelier model** (also called the **normal model**) assumes the underlying F follows an **arithmetic Brownian motion**. Under this assumption, F is *normally*

distributed and can take negative values.
The European call and put prices are:

$$C = \mathrm{DF}(T) \cdot \left[(F - K) \cdot \Phi(d) + \sigma_n \sqrt{T} \cdot \phi(d)\right],$$
$$P = \mathrm{DF}(T) \cdot \left[(K - F) \cdot \Phi(-d) + \sigma_n \sqrt{T} \cdot \phi(d)\right],$$

where

$$d = \frac{F - K}{\sigma_n \sqrt{T}},$$

σ_n is the normal volatility (expressed in absolute terms, e.g. basis points per annum), Φ and ϕ are the cumulative and density functions of the standard normal distribution, and $\mathrm{DF}(T)$ is the discount factor to expiry.

	Black (log-normal)	**Bachelier (normal)**
Underlying process	Geometric Brownian motion	Arithmetic Brownian motion
Distribution of F	Log-normal	Normal
Negative rates	Not possible	Possible
Volatility units	Relative (%)	Absolute (bps/year)
Use case	Positive rate environments	Any environment
At-the-money relation	$\sigma_n \approx \sigma \cdot F$	$\sigma \approx \sigma_n / F$

Example 73: Pricing a caplet with the Bachelier model

Price the same caplet as in Example 72, but using the Bachelier model with normal volatility $\sigma_n = 0.70\%$ (70 bps per annum).
Parameters: $N = 1{,}000{,}000$ €, $F = 3.50\%$, $K = 3.00\%$, $\Delta t = 0.25$, $T = 0.5$y, $\mathrm{DF}(0.75) = 0.974$.

Solution. Step 1 — Compute d:

$$d = \frac{F - K}{\sigma_n \sqrt{T}} = \frac{0.0350 - 0.0300}{0.0070 \cdot \sqrt{0.5}} = \frac{0.0050}{0.00495} = 1.010.$$

Step 2 — Normal distribution values:

$$\Phi(1.010) = 0.8438, \qquad \phi(1.010) = 0.2396.$$

Step 3 — Undiscounted call price (on the rate):

$$\begin{aligned} C_{\text{rate}} &= (F - K) \cdot \Phi(d) + \sigma_n \sqrt{T} \cdot \phi(d) \\ &= 0.0050 \cdot 0.8438 + 0.00495 \cdot 0.2396 \\ &= 0.004219 + 0.001186 = 0.005405. \end{aligned}$$

Step 4 — Caplet price:

$$\text{Caplet} = N \cdot \Delta t \cdot \mathrm{DF}(T_i) \cdot C_{\text{rate}} = 1{,}000{,}000 \cdot 0.25 \cdot 0.974 \cdot 0.005405 = 1{,}316\,€.$$

The Bachelier price (1,316 €) is close to the Black price (1,295 € from Example 72). The small difference arises because the two models make different distributional assumptions; the normal volatility $\sigma_n = 0.70\%$ was calibrated to be approximately consistent with the log-normal volatility $\sigma = 20\%$ at the forward rate of 3.50% (using the ATM relation $\sigma_n \approx \sigma \cdot F = 0.20 \cdot 0.035 = 0.70\%$).

14.5 Implied volatility and the volatility surface

The pricing models presented above all require a volatility parameter σ (or σ_n for Bachelier). But volatility is not directly observable in the market — instead, we observe option prices. The market convention is to quote options not in terms of price, but in terms of implied volatility: the volatility that, when inserted into the pricing formula, reproduces the observed market price.

14.5.1 Implied volatility

Definition 94: Implied volatility

The **implied volatility** σ_{imp} of an interest rate option is the volatility parameter that, when inserted into the Black (or Bachelier) model, reproduces the option's observed market price.

Implied volatility is not directly observable; it is *backed out* from market prices by numerically inverting the pricing formula. It represents the market's consensus view of future rate uncertainty. Importantly, implied volatility is a *quote convention* rather than a model parameter: it allows market participants to communicate option prices in a standardised way that is independent of notional, strike, and maturity.

Remark (Normal vs. log-normal volatility). There are two market conventions for quoting implied volatility:

- **Log-normal (Black) volatility:** The standard convention when rates are positive. The forward rate is assumed to follow a geometric Brownian motion.
- **Normal (Bachelier) volatility:** The forward rate follows an arithmetic Brownian motion, where absolute changes are normally distributed. This convention handles negative rates naturally and has become increasingly popular since 2014.

The two can be converted approximately using $\sigma_n \approx \sigma \cdot F$ (at-the-money). For example, a log-normal vol of 20% at a forward rate of 3% corresponds to a normal vol of approximately 0.60% (60 bps).

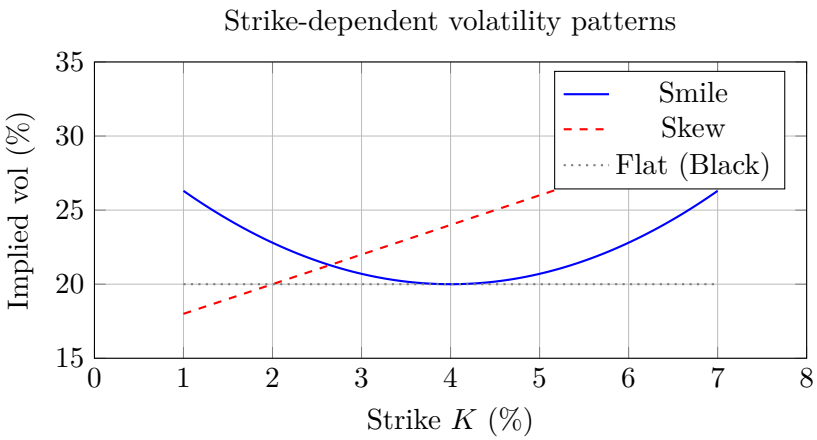
14.5.2 The volatility surface: strike dimension

If the Black model's assumption of constant volatility were correct, all options on the same underlying with the same expiry would trade at the same implied volatility, regardless of strike. In practice, this is not the case. Implied volatility varies systematically with strike, creating patterns known as the smile or skew.

Definition 95: Volatility smile and skew

For fixed expiry T_0, implied volatility $\sigma_{\text{imp}}(K)$ varies with strike K:

- **Smile** (U-shaped): Both out-of-the-money puts and calls have higher implied vol than at-the-money options. This reflects **fat tails** — the market prices extreme moves as more likely than the log-normal model predicts.
- **Skew** (monotonic): Implied vol increases or decreases consistently with strike. Common in interest rate caps: high strikes often have higher vol (upward skew), reflecting asymmetric rate uncertainty.



Example 74: Reading the smile

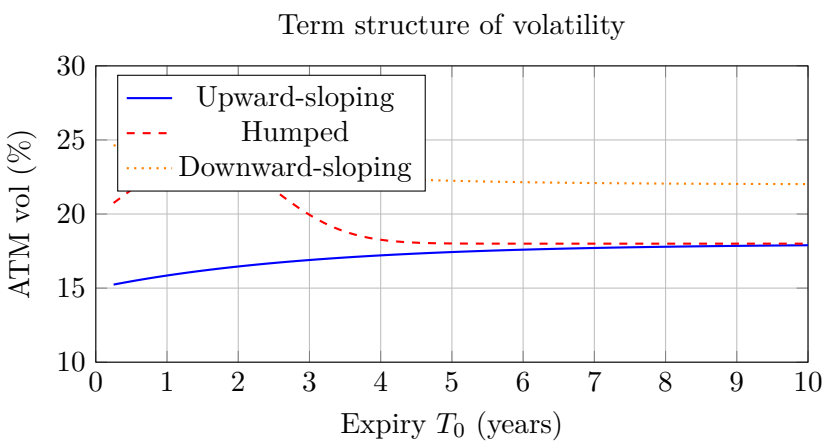
Consider 1y caplets on 3m EURIBOR with forward rate $F = 3.00\%$. The market quotes the following implied volatilities:

Strike K	**Moneyness**	$\sigma_{\mathbf{imp}}$
2.00%	−100 bps (OTM)	22%
2.50%	−50 bps (OTM)	20%
3.00%	ATM	19%
3.50%	+50 bps (ITM)	21%
4.00%	+100 bps (ITM)	24%

To price a caplet with strike $K = 2.00\%$, we use $\sigma = 22\%$ in the Black formula, *not* the at-the-money volatility of 19%. Using the wrong volatility would misprice the option. The smile tells us that the market expects extreme rate movements (both up and down) to be more likely than the log-normal model predicts.

14.5.3 The volatility surface: time dimension

Implied volatility also varies with the option's expiry. For a fixed strike (typically at-the-money), longer-dated options often have different implied volatilities than short-dated ones.



Common patterns are:

- **Upward-sloping:** Long-dated options have higher vol, reflecting greater uncertainty over longer horizons.

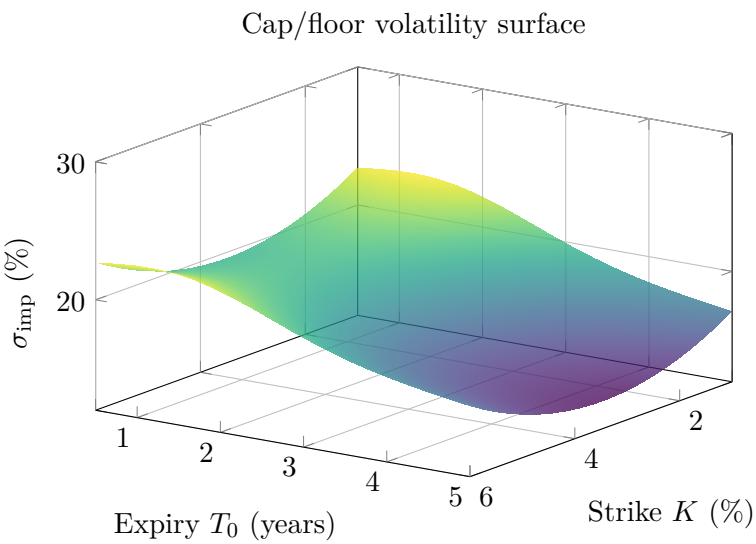
- **Humped:** Volatility peaks at intermediate maturities (e.g. 1–2y), often corresponding to the horizon of monetary policy uncertainty, then declines as rates are expected to mean-revert.
- **Downward-sloping:** Short-dated options have higher vol, reflecting imminent events (central bank meetings, data releases).

14.5.4 The full volatility surface

Definition 96: Volatility surface

The **volatility surface** $\sigma_{\text{imp}}(K, T_0)$ is a two-dimensional function that maps each (strike, expiry) pair to its implied volatility. For swaptions, the surface is three-dimensional: $\sigma_{\text{imp}}(K, T_0, T_n)$, where T_n is the tenor of the underlying swap.

The volatility surface is constructed by:
1. Collecting market prices for liquid options at a grid of (K, T_0) points.
2. For each option, inverting the Black (or Bachelier) formula numerically to extract σ_{imp}.
3. Interpolating between observed points to create a smooth surface (using linear interpolation, cubic splines, or model-based methods like SABR).



To price an option with strike K and expiry T_0, using the volatility surface:
1. Look up (or interpolate) $\sigma_{\text{imp}}(K, T_0)$ from the surface.
2. Insert this volatility into the Black or Bachelier formula.
3. Compute the option price.

The surface captures the market's complete view of rate uncertainty across all strikes and maturities. Slicing the surface at constant expiry gives the smile/skew for that maturity; slicing at constant strike gives the term structure for that strike.

14.6 Embedded options in mortgages

Mortgages — the most widely held fixed income instruments by volume — frequently contain embedded options that significantly affect their valuation and risk characteristics. Unlike the standalone options discussed above, these options are *built into* the loan contract and cannot be separated or traded independently.

Definition 97: Embedded option

An **embedded option** is an option-like feature that is part of a bond or loan contract, rather than a standalone traded instrument. The holder of the embedded option is typically the borrower. The presence of an embedded option modifies the instrument's cash flow profile and risk characteristics.

14.6.1 Embedded caps and floors in variable-rate mortgages

Many variable-rate (adjustable-rate) mortgages include contractual limits on the reference rate:

- An **embedded cap** limits the maximum rate the borrower can be charged. If the reference rate exceeds the cap level K_{cap}, the borrower pays only K_{cap}. From the borrower's perspective, this is equivalent to holding a long position in an interest rate cap with strike K_{cap}, paid for implicitly through a higher spread.
- An **embedded floor** sets a minimum rate the borrower must pay, even if the reference rate falls to zero or below. This is equivalent to the borrower being short a floor with strike K_{floor}. The lender benefits from this protection.
- The combination of an embedded cap and an embedded floor creates an **embedded collar**: the borrower's effective rate is confined to the range $[K_{\text{floor}}, K_{\text{cap}}]$.

Example 75: Valuation impact of an embedded cap

A 20y variable-rate mortgage has notional $N = 300{,}000$ €, pays 3m EURIBOR + 1.50% quarterly, and includes a contractual cap at 5.00% (meaning the total rate cannot exceed 5%, i.e. the EURIBOR component is capped at 3.50%).
If a standalone 20y cap with strike 3.50% on 3m EURIBOR and notional 300,000 € costs 12,500 €, explain how this affects the mortgage's value.

Solution. From the borrower's perspective, the mortgage with embedded cap is equivalent to:
1. A plain variable-rate mortgage at EURIBOR + 1.50% (no cap), *plus*
2. A long position in a cap on 3m EURIBOR with strike 3.50%.

The cap ensures that whenever EURIBOR exceeds 3.50% (the cap is in the money, Definition 88), the excess is compensated, keeping the total rate at 5%. The value of this protection is approximately 12,500 € — the price of the equivalent standalone cap.
From the lender's perspective, granting the cap reduces the mortgage's value by 12,500 €. The lender recovers this cost through the 1.50% spread, which is higher than it would be for an uncapped mortgage with the same credit risk.

14.6.2 Prepayment options in fixed-rate mortgages

Definition 98: Prepayment option

A **prepayment option** gives the borrower the right to repay the outstanding principal of a loan before its contractual maturity, typically without (or with a limited) penalty. This is economically equivalent to a **call option on the bond**: the borrower "calls back" the debt at par.

Prepayment is rational when interest rates fall: the borrower can refinance at a lower rate and use the proceeds to repay the existing (higher-rate) mortgage. This behaviour creates

significant challenges for the lender:

- **Reinvestment risk:** The lender receives cash when rates are low and must reinvest at lower rates.
- **Negative convexity:** A mortgage portfolio exhibits **negative convexity** — its price increases less than expected when rates fall (because falling rates trigger prepayments, shortening the portfolio's effective life) and decreases more than expected when rates rise. In the language of the Greeks (Section 10.4), the portfolio has negative gamma with respect to the interest rate.
- **Uncertain cash flows:** Prepayment makes cash flow projection (Section 2.5) inherently uncertain: the amounts and timing of future principal repayments depend on borrower behaviour, which is influenced by rate levels, housing market conditions, and personal circumstances.

CHAPTER 15

Risk Measurement

Duration and convexity (Chapter 9) measure a bond's sensitivity to yield changes; the Greeks (Chapter 10) generalise these ideas to any derivative and any risk factor. But measuring sensitivity is not the same as measuring risk: risk management requires answering a more fundamental question — how much can a portfolio *lose* over a given horizon? This chapter introduces the two most important risk measures: Value at Risk (VaR) and Expected Shortfall (ES).

15.1 Value at Risk (VaR)

Definition 99: Value at Risk (VaR)

The **Value at Risk** VaR_α at confidence level α over holding period h is the loss level ℓ such that the probability of the portfolio losing more than ℓ is at most $1-\alpha$:

$$\mathbb{P}[\Delta V \leq -\mathrm{VaR}_\alpha] = 1-\alpha,$$

where ΔV is the portfolio's profit/loss over h.
Equivalently, VaR_α is the negative of the $(1-\alpha)$-quantile of the P&L distribution:

$$\mathrm{VaR}_\alpha = -q_{1-\alpha}(\Delta V).$$

Common parameter choices are $\alpha = 99\%$ and $h = 1\mathtt{d}$ (Basel regulatory standard) or $\alpha = 95\%$ and $h = 10\mathtt{d}$. A $1\mathtt{d}$ 99% VaR of 500,000 € means: "On 99% of trading days, we expect to lose no more than 500,000 €. On the remaining 1% of days, losses may exceed this amount".

15.1.1 Parametric (variance–covariance) VaR

If portfolio returns are normally distributed with mean μ and standard deviation σ:

$$\mathrm{VaR}_\alpha = -(\mu - z_\alpha \cdot \sigma),$$

where z_α is the α-quantile of the standard normal distribution (e.g. $z_{99\%} = 2.326$, $z_{95\%} = 1.645$).
For a bond portfolio, the P&L can be linked to yield changes using duration (Definition 55):

$$\Delta V \approx -D_{\mathrm{mod}} \cdot V \cdot \Delta y,$$

where V is the portfolio value and Δy is the yield change. If $\Delta y \sim \mathcal{N}(0, \sigma_y^2)$, then:

$$\mathrm{VaR}_\alpha = D_{\mathrm{mod}} \cdot V \cdot z_\alpha \cdot \sigma_y.$$

Example 76: Parametric VaR of a bond portfolio

A bond portfolio has market value $V = 10{,}000{,}000\,€$ and modified duration $D_{\mathrm{mod}} = 5.2$. Daily yield volatility is $\sigma_y = 5\,\mathrm{bps}$ (0.0005). Compute the 1d 99% VaR.

Solution.

$$\mathrm{VaR}_{99\%} = 5.2 \cdot 10{,}000{,}000 \cdot 2.326 \cdot 0.0005 = 60{,}476\,€.$$

On 99% of trading days, the portfolio's loss should not exceed approximately 60,500 €. On the worst 1% of days, losses could be larger.

15.1.2 Historical VaR

Definition 100: Historical VaR

Historical VaR uses the empirical distribution of past P&L observations rather than assuming normality. Given n historical return observations sorted in ascending order, the $(1-\alpha)$-VaR is the $\lfloor n(1-\alpha) \rfloor$-th worst observation.

Example 77: Historical VaR

A trader has 250 days of historical P&L data for a bond portfolio. The five worst daily P&Ls were: −120,000 €, −95,000 €, −82,000 €, −75,000 €, −68,000 €. Compute the 1d 99% and 95% historical VaR.

Solution. 99% **VaR:** At $\alpha = 99\%$, we need the $(1 - 0.99) \cdot 250 = 2.5$-th worst observation. Interpolating between the 2nd and 3rd worst:

$$\mathrm{VaR}_{99\%} \approx \frac{95{,}000 + 82{,}000}{2} = 88{,}500\,€.$$

(More precisely, the 99% VaR lies between 82,000 € and 95,000 €.)
95% **VaR:** At $\alpha = 95\%$, we need the $(0.05) \cdot 250 = 12.5$-th worst observation. This would be approximately the 12th–13th worst daily loss.
Historical VaR makes no distributional assumptions, which is an advantage when returns are fat-tailed (as they typically are in practice). However, it relies on having a sufficiently long and representative history.

15.1.3 Monte Carlo VaR

Definition 101: Monte Carlo VaR

Monte Carlo VaR generates a large number M of simulated scenarios for risk factors (yields, spreads, FX rates, etc.), revalues the portfolio under each scenario, and computes the $(1-\alpha)$-quantile of the resulting P&L distribution.

Monte Carlo VaR is the most flexible approach: it can handle non-linear instruments (options), multiple risk factors, complex correlations, and non-normal distributions. Its main

drawback is computational cost.

15.2 Expected Shortfall (ES)

VaR answers "What is the maximum loss at a given confidence level?" but it says nothing about the *magnitude* of losses beyond VaR. For this, we need Expected Shortfall.

Definition 102: Expected Shortfall (ES)

The **Expected Shortfall** (also called **Conditional VaR** or **CVaR**) at confidence level α is the expected loss *conditional* on the loss exceeding VaR:

$$\mathrm{ES}_\alpha = -\,\mathbb{E}\big[\Delta V \mid \Delta V \leq -\,\mathrm{VaR}_\alpha\big].$$

ES answers: "If we do have a bad day (in the worst $(1-\alpha)\%$ of outcomes), how bad is it *on average*?"

ES is always greater than or equal to VaR: $\mathrm{ES}_\alpha \geq \mathrm{VaR}_\alpha$. Under normality with mean zero:

$$\mathrm{ES}_\alpha = \sigma \cdot \frac{\phi(z_\alpha)}{1-\alpha},$$

where ϕ is the standard normal density.

Remark (ES vs. VaR: regulatory perspective). The Basel III/IV framework has moved from VaR to ES as the primary risk measure for market risk capital requirements (the **Fundamental Review of the Trading Book**, FRTB). The reasons include:

- **Coherence:** ES is a **coherent risk measure** (it satisfies subadditivity: $\mathrm{ES}(A+B) \leq \mathrm{ES}(A) + \mathrm{ES}(B)$), meaning diversification always reduces risk. VaR is *not* coherent — there exist cases where diversification increases VaR.
- **Tail sensitivity:** ES captures the severity of tail losses, not just their frequency. Two portfolios with the same VaR can have very different ES if one has heavier tails.
- **Risk management incentives:** VaR can be "gamed" by concentrating risk just beyond the VaR threshold. ES discourages this because it penalises the entire tail.

Example 78: Parametric ES

Using the same portfolio as in Example 76 ($V = 10{,}000{,}000\,€$, $D_{\mathrm{mod}} = 5.2$, $\sigma_y = 5\,\mathrm{bps}$), compute the `1d` 99% Expected Shortfall under normality.

Solution. The portfolio's daily P&L standard deviation is:

$$\sigma_V = D_{\mathrm{mod}} \cdot V \cdot \sigma_y = 5.2 \cdot 10{,}000{,}000 \cdot 0.0005 = 26{,}000\,€.$$

Under normality (mean zero):

$$\mathrm{ES}_{99\%} = \sigma_V \cdot \frac{\phi(z_{99\%})}{1-0.99} = 26{,}000 \cdot \frac{\phi(2.326)}{0.01}.$$

The standard normal density at $z = 2.326$ is $\phi(2.326) = 0.02665$:

$$\mathrm{ES}_{99\%} = 26{,}000 \cdot \frac{0.02665}{0.01} = 69{,}290\,€.$$

Compare with $\mathrm{VaR}_{99\%} = 60{,}476\,€$ (Example 76). The ES is about 14.6% higher than VaR, reflecting the average severity of losses in the worst 1% of scenarios. Under normality, the ES/VaR ratio at 99% is approximately $\phi(2.326)/(0.01 \cdot 2.326) = 1.146$.

Example 79: Historical ES

Using the historical data from Example 77, compute the 99% Expected Shortfall.

Solution. At $\alpha = 99\%$, the worst 1% of 250 observations corresponds to the 2.5 worst days. Taking the average of the two worst observations:

$$\mathrm{ES}_{99\%} \approx \frac{120{,}000 + 95{,}000}{2} = 107{,}500\,€.$$

(More precisely, we could weight the 2nd and 3rd worst observations to account for the fractional 2.5.) The historical ES is 107,500 €, substantially larger than the historical VaR of approximately 88,500 €, revealing that the worst losses are concentrated: when things go badly, they go *very* badly.

15.3 Limitations and practical considerations

Remark (Limitations of VaR and ES). Both VaR and ES have important limitations:

- **Model risk:** Parametric approaches assume a distribution (typically normal) that may underestimate tail risks. In practice, financial returns exhibit **fat tails** (extreme events occur more frequently than the normal distribution predicts) and **volatility clustering** (periods of high volatility tend to persist).
- **Stationarity assumption:** Historical approaches assume the past is representative of the future, which fails during regime changes (e.g. the transition from low to high interest rate environments).
- **Horizon scaling:** VaR is often scaled from daily to multi-day using the "square-root-of-time" rule ($\mathrm{VaR}_h = \mathrm{VaR}_1 \cdot \sqrt{h}$), which assumes independent, identically distributed returns — an assumption that fails during crises.
- **Correlation instability:** In stressed markets, correlations between risk factors often increase dramatically (**correlation breakdown**), making diversification less effective than models suggest.
- **Liquidity risk:** Both measures assume positions can be liquidated at current market prices. During crises, **market liquidity** may evaporate, causing actual losses to exceed model predictions.

Risk measures should therefore be complemented by **stress testing** (evaluating portfolio losses under extreme but plausible scenarios) and **sensitivity analysis** (measuring exposure to individual risk factors).

Feature	VaR	ES
Interpretation	Maximum loss at confidence level	Average loss beyond VaR
Coherence	No (not subadditive)	Yes
Tail sensitivity	None (ignores tail shape)	High
Regulatory use	Basel II/III (being replaced)	Basel IV / FRTB
Computation	Simpler	Requires full tail

15.4 Shocks and scenarios

The VaR methods introduced so far rely on either a distributional assumption (parametric) or historical observations (historical simulation). A third, widely used approach constructs explicit scenarios — hypothetical market movements — and evaluates the portfolio under each one.

Definition 103: Shock

A **shock** is a hypothetical or real change in a single risk factor. It is expressed as an absolute or relative shift from the current level:

$$x_{\text{shocked}} = x + \Delta x \quad \text{(absolute)}, \qquad x_{\text{shocked}} = x \cdot (1 + \Delta x) \quad \text{(relative)}.$$

In interest rate risk, shocks are typically expressed in **basis points** (bps), where $1\,\text{bps} = 0.01\% = 0.0001$.

For example, a $+100\,\text{bps}$ shock to the 5y rate means an absolute increase of 1% (i.e., if the rate is 3%, it becomes 4%).

Definition 104: Scenario

A **scenario** is a vector of simultaneous shocks to all relevant risk factors:

$$\mathbf{s} = (\Delta x_1, \Delta x_2, \ldots, \Delta x_k),$$

where k is the number of risk factors and Δx_i is the shock to the i-th factor.

Definition 105: Scenario set

A **scenario set** is a collection of M scenarios $\{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_M\}$ used for risk analysis. Common types include:

- **Historical scenarios:** Derived from actual observed changes in risk factors over a historical window (e.g. 10,000 daily observations).
- **Simulated (Monte Carlo) scenarios:** Generated from a statistical model calibrated to historical data (e.g. a multivariate normal distribution with estimated means, volatilities, and correlations).
- **Stress scenarios:** Designed by risk managers to represent extreme but plausible events (e.g. a simultaneous 300 bps increase in rates and 20% currency depreciation).

15.5 Scenario-based VaR

Given a scenario set and a method to compute the portfolio's P&L under each scenario, the VaR is simply a quantile of the resulting P&L distribution.

15.5.1 Full revaluation vs. sensitivity-based approach

There are two methods for computing the P&L under each scenario:

- **Full revaluation:** Reprice every instrument in the portfolio from scratch using the shocked risk factors. This is exact but computationally expensive for large portfolios.
- **Sensitivity-based (Taylor) approach:** Approximate the P&L using the Greeks (Chapter 10, Section 10.4):
$$\Delta V_i \approx \Delta_i \cdot \Delta x + \tfrac{1}{2}\,\Gamma_i \cdot (\Delta x)^2,$$
for each instrument i, then sum across the portfolio. This is fast but approximate.

Scenario-based VaR

Given M scenarios, each producing a portfolio P&L Π_m $(m = 1, \ldots, M)$, the α-confidence VaR is:
$$\mathrm{VaR}_\alpha = -\Pi_{(\lceil M(1-\alpha)\rceil)},$$
where $\Pi_{(k)}$ denotes the k-th smallest (worst) P&L. At $\alpha = 99.9\%$ with $M = 10{,}000$ scenarios, the VaR is the negative of the 10th worst P&L.

Example 80: Scenario-based VaR with Taylor approximation

A portfolio contains three contracts with the following Greeks with respect to a single interest rate factor:

Contract	Delta (Δ_i)	Gamma (Γ_i)
A	$-200{,}000$	$+50{,}000{,}000$
B	$+350{,}000$	$-20{,}000{,}000$
C	$-500{,}000$	$+80{,}000{,}000$
Portfolio	$\mathbf{-350{,}000}$	$\mathbf{+110{,}000{,}000}$

Three scenarios are given: $\Delta x \in \{-0.02,\ 0,\ +0.02\}$. Compute the portfolio P&L under each scenario and identify the worst loss.

Solution. Using the Taylor expansion (Section 10.5) at the portfolio level:
$$\Delta V_{\text{port}} = \Delta_{\text{port}} \cdot \Delta x + \tfrac{1}{2}\,\Gamma_{\text{port}} \cdot (\Delta x)^2.$$

Scenario 1 ($\Delta x = -0.02$):
$$\Delta V = (-350{,}000)(-0.02) + \tfrac{1}{2}(110{,}000{,}000)(0.02)^2 = 7{,}000 + 22{,}000 = +29{,}000\,€.$$

Scenario 2 ($\Delta x = 0$):
$$\Delta V = 0\,€.$$

Scenario 3 ($\Delta x = +0.02$):
$$\Delta V = (-350{,}000)(+0.02) + \tfrac{1}{2}(110{,}000{,}000)(0.02)^2 = -7{,}000 + 22{,}000 = +15{,}000\,€.$$

All three scenarios produce non-negative P&L. The positive gamma ($\Gamma_{\text{port}} > 0$) ensures the portfolio benefits from large moves in *either* direction, while the negative delta means it particularly benefits from rate decreases. The worst outcome is scenario 2 (no change), with zero P&L.

15.6 Sensitivity analysis

While VaR provides a single summary number, sensitivity analysis offers a more detailed view of the portfolio's risk profile by examining how its value responds to a range of predetermined shocks.

Definition 106: Sensitivity ladder (bump analysis)

A **sensitivity ladder** (or **bump analysis**) is a table showing the portfolio's estimated P&L for a series of parallel shifts in the risk factor, typically ranging from -200 bps to $+200$ bps.

Example 81: Sensitivity ladder for a multi-currency portfolio

A portfolio of EUR, CZK, and HUF interest rate instruments has portfolio delta $\Delta_{\text{port}} = +2{,}263{,}000$ (per unit shift) and gamma $\Gamma_{\text{port}} = -1{,}200{,}000{,}000$. All amounts are in EUR after FX conversion.
Compute the sensitivity ladder for parallel shifts from -200 bps to $+200$ bps.

Solution. Using the Taylor approximation $\Delta V \approx \Delta_{\text{port}} \cdot \Delta x + \frac{1}{2}\Gamma_{\text{port}} \cdot (\Delta x)^2$:

Shift	Δx	Delta effect	Gamma effect	Total P&L
-200 bps	-0.020	$-45{,}260$ €	$-240{,}000$ €	$-285{,}260$ €
-100 bps	-0.010	$-22{,}630$ €	$-60{,}000$ €	$-82{,}630$ €
-50 bps	-0.005	$-11{,}315$ €	$-15{,}000$ €	$-26{,}315$ €
$+50$ bps	$+0.005$	$+11{,}315$ €	$-15{,}000$ €	$-3{,}685$ €
$+100$ bps	$+0.010$	$+22{,}630$ €	$-60{,}000$ €	$-37{,}370$ €
$+200$ bps	$+0.020$	$+45{,}260$ €	$-240{,}000$ €	$-194{,}740$ €

Observations:

1. The portfolio has positive delta (benefits from rate increases through the linear term) but strongly negative gamma ($\Gamma < 0$). The gamma effect is always negative regardless of direction, meaning the portfolio *loses* from large moves.
2. The P&L is *asymmetric*: the loss from -200 bps ($-285{,}260$ €) exceeds the loss from $+200$ bps ($-194{,}740$ €). This asymmetry arises because for negative shocks, both the delta and gamma effects are negative and reinforce each other.
3. A symmetry check — $|\text{P\&L}_{+s}|$ vs. $|\text{P\&L}_{-s}|$ — immediately reveals the direction of the portfolio's exposure. If the two are approximately equal, delta is small relative to gamma; if they differ substantially, delta dominates.

15.7 Multi-currency portfolio risk

When a portfolio contains instruments denominated in different currencies, all P&L contributions must be converted to a common base currency (or reporting currency) before they can be aggregated.

Definition 107: Base currency P&L conversion

For an instrument denominated in a foreign currency c, the P&L in the base currency (e.g. EUR) is:

$$\Delta V_{\text{EUR}} = \Delta V_c \cdot \text{FX}_c,$$

where ΔV_c is the P&L in local currency and FX_c is the exchange rate (units of foreign currency c per one EUR, inverted; or equivalently, EUR per one unit of currency c).

This approach makes the simplifying assumption that the FX rate remains constant during the shock — a reasonable first-order approximation when the primary risk factors are interest rates. For a more precise analysis, FX rate changes should be included as additional risk factors.

Multi-currency portfolio P&L

For a portfolio with instruments in C currencies, the aggregate P&L in the base currency under a scenario $\mathbf{s}$ is:

$$\Delta V_{\text{port}}^{\text{EUR}} = \sum_{c=1}^{C} \left[\sum_{j \in \mathcal{P}_c} \left(\Delta_j \cdot \Delta x_c + \tfrac{1}{2} \Gamma_j \cdot (\Delta x_c)^2 \right) \right] \cdot \text{FX}_c,$$

where $\mathcal{P}_c$ is the set of instruments in currency c and Δx_c is the shock to currency c's interest rate.

Example 82: Multi-currency VaR aggregation

A portfolio has interest rate contracts in three currencies. Under a particular stress scenario, the shocks and portfolio-level Greeks per currency are:

Currency	Delta (local)	Gamma (local)	Shock	FX to EUR
EUR	−1,500,000	+200,000,000	−0.015	1.0000
CZK	−30,000,000	+5,000,000,000	−0.020	0.0400
HUF	−500,000,000	+80,000,000,000	−0.018	0.0026

Compute the portfolio P&L in EUR.

Solution. EUR contribution:

$$\begin{aligned} \Delta V_{\text{EUR}} &= (-1{,}500{,}000)(-0.015) + \tfrac{1}{2}(200{,}000{,}000)(0.015)^2 \\ &= 22{,}500 + 22{,}500 = 45{,}000\,€. \end{aligned}$$

CZK contribution (in CZK, then converted):

$$\begin{aligned} \Delta V_{\text{CZK}} &= (-30{,}000{,}000)(-0.020) + \tfrac{1}{2}(5{,}000{,}000{,}000)(0.020)^2 \\ &= 600{,}000 + 1{,}000{,}000 = 1{,}600{,}000 \text{ CZK}. \\ \Delta V_{\text{CZK}}^{\text{EUR}} &= 1{,}600{,}000 \cdot 0.04 = 64{,}000\,€. \end{aligned}$$

HUF contribution (in HUF, then converted):

$$\begin{aligned} \Delta V_{\text{HUF}} &= (-500{,}000{,}000)(-0.018) + \tfrac{1}{2}(80{,}000{,}000{,}000)(0.018)^2 \\ &= 9{,}000{,}000 + 12{,}960{,}000 = 21{,}960{,}000 \text{ HUF}. \\ \Delta V_{\text{HUF}}^{\text{EUR}} &= 21{,}960{,}000 \cdot 0.0026 = 57{,}096\,€. \end{aligned}$$

Total portfolio P&L:

$$\Delta V_{\text{port}}^{\text{EUR}} = 45{,}000 + 64{,}000 + 57{,}096 = +166{,}096\,€.$$

In this scenario all currencies experienced negative rate shocks, and the portfolio — which has negative delta in all three currencies — gains value. Note that the CZK contribution (64,000 €) is significant despite the small FX rate, because the local-currency delta is large.

15.8 Correlation and diversification

The risk of a multi-factor portfolio depends critically on the correlation between its risk factors.

Definition 108: Correlation in risk management

The **correlation** ρ_{ij} between risk factors x_i and x_j measures the degree to which their movements are related:

$$\rho_{ij} = \frac{\text{Cov}(\Delta x_i, \Delta x_j)}{\sigma_i \cdot \sigma_j},$$

where σ_i and σ_j are the standard deviations of the respective factor changes. Correlation ranges from -1 (perfect inverse co-movement) to $+1$ (perfect co-movement).

Correlation has a direct impact on portfolio risk:

- **Positive correlation** ($\rho > 0$): Risk factors tend to move in the same direction. If the portfolio is exposed in the same direction to both factors, losses tend to occur simultaneously, *amplifying* overall risk. Diversification benefit is limited.
- **Negative correlation** ($\rho < 0$): Risk factors tend to move in opposite directions. Losses on one factor are partially offset by gains on the other, providing a natural hedge and *reducing* overall risk.
- **Zero correlation** ($\rho = 0$): No systematic relationship. Risk combines according to the square-root rule: $\sigma_{\text{port}} = \sqrt{\sigma_1^2 + \sigma_2^2}$.

Remark (Correlation breakdown in stress scenarios). A critical challenge in risk management is that correlations estimated from normal market conditions may not hold during stress events. During crises, correlations between asset classes and currencies often **increase sharply** — a phenomenon known as **correlation breakdown** or **contagion**. This means that diversification benefits, which rely on imperfect correlations, tend to disappear precisely when they are most needed.
For example, in the worst scenarios of a multi-currency portfolio, one typically observes *all* currencies being shocked in the same direction simultaneously, even if historical correlations are moderate. This is why stress testing (Section 15.10) with coordinated extreme scenarios is an essential complement to correlation-based risk models.

15.9 Fat tails and non-normality

The parametric VaR approach (Section 15.5) assumes normally distributed returns. In practice, financial returns systematically deviate from normality.

Definition 109: Fat tails (leptokurtosis)

A distribution is said to have **fat tails** (or to be **leptokurtic**) if extreme events occur more frequently than predicted by a normal distribution with the same mean and variance. Formally, the **excess kurtosis**

$$\kappa = \frac{\mathbb{E}[(\Delta x - \mu)^4]}{\sigma^4} - 3$$

is positive for fat-tailed distributions (it is zero for the normal distribution).

Financial returns exhibit fat tails almost universally: daily interest rate changes, FX returns, and equity returns all produce more extreme moves than the normal distribution predicts. A practical consequence is that the ratio of high-confidence VaR to low-confidence VaR is larger than the normal model implies:

VaR ratio	Normal distribution	Typical financial data
$\mathrm{VaR}_{99.9\%}$ / $\mathrm{VaR}_{95\%}$	1.88	3.0 – 4.0
$\mathrm{VaR}_{99\%}$ / $\mathrm{VaR}_{95\%}$	1.41	1.6 – 2.2

A ratio of $\mathrm{VaR}_{99.9\%}$ / $\mathrm{VaR}_{95\%}$ significantly exceeding 2 is a clear indicator of a fat-tailed P&L distribution. This has important implications for risk capital: models based on normality will systematically *underestimate* the capital required to cover extreme losses.

Remark (Implications for risk management). Fat tails mean that:

- **Parametric VaR underestimates tail risk.** The normal distribution predicts a 99.9% loss as roughly 3.09 standard deviations from the mean; empirical data often shows 4–5 standard deviations.
- **Expected Shortfall is more robust.** Because ES averages over the entire tail, it captures the severity of extreme events more accurately than VaR, which only identifies the boundary.
- **Historical or scenario-based methods are preferred** for high-confidence levels (99% and above), as they make no distributional assumptions and naturally capture fat tails present in the data.

15.10 Stress testing

While VaR and ES are backward-looking (based on historical patterns or calibrated models), stress testing is forward-looking: it asks "what if?" questions about specific extreme events.

Definition 110: Stress test

A **stress test** is the evaluation of a portfolio's P&L under one or more extreme but plausible scenarios designed to capture risks that may not be adequately reflected in statistical risk measures. Stress scenarios may be:

- **Historical:** Based on actual past crises (e.g. the 2008 financial crisis, the 2020 COVID shock, the 2022 rate hiking cycle).
- **Hypothetical:** Designed by risk managers to test specific vulnerabilities (e.g. a simultaneous 500 bps rate increase across all maturities with a 30% currency depreciation).
- **Reverse stress tests:** Starting from a severe outcome (e.g. the portfolio losing

10,000,000 €) and working backwards to identify which scenarios could cause it.

Stress testing complements VaR and ES by exploring scenarios that may lie outside the range of historical experience or statistical models. It is particularly valuable for identifying **concentration risk** (excessive exposure to a single factor or direction) and **correlation risk** (the danger that diversification fails under stress).

Example 83: Stress testing a multi-currency portfolio

A risk manager designs three stress scenarios for a portfolio with EUR, CZK, and HUF exposures:

Scenario	EUR shock	CZK shock	HUF shock
Moderate stress	−100 bps	−150 bps	−150 bps
Severe stress	−200 bps	−400 bps	−450 bps
Extreme / crisis	−350 bps	−600 bps	−600 bps

Using portfolio-level deltas $\Delta_{\text{EUR}} = -1{,}200{,}000$, $\Delta_{\text{CZK}} = -800{,}000$, $\Delta_{\text{HUF}} = -600{,}000$ (all expressed in EUR), and assuming gamma effects are relatively small, estimate the losses.

Solution. Using the delta-only approximation $\Delta V \approx \sum_c \Delta_c \cdot \Delta x_c$:
Moderate stress:

$$\begin{aligned}\Delta V &\approx (-1{,}200{,}000)(-0.01) + (-800{,}000)(-0.015) + (-600{,}000)(-0.015)\\ &= 12{,}000 + 12{,}000 + 9{,}000 = +33{,}000\,€.\end{aligned}$$

Severe stress:

$$\begin{aligned}\Delta V &\approx (-1{,}200{,}000)(-0.02) + (-800{,}000)(-0.04) + (-600{,}000)(-0.045)\\ &= 24{,}000 + 32{,}000 + 27{,}000 = +83{,}000\,€.\end{aligned}$$

Extreme stress:

$$\begin{aligned}\Delta V &\approx (-1{,}200{,}000)(-0.035) + (-800{,}000)(-0.06) + (-600{,}000)(-0.06)\\ &= 42{,}000 + 48{,}000 + 36{,}000 = +126{,}000\,€.\end{aligned}$$

In this example, the portfolio has negative delta across all currencies, so negative shocks produce *gains*. The stress scenarios reveal that the portfolio's primary risk is from *positive* rate moves. The risk manager should also test scenarios with positive shocks — e.g. a rising rate environment — which would produce losses of similar magnitude.
This illustrates a key purpose of stress testing: it reveals the *direction* of risk and helps identify which scenarios are truly dangerous for a particular portfolio.

Remark (Data quality and validation). In practice, the accuracy of any risk measure depends critically on the quality of the underlying data. Professional risk

management requires systematic **data validation** before computing Greeks or VaR. Common checks include:

- **Completeness:** Verify that all required valuation data (current value, shifted values for Greek computation) is present. Missing values must be identified and handled — typically by excluding affected instruments.
- **Greek consistency:** Compute Greeks using two different perturbation sizes (e.g. ε and 2ε; see Definition 74) and verify that the results are consistent. Large discrepancies signal numerical instability or data corruption.
- **Taylor approximation quality:** Use the computed Greeks to predict a known shifted value (e.g. $V(x + 2\varepsilon)$) and compare with the actual value. If the prediction error exceeds a threshold (e.g. 10%), the instrument's data may be unreliable and should be excluded.
- **Outlier detection:** Flag instruments with extreme Greeks (e.g. using the **interquartile range** method with a 3× threshold). Outliers should be investigated — they may reflect legitimate leveraged positions or data errors.
- **Monotonicity:** For instruments expected to have monotonic price–rate relationships (e.g. plain bonds), verify that $V(x - \varepsilon) > V(x) > V(x + \varepsilon)$ (for instruments with negative delta). Non-monotonic behaviour may indicate complex embedded optionality or data issues.

A robust approach is to compute VaR under different data cleaning assumptions (e.g. raw data, standard cleaning, conservative cleaning) and verify that the results are *stable*. If VaR changes by less than 1–2% across cleaning approaches, the result is considered robust. This **sensitivity analysis of the risk measure itself** provides confidence in the final number.

APPENDIX

Statistical Foundations

A

This appendix collects the essential statistical tools required for a rigorous understanding of fixed income valuation and risk management. We provide precise definitions, key theorems, and worked examples as a reference. The material is structured to support self-contained study while maintaining the mathematical rigour necessary for quantitative finance.

A.1 Probability theory

A.1.1 Sample spaces and events

Definition 111: Probability space

A **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$ consists of:

- A **sample space** Ω (the set of all possible outcomes).
- A **σ-algebra** $\mathcal{F}$ (a collection of subsets of Ω representing events).
- A **probability measure** $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying:
  1. $\mathbb{P}(\Omega) = 1$ (total probability).
  2. For disjoint events $A_1, A_2, \ldots$: $\mathbb{P}(A_1 \cup A_2 \cup \cdots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots$ (countable additivity).

Definition 112: Conditional probability

The **conditional probability** of event A given event B (with $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Definition 113: Independence

Events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Equivalently, if $\mathbb{P}(B) > 0$, then $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

A.1.2 Random variables

Definition 114: Random variable

A **random variable** X is a measurable function $X : \Omega \to \mathbb{R}$ that assigns a real number to each outcome. A random variable is:
- **Discrete** if it takes countably many values.
- **Continuous** if its distribution can be described by a probability density function.

Definition 115: Cumulative distribution function (CDF)

The **CDF** of a random variable X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Properties:
- F_X is non-decreasing.
- $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to\infty} F_X(x) = 1$.
- F_X is right-continuous.

Definition 116: Probability density function (PDF)

For a continuous random variable X, the **PDF** $f_X(x)$ satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

Equivalently, $f_X(x) = F'_X(x)$ wherever the derivative exists. The PDF satisfies:
- $f_X(x) \geq 0$ for all x.
- $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$ (total probability).

A.1.3 Expectation and moments

Definition 117: Expected value

The **expected value** (or **mean**) of a random variable X is

$$\mathbb{E}[X] = \begin{cases} \sum_x x \cdot \mathbb{P}(X = x) & \text{(discrete)}, \\ \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx & \text{(continuous)}. \end{cases}$$

Theorem A.1 (Properties of expectation)

Let X and Y be random variables and a, b constants.
1. **Linearity:** $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.
2. **Constant:** $\mathbb{E}[c] = c$.
3. **Non-negativity:** If $X \geq 0$, then $\mathbb{E}[X] \geq 0$.
4. **Law of iterated expectations:** $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.

Definition 118: Variance and standard deviation

The **variance** of X is

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The **standard deviation** is $\sigma_X = \sqrt{\mathrm{Var}(X)}$.

Theorem A.2 (Properties of variance)

1. $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ (variance is unaffected by shifts, scales quadratically).
2. If X and Y are independent: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.
3. $\mathrm{Var}(X) \geq 0$, with equality if and only if X is constant.

Example 84: Computing expectation and variance

Let X be a random variable with the following discrete distribution:

x	1	2	3
$\mathbb{P}(X = x)$	0.2	0.5	0.3

Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution.

$$\mathbb{E}[X] = 1 \cdot 0.2 + 2 \cdot 0.5 + 3 \cdot 0.3 = 0.2 + 1.0 + 0.9 = 2.1.$$

$$\mathbb{E}[X^2] = 1^2 \cdot 0.2 + 2^2 \cdot 0.5 + 3^2 \cdot 0.3 = 0.2 + 2.0 + 2.7 = 4.9.$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 4.9 - (2.1)^2 = 4.9 - 4.41 = 0.49.$$

Thus, $\sigma_X = \sqrt{0.49} = 0.7$.

A.1.4 Covariance and correlation

Definition 119: Covariance

The **covariance** between random variables X and Y is

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Theorem A.3 (Properties of covariance)

1. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
2. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$ (symmetry).
3. $\mathrm{Cov}(aX + b, cY + d) = ac \cdot \mathrm{Cov}(X, Y)$.
4. If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$ (the converse is not always true).
5. $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$.

Definition 120: Correlation

The **correlation coefficient** ρ_{XY} between X and Y is

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y},$$

where $\sigma_X = \sqrt{\mathrm{Var}(X)}$ and $\sigma_Y = \sqrt{\mathrm{Var}(Y)}$.

Correlation satisfies $-1 \leq \rho_{XY} \leq 1$:
- $\rho_{XY} = 1$: Perfect positive linear relationship.
- $\rho_{XY} = -1$: Perfect negative linear relationship.
- $\rho_{XY} = 0$: No linear relationship (uncorrelated).

Example 85: Portfolio variance with correlation

A portfolio consists of two risk factors X and Y with $\sigma_X = 2$, $\sigma_Y = 3$, and correlation $\rho = 0.6$. The portfolio is $V = X + Y$. Compute $\mathrm{Var}(V)$.

Solution.

$$\mathrm{Cov}(X,Y) = \rho\sigma_X\sigma_Y = 0.6 \cdot 2 \cdot 3 = 3.6.$$

$$\mathrm{Var}(V) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y) = 4 + 9 + 2 \cdot 3.6 = 20.2.$$

Standard deviation: $\sigma_V = \sqrt{20.2} \approx 4.49$.
Compare this with the case of independence ($\rho = 0$):

$$\mathrm{Var}(V) = 4 + 9 = 13 \quad \Rightarrow \quad \sigma_V = \sqrt{13} \approx 3.61.$$

Positive correlation *increases* portfolio risk.

A.2 Important probability distributions

A.2.1 The normal (Gaussian) distribution

Definition 121: Normal distribution

A random variable X follows a **normal distribution** with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The **standard normal distribution** has $\mu = 0$ and $\sigma = 1$, denoted $Z \sim \mathcal{N}(0,1)$.

Theorem A.4 (Properties of the normal distribution)

1. If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.
2. If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$ (standardisation).

3. If $X_1, \ldots, X_n$ are independent normal variables with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then

$$\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2\right).$$

4. Approximately 68% of values lie within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, and 99.7% within $\mu \pm 3\sigma$ (empirical rule).

Definition 122: Cumulative standard normal function

The CDF of the standard normal distribution is denoted $\Phi(z)$:

$$\Phi(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt.$$

Key values:

z	$\Phi(z)$
0.00	0.5000
1.00	0.8413
1.645	0.9500
1.96	0.9750
2.00	0.9772
2.326	0.9900
3.00	0.9987

Remark (Normal distribution in finance). The normal distribution is ubiquitous in quantitative finance:

- **Black–Scholes model** (Chapter 14): Assumes log-normal asset prices, equivalent to normally distributed log-returns.
- **Bachelier model** (Chapter 14): Assumes interest rates are normally distributed.
- **Value at Risk (VaR)** (Chapter 15): Parametric VaR uses quantiles of the normal distribution to estimate tail losses.
- **Monte Carlo simulation** (Chapter 15): Scenarios are often generated from multivariate normal distributions.

Example 86: Computing normal probabilities

Let $X \sim \mathcal{N}(100, 25)$ (mean 100, variance 25, so $\sigma = 5$). Compute $\mathbb{P}(X \leq 110)$.

Solution. Standardise:

$$Z = \frac{X - 100}{5} \sim \mathcal{N}(0, 1).$$

$$\mathbb{P}(X \leq 110) = \mathbb{P}\left(\frac{X - 100}{5} \leq \frac{110 - 100}{5}\right) = \mathbb{P}(Z \leq 2) = \Phi(2) = 0.9772.$$

There is a 97.72% probability that X is at most 110.

A.2.2 The log-normal distribution

Definition 123: Log-normal distribution

A random variable X is **log-normally distributed** if $\ln(X)$ is normally distributed. If $\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$, then X has PDF

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0.$$

Theorem A.5 (Moments of the log-normal distribution)

If X is log-normal with $\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$, then:

$$\mathbb{E}[X] = e^{\mu+\sigma^2/2}, \qquad \mathrm{Var}(X) = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1).$$

Remark (Log-normal in option pricing). The Black–Scholes model (Chapter 14) assumes asset prices follow a geometric Brownian motion, implying S_T is log-normally distributed. This ensures $S_T > 0$ always, which is economically sensible (prices cannot be negative). However, the log-normal assumption breaks down for interest rates when they can become negative, motivating the Bachelier (normal) model.

A.2.3 Quantiles and percentiles

Definition 124: Quantile

The **α-quantile** (or **100α-th percentile**) of a random variable X with CDF F_X is the value q_α such that

$$\mathbb{P}(X \leq q_\alpha) = \alpha.$$

For the standard normal distribution, we write $z_\alpha = \Phi^{-1}(\alpha)$.

Example 87: Value at Risk using normal quantiles

A portfolio has daily P&L normally distributed with mean $\mu = 0$ and standard deviation $\sigma = 10{,}000$ €. Compute the 95% Value at Risk (VaR).

Solution. The 95% VaR is the loss level exceeded with 5% probability. Since losses correspond to negative P&L, we seek $q_{0.05}$:

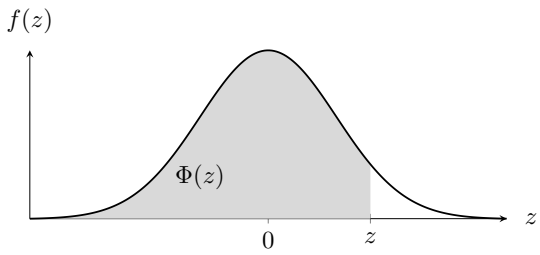
$$\mathbb{P}(X \leq q_{0.05}) = 0.05 \quad \Rightarrow \quad q_{0.05} = \mu + \sigma \cdot z_{0.05} = 0 + 10{,}000 \cdot (-1.645) = -16{,}450\,€.$$

The 95% VaR is 16,450 €. There is a 5% chance of losing more than this amount in a single day.

A.3 Standard normal distribution table

What the table shows

The table on the following page gives the **cumulative probability** $\Phi(z) = \mathbb{P}(Z \leq z)$ for a standard normal random variable $Z \sim \mathcal{N}(0, 1)$. Each entry is the area under the standard normal curve from $-\infty$ to z, i.e. the probability that Z takes a value *at most* z. The shaded region in the diagram below illustrates this area for a generic $z > 0$.



How to use the table

1. **Find the row** corresponding to the first decimal place of z (e.g. for $z = 1.64$, use the row 1.6).
2. **Find the column** corresponding to the second decimal place (e.g. for $z = 1.64$, use column 0.04).
3. **Read off** the entry: it gives $\Phi(z) = \mathbb{P}(Z \leq z)$.

Derived quantities and symmetry

Four identities cover every practical case:

- **Right tail:** $\mathbb{P}(Z > z) = 1 - \Phi(z)$.
- **Negative argument:** $\Phi(-z) = 1 - \Phi(z)$ (symmetry of the normal distribution).
- **Symmetric interval:** $\mathbb{P}(-z \leq Z \leq z) = 2\Phi(z) - 1$.
- **General normal:** for $X \sim \mathcal{N}(\mu, \sigma^2)$, standardise via $Z = (X - \mu)/\sigma$ and then apply the table.

Worked example. $\mathbb{P}(Z \leq 1.96) = \Phi(1.96) = 0.9750$, so $\mathbb{P}(Z > 1.96) = 0.0250$, $\mathbb{P}(Z \leq -1.96) = 1 - 0.9750 = 0.0250$, and $\mathbb{P}(-1.96 \leq Z \leq 1.96) = 0.9500$.

Inverse look-up: from probability to z

Often one needs to find z given a target probability α — i.e. compute $z_\alpha = \Phi^{-1}(\alpha)$. Scan the body of the table for the entry closest to α and read off the corresponding row and column. The table below collects the most frequently needed values directly.

Confidence level $1 - \alpha$	80%	90%	95%	97.5%	99%	99.9%
One-sided z_α	0.842	1.282	1.645	1.960	2.326	3.090
Two-sided $z_{\alpha/2}$	1.282	1.645	1.960	2.240	2.576	3.291

The *one-sided* z_α is used in one-tailed hypothesis tests and in VaR calculations (e.g. 95% VaR uses $z_{0.05} = -1.645$). The *two-sided* $z_{\alpha/2}$ is used in confidence intervals and two-tailed tests (e.g. a 95% confidence interval uses ± 1.960).

The empirical rule

For quick mental calculations, the following three values are essential:

$$\mathbb{P}(|Z| \leq 1) \approx 68.3\%, \qquad \mathbb{P}(|Z| \leq 2) \approx 95.4\%, \qquad \mathbb{P}(|Z| \leq 3) \approx 99.7\%.$$

These correspond to the $\mu \pm \sigma$, $\mu \pm 2\sigma$, and $\mu \pm 3\sigma$ bands from the properties of the normal distribution. In risk management, events beyond $\pm 3\sigma$ are often labelled *tail events*; the table confirms they occur with probability less than 0.3% under normality.

Table A.1: Cumulative standard normal probabilities $\Phi(z) = \mathbb{P}(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9971	0.9972	0.9973
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990

APPENDIX

B Nelson–Siegel and Svensson Models

This appendix introduces two parametric models for constructing smooth yield curves from market data. These models are practical tools used daily by central banks, investment banks, and risk managers to interpolate yields, price bonds, and perform scenario analysis.

B.1 Why parametric yield curve models?

Market data provides yields at discrete maturities (e.g., 2Y, 5Y, 10Y bonds). To value instruments with intermediate maturities or compute forward rates, we need a continuous curve. Parametric models offer key advantages over simple interpolation (linear, cubic splines):

- **Economic interpretation:** Parameters capture meaningful features (long-term level, slope, curvature) rather than arbitrary spline knots.
- **Smoothness:** Continuous derivatives avoid artificial kinks in forward rates.
- **Parsimony:** 4–6 parameters summarise the entire curve, facilitating communication and analysis.
- **Extrapolation:** Reliable behaviour beyond observed maturities (splines often diverge).
- **Scenario analysis:** Easy to generate plausible stress scenarios by perturbing parameters.

The trade-off: parametric models may not fit market prices as tightly as flexible splines, especially for bonds with idiosyncratic features.

B.2 The Nelson–Siegel model

B.2.1 Mathematical specification

Nelson–Siegel model

The Nelson–Siegel model represents the spot rate (yield to maturity) as a weighted sum of three factors:

$$r(T) = \beta_0 + \beta_1 \cdot \frac{1 - e^{-T/\tau}}{T/\tau} + \beta_2 \cdot \left(\frac{1 - e^{-T/\tau}}{T/\tau} - e^{-T/\tau} \right), \tag{B.1}$$

where:

- β_0: **Long-term level** ($\lim_{T\to\infty} r(T) = \beta_0$).

- β_1: **Short-term component (slope)**. Controls $r(0) - \beta_0$.
- β_2: **Medium-term component (curvature)**. Creates a hump or trough.
- $\tau > 0$: **Decay parameter**. Determines where slope and curvature effects peak (typically $\tau \in [1, 3]$ for annual data).

The discount factor is $D(T) = e^{-r(T)\cdot T}$, and the instantaneous forward rate is:

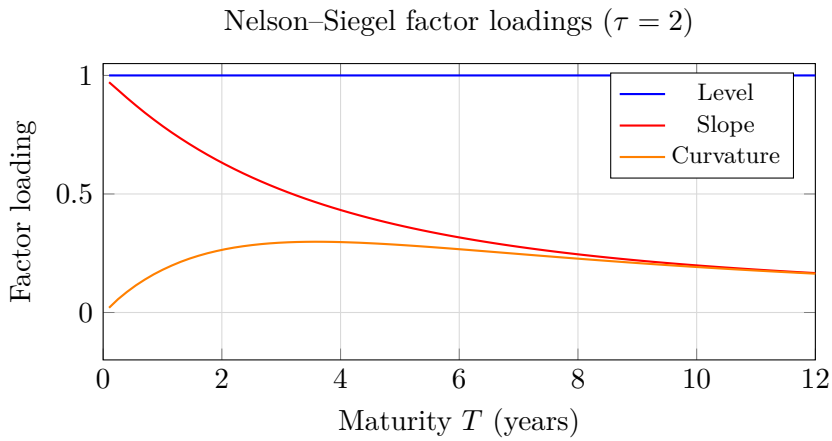
$$f(T) = \beta_0 + \beta_1 e^{-T/\tau} + \beta_2 \frac{T}{\tau} e^{-T/\tau}. \tag{B.2}$$

B.2.2 Intuition: factor loadings

Rewrite equation (B.1) as:

$$r(T) = \beta_0 \cdot \underbrace{1}_{\text{Level}} + \beta_1 \cdot \underbrace{\frac{1 - e^{-T/\tau}}{T/\tau}}_{\text{Slope}} + \beta_2 \cdot \underbrace{\left[\frac{1 - e^{-T/\tau}}{T/\tau} - e^{-T/\tau}\right]}_{\text{Curvature}}.$$

Each factor (level, slope, curvature) has a *loading* that varies with maturity:



- **Level:** Constant loading = 1. Affects all maturities equally.
- **Slope:** Loading starts at 1 (short end) and decays to 0 (long end). Positive β_1 makes short rates higher than long rates (inverted curve); negative β_1 produces upward slope.
- **Curvature:** Loading starts at 0, peaks near $T = 2\tau$, then decays. Positive β_2 creates a hump; negative β_2 creates a trough.

B.2.3 Practical example

Example 88: Nelson–Siegel model computation

uppose we estimate $\beta_0 = 3.5\%$, $\beta_1 = -1.8\%$, $\beta_2 = 1.2\%$, $\tau = 2.0$ from euro-area bond data. Compute yields and discount factors at selected maturities.

Solution. For each maturity T, compute $\lambda = T/\tau = T/2$, then:

$$r(T) = 3.5 + (-1.8 + 1.2) \cdot \frac{1 - e^{-\lambda}}{\lambda} - 1.2e^{-\lambda}.$$

T	λ	$\frac{1-e^{-\lambda}}{\lambda}$	$e^{-\lambda}$	$r(T)$ (%)	$D(T)$
1	0.50	0.787	0.607	2.77	0.9727
2	1.00	0.632	0.368	3.08	0.9404
5	2.50	0.367	0.082	3.38	0.8448
10	5.00	0.199	0.007	3.49	0.7050
30	15.0	0.067	0.000	3.46	0.3569

The curve is upward-sloping (from 2.77% to 3.49%), flattening at long maturities toward $\beta_0 = 3.5\%$.

B.3 The Svensson extension

The Nelson–Siegel model can produce at most one hump or trough. Real yield curves sometimes exhibit more complex shapes (e.g., humps at both 5Y and 15Y). The Svensson model adds a second curvature term:

$$r(T) = \beta_0 + \beta_1 \cdot \frac{1 - e^{-T/\tau_1}}{T/\tau_1} + \beta_2 \cdot \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} - e^{-T/\tau_1} \right) + \beta_3 \cdot \left(\frac{1 - e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2} \right), \quad \text{(B.3)}$$

where β_3 and $\tau_2 > 0$ are the additional curvature parameter and decay rate. Choosing $\tau_1 \neq \tau_2$ allows two humps at different maturities.
The Svensson model is employed by the ECB, Bundesbank, and other central banks for official yield curve estimation when market data exhibits non-standard shapes.

B.4 Calibration: fitting the models to data

B.4.1 Setup and objective

Given market prices $\{P_1, \ldots, P_J\}$ of J liquid bonds, we seek parameters $\boldsymbol{\theta}$ that minimise pricing errors. For bond j with cash flows $(c_{j,1}, t_{j,1}), \ldots, (c_{j,n_j}, t_{j,n_j})$, the model price is:

$$P_j^{\text{model}}(\boldsymbol{\theta}) = \sum_{i=1}^{n_j} c_{j,i} \cdot D(t_{j,i}; \boldsymbol{\theta}), \quad D(t; \boldsymbol{\theta}) = e^{-r(t;\boldsymbol{\theta}) \cdot t}.$$

The calibration problem is:

$$\min_{\boldsymbol{\theta}} \sum_{j=1}^{J} w_j \left(P_j^{\text{market}} - P_j^{\text{model}}(\boldsymbol{\theta}) \right)^2, \quad \text{(B.4)}$$

where w_j are weights (common choice: $w_j = 1/D_{\text{mod},j}^2$, where $D_{\text{mod},j}$ is the modified duration of bond j; this weighting equalises yield errors across maturities).
For Nelson–Siegel, $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \tau)$. For Svensson, $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$.

B.5 Assumptions, limitations, and practical guidance

B.5.1 Key assumptions

The models assume: (1) the yield curve belongs to a low-dimensional family of exponential decay functions; (2) market prices reflect fundamental interest rate expectations without large liquidity premia or idiosyncratic distortions; (3) bond cash flows are known and deterministic (no embedded options or default risk).

B.5.2 Limitations

Imperfect fit. Parametric models may not match all market prices exactly (typical errors: 0.5–2 basis points for liquid government bonds). Bonds with liquidity premia or embedded options may exhibit larger deviations. Svensson improves fit but requires more liquid benchmarks.
Short-end rigidity. The models constrain $r(T)$ to converge smoothly as $T \to 0$, which may miss sharp kinks in overnight rates during market stress or central bank interventions.
Calibration sensitivity. Nonlinear optimisation can be sensitive to initial guesses, particularly for Svensson's six parameters. Grid search mitigates this but increases computational cost.
No arbitrage-free guarantee. Unlike bootstrapping, parametric models do not exactly replicate market prices, creating small theoretical arbitrages (typically < 1 bps, negligible in practice).

B.5.3 When to use these models

Nelson–Siegel: Standard yield curve shapes (upward-sloping, inverted, single hump). Ideal for daily curve estimation, scenario analysis, and communication. Recommended for most applications.
Svensson: Complex shapes (multiple humps), precise fit to benchmarks, official central bank curves. Requires dense bond coverage across maturities.
Curve construction: When bootstrapping is impractical (sparse or irregular data), fit a parametric model to available liquid bonds and extract the full discount curve.
Risk management: Generate smooth stressed curves by perturbing parameters ($\Delta\beta_0$ for parallel shifts, $\Delta\beta_1$ for steepening, $\Delta\beta_2$ for curvature changes).
These models balance simplicity, interpretability, and realism, making them essential tools in fixed income analysis.

APPENDIX

Notation

The following table collects the main symbols and conventions used throughout this booklet. Symbols are grouped by category.

General conventions

Symbol	Meaning
t	Time (in years from today)
T	Maturity date; time to maturity
Δt	Year fraction (accrual period)
N	Principal / face value / nominal / par value
r	Interest rate (per annum, unless stated otherwise)
y	Yield to maturity (YTM)
n	Compounding frequency (times per year), or number of payments
m	Total number of periodic payments
p	Annual default probability
R	Recovery rate (fraction of par recovered in default)

Time notation

Notation	Meaning
`3y`	3 years
`6m`	6 months
`1w`	1 week
`90d`	90 days

Currency notation

Notation	Usage
100 €	Euro amount in formulas and tables
EUR 100	Euro amount in running text
100 $	US dollar amount in formulas and tables
USD 100	US dollar amount in running text
1 bps	One basis point = 0.01% = 0.0001

Time value of money

Symbol	Meaning
FV	Future value
PV	Present value
NPV	Net present value
$\mathrm{DF}(t)$	Discount factor for maturity t
B	Bond price (dirty price unless stated otherwise)
B_{clean}	Clean price
B_{dirty}	Dirty price (= invoice price)
AI	Accrued interest

Interest rates and curves

Symbol	Meaning
$r(T)$	Spot rate (zero rate) for maturity T
r_{nom}	Nominal interest rate
r_{real}	Real interest rate
$f(T_1, T_2)$	Forward rate for the period $[T_1, T_2]$ (primary notation)
f_{T_1,T_2}	Forward rate (alternative subscript notation)
$f(T)$	Instantaneous forward rate at maturity T
c	Coupon rate (per annum)
C	Coupon payment amount
s	Credit spread, or floating-rate note spread
r_{par}, c_{par}	Par yield / par coupon rate
K	Strike rate (for FRAs, swaps, options)
L_i	Reference rate (e.g. EURIBOR, SOFR) for period i
π	Expected inflation rate

Risk measures

Symbol	Meaning
D_{Mac}	Macaulay duration (years)
D_{mod}	Modified duration
$\mathcal{C}$	Convexity (years^2)
Δ (delta)	Price sensitivity to underlying risk factor
Γ (gamma)	Second-order price sensitivity (convexity)
$\mathcal{V}$ (vega)	Sensitivity to volatility
ρ (rho)	Sensitivity to interest rate
θ (theta)	Sensitivity to time (time decay)
VaR_α	Value at Risk at confidence level α
ES_α	Expected Shortfall at confidence level α

Options

Symbol	Meaning
σ	Implied volatility (log-normal / Black)
σ_n	Implied volatility (normal / Bachelier)
$\Phi(\cdot)$	Cumulative standard normal distribution function
$\phi(\cdot)$	Standard normal density function
F	Forward rate (in Black model context)
S_0	Forward swap rate (in swaption context)
A	Annuity factor / present value of an annuity
$a_{m\rceil r}$	Present value interest factor of an annuity

FX

Symbol	Meaning
S	FX spot rate (quote currency per unit of base currency)
$F(T)$	FX forward rate for maturity T
b	Cross-currency basis spread

Parametric yield curve models

Symbol	Meaning
β_0	Nelson–Siegel / Svensson: long-term level parameter
β_1	Nelson–Siegel / Svensson: short-term (slope) parameter
β_2	Nelson–Siegel / Svensson: first curvature parameter
β_3	Svensson: second curvature parameter
τ	Nelson–Siegel: decay parameter
τ_1	Svensson: first decay parameter
τ_2	Svensson: second decay parameter
$\boldsymbol{\theta}$	Vector of model parameters
w_j	Weighting factor for bond j in calibration

Probability and statistics

Symbol	Meaning
$\mathbb{E}[\cdot]$	Expectation operator
$\mathbb{P}[\cdot]$	Probability measure
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2
z_α	α-quantile of the standard normal distribution
$\mathrm{Cov}(X, Y)$	Covariance between random variables X and Y
$\mathrm{Var}(X)$	Variance of random variable X
ρ	Correlation coefficient

APPENDIX

Glossary of Definitions

D

This glossary provides an alphabetical index of all formal definitions introduced throughout the booklet. Each entry includes the term and a reference to the definition location.

Term	Definition
Amortised loan	52
Annuity (ordinary, fixed)	49
Arbitrage	11
Asset	60
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Continuous compounding	9
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Correlation	120
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Zero-coupon bond (ZCB)	30

Further Reading

The material in this booklet draws on a well-established body of literature in fixed income mathematics, derivatives pricing, and quantitative risk management. The references below are organised thematically and represent the foundational texts from which the core ideas, definitions, and methods presented here originate.

Fixed income fundamentals

[1] **J.C. Hull**, *Options, Futures, and Other Derivatives*, 11th ed., Pearson, 2022.

The standard graduate-level textbook covering derivatives pricing, including bond mathematics, forward rates, swaps, caps/floors, swaptions, and the Black model. Chapters 1–7 and 33–34 of Hull correspond closely to the material in our Chapters 1–5, 11–12, and 14. Hull's treatment of duration, convexity, and the yield curve is the basis for much of the exposition in Chapters 9 and 7.

[2] **P. Veronesi**, *Fixed Income Securities: Valuation, Risk, and Risk Management*, Wiley, 2010.

A comprehensive, mathematically rigorous treatment of fixed income from a finance professor's perspective. Veronesi covers bond pricing, term structure models, duration/convexity, floating-rate notes, and interest rate derivatives with careful attention to both theory and market practice. This text informed our treatment of the spot/forward rate framework (Chapter 5), curve construction (Chapter 7), and the par yield concept (Definition 25).

[3] **B. Tuckman and A. Serrat**, *Fixed Income Securities: Tools for Today's Markets*, 3rd ed., Wiley, 2011.

A practitioner-oriented text that excels at connecting mathematical models to real market conventions. Tuckman and Serrat's treatment of day-count conventions (our Section 1.6), repo markets, key-rate duration, and the OIS curve transition informed our Chapters 1, 7, and 9. The discussion of the OIS vs. LIBOR transition (Definition 40) draws heavily on their exposition.

[4] **F.J. Fabozzi**, *Bond Markets, Analysis, and Strategies*, 10th ed., Pearson, 2021.

The classic reference for bond market practitioners. Fabozzi covers bond pricing, yield measures, duration, convexity, the term structure, mortgage-backed securities, and credit analysis. Our treatment of clean/dirty prices (Definition 22), pull to par (Definition 26), and credit spreads (Definition 28) follows the conventions established in this text.

Yield curve modelling

[5] **C.R. Nelson and A.F. Siegel**, "Parsimonious Modeling of Yield Curves", *Journal of Business*, 60(4):473–489, 1987.

The original paper introducing the Nelson–Siegel parametric yield curve model (Appendix B). This three-factor model (level, slope, curvature) remains the workhorse for central bank yield curve estimation worldwide.

[6] **L.E.O. Svensson**, "Estimating Forward Interest Rates with the Extended Nelson & Siegel Method", Sveriges Riksbank Working Paper No. 15, 1994.

The extension of Nelson–Siegel with a second curvature term (Equation (B.3) in Appendix B), used by the ECB, Bundesbank, and many other central banks for official yield curve estimation.

Derivatives pricing and the Black model

[7] **F. Black**, "The Pricing of Commodity Contracts", *Journal of Financial Economics*, 3(1–2):167–179, 1976.

The foundational paper adapting the Black–Scholes framework to forward prices and futures, yielding "Black's formula" (Black-76). This is the pricing model presented in Section 14.4.2 for caplets and swaptions.

[8] **P.S. Hagan, D. Kumar, A.S. Lesniewski, and D.E. Woodward**, "Managing Smile Risk", *Wilmott Magazine*, pp. 84–108, January 2002.

The paper introducing the SABR stochastic volatility model, which has become the industry standard for cap/floor and swaption smile modelling. The volatility surface and smile dynamics discussed in Section 14.5 motivate the need for models of this type.

Risk measurement

[9] **A.J. McNeil, R. Frey, and P. Embrechts**, *Quantitative Risk Management: Concepts, Techniques and Tools*, 2nd ed., Princeton University Press, 2015.

The definitive mathematical treatment of VaR, Expected Shortfall, coherent risk measures, and extreme value theory. Chapter 15 draws on this text's framework for risk measurement, particularly the definitions of VaR (Definition 99) and ES (Definition 102), the discussion of coherence, and the comparison of parametric, historical, and Monte Carlo approaches.

[10] **P. Jorion**, *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed., McGraw-Hill, 2006.

The classic practitioner reference on VaR methodology, covering parametric, historical, and Monte Carlo VaR, backtesting, and stress testing. Jorion's exposition informed our treatment of the three VaR methods and the discussion of limitations in Chapter 15.

Mathematical finance and stochastic calculus

[11] **S.E. Shreve**, *Stochastic Calculus for Finance II: Continuous-Time Models*, Springer, 2004.

A formal introduction to stochastic calculus, risk-neutral pricing, and the Black–Scholes–Merton framework, aimed at mathematicians entering quantitative finance. While this booklet does not require stochastic calculus, Shreve's text provides the natural next step for readers who wish to understand the measure-theoretic foundations of the pricing formulas in Chapters 14 and 5.

[12] **D. Brigo and F. Mercurio**, *Interest Rate Models — Theory and Practice: With Smile, Inflation and Credit*, 2nd ed., Springer, 2006.

The most comprehensive reference on interest rate modelling, covering short-rate models (Vasicek, CIR, Hull–White), the HJM framework, the LIBOR market model, and smile modelling (SABR, local vol). This text is the natural continuation for readers who wish to go beyond the introductory level of our Chapter 14 and Appendix B.

Probability and statistics

[13] **S.M. Ross**, *A First Course in Probability*, 10th ed., Pearson, 2019.

A clear and accessible introduction to probability theory covering sample spaces, random variables, expectation, variance, covariance, and the key distributions used throughout quantitative finance. The definitions and theorems in Appendix A follow the conventions established in this text, including the treatment of the normal distribution (Definition 121), the log-normal distribution (Definition 123), and the standard normal table (Table A.1).

[14] **P. Billingsley**, *Probability and Measure*, 3rd ed., Wiley, 1995.

The measure-theoretic foundation underpinning the probability space framework (Definition 111) presented in Appendix A. Recommended for readers who wish to understand the rigorous mathematical basis for risk-neutral pricing and martingale theory as used in Chapter 14.

FX and cross-currency markets

[15] **K. Nomura (Bank for International Settlements)**, "Cross-currency basis swaps", *BIS Quarterly Review*, March 2008; and **C. Borio, R. McCauley, P. McGuire, and V. Sushko**, "Covered interest parity lost: understanding the cross-currency basis", *BIS Quarterly Review*, September 2016.

These BIS publications document the empirical breakdown of Covered Interest Parity (our Chapter 13) and the emergence of persistent cross-currency basis spreads (Definition 86). They provide the institutional context for the CIP violations discussed in the remark following Definition 86.

Note. References [1]–[4] are the primary sources for the "textbook" material (definitions, examples, conventions) in Chapters 1–8. References [5]–[6] are the original research papers behind the Nelson–Siegel and Svensson models (Appendix B). References [7]–[8] are the original research papers behind specific pricing models and formulas in Chapter 14. References [9]–[10] underpin Chapter 15. References [11]–[12] provide the mathematical finance foundations for readers wishing to go deeper. References [13]–[14] support the statistical material in Appendix A. Reference [15] gives the empirical backdrop for Chapter 13.